

Single Variable Calculus: 240+ Problems

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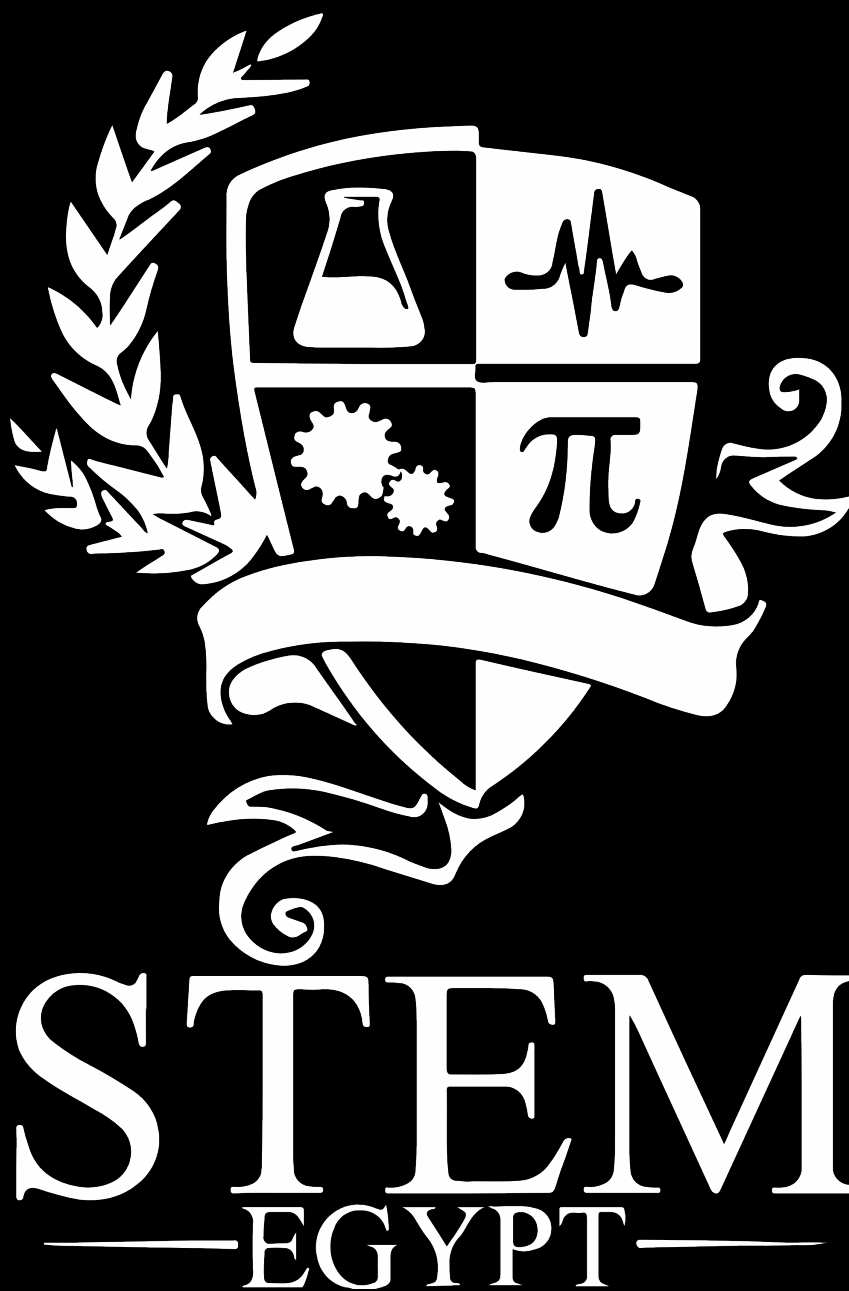


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1 Summary

1.1 Common Limit Identities

1. $\lim_{x \rightarrow a} \frac{x^b - a^b}{x^c - a^c} = \frac{b}{c} a^{b-c}$
2. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
3. $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1$
4. $\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$
5. $\lim_{x \rightarrow 0} \frac{\sin(ax)}{\sin(bx)} = \frac{a}{b}$
6. $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$
7. $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
8. $\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1$
9. $\lim_{x \rightarrow 0} \frac{\tan(ax)}{bx} = \frac{a}{b}$
10. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$
11. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
12. $\lim_{x \rightarrow \infty} e^x = \infty$
13. $\lim_{x \rightarrow -\infty} e^x = 0$
14. $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$
15. $\lim_{x \rightarrow 0} (1+ax)^{b/x} = e^{ab}$
16. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$
17. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
18. $\lim_{x \rightarrow \infty} \ln x = \infty$
19. $\lim_{x \rightarrow 0^+} \ln x = -\infty$
20. $\lim_{x \rightarrow 1} \frac{\log_a x}{x-1} = \frac{1}{\ln a}$
21. $\lim_{x \rightarrow 0^+} x \log_a x = 0$

22. Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$,
 then $\lim_{x \rightarrow a} g(x) = L$.

where a, b, c are constants and x is a variable.

1.2 Differentiation Rules and Applications

1. Derivative Definition:

$$\frac{d}{dx} [f(x)] = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

2. Power Rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

3. Product Rule:

$$\begin{aligned}\frac{d}{dx} [f(x)g(x)] &= f'(x)g(x) + f(x)g'(x) \\ \frac{d}{dx} [f(x)h(x)g(x)] &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x) \\ \frac{d}{dx} (f_1 f_2 \dots f_n) &= f_1' f_2 f_3 \dots f_n + f_1 f_2' f_3 \dots f_n + \dots + f_1 f_2 f_3 \dots f_n'\end{aligned}$$

4. Quotient Rule:

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x)^2)}$$

5. Chain Rule:

$$\frac{d}{dx} [(f \circ g)(x)] = \frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$$

6. Inverse Function Rule:

$$\frac{d}{dx} [f^{-1}(x)] = \frac{1}{f'(f^{-1}(x))}$$

7. The First Derivative Theorem for Local Extreme Values:

If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c , then

$$f'(c) = 0$$

8. Rolle's Theorem:

Suppose that $y = f(x)$ is continuous at every point of the closed interval $[a, b]$ and differentiable at every point of its interior (a, b) . If $f(a) = f(b)$, then there is at least one number c in (a, b) at which $f'(c) = 0$.

9. The Mean Value Theorem:

Suppose $y = f(x)$ is continuous on a closed interval $[a, b]$ and differentiable on the interval's interior (a, b) . Then there is at least one point c in (a, b) at which

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

10. L'Hôpital's rule:

Suppose that $f(a) = g(a) = 0$, that f and g are differentiable on an open interval I containing a , and that $g'(x) \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of this equation exists.

11. Line Equations:

- **Tangent Line:** The equation of the line tangent to the curve at a point (x_1, y_1) is given by:

$$y - y_1 = f'(x_1) \cdot (x - x_1)$$

- **Normal Line:** The equation of the line normal to the tangent of the curve at the point (x_1, y_1) is given by:

$$y - y_1 = \frac{-1}{f'(x_1)} \cdot (x - x_1)$$

1.3 Common Derivatives

1. $\frac{d}{dx}(a) = 0$
2. $\frac{d}{dx}(x) = 1$
3. $\frac{d}{dx}[\sin(x)] = \cos(x)$
4. $\frac{d}{dx}[\cos(x)] = -\sin(x)$
5. $\frac{d}{dx}[\tan(x)] = \sec^2(x)$
6. $\frac{d}{dx}[\cot(x)] = -\csc^2(x)$
7. $\frac{d}{dx}[\sec(x)] = \sec(x) \tan(x)$
8. $\frac{d}{dx}[\csc(x)] = -\csc(x) \cot(x)$
9. $\frac{d}{dx}[\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}}$
10. $\frac{d}{dx}[\cos^{-1}(x)] = -\frac{1}{\sqrt{1-x^2}}$
11. $\frac{d}{dx}[\tan^{-1}(x)] = \frac{1}{1+x^2}$
12. $\frac{d}{dx}[\cot^{-1}(x)] = -\frac{1}{1+x^2}$
13. $\frac{d}{dx}[\sec^{-1}(x)] = \frac{1}{x\sqrt{x^2-1}}$
14. $\frac{d}{dx}[\csc^{-1}(x)] = -\frac{1}{x\sqrt{x^2-1}}$
15. $\frac{d}{dx}(a^x) = a^x \ln a$
16. $\frac{d}{dx}(e^x) = e^x$
17. $\frac{d}{dx}[\log_a(x)] = \frac{1}{x \ln(a)}$
18. $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$
19. $\frac{d}{dx}[\ln|x|] = \frac{1}{x}$
20. $\frac{d}{dx}(|x|) = \frac{x}{|x|}$

where a is an arbitrary constant and x is the variable.

1.4 Common General Derivatives Forms

For a real function $g(x)$, we can generalize the following using the chain rule:

1. $\frac{d}{dx} [|g(x)|] = \frac{g'(x)g(x)}{|g(x)|}$
2. $\frac{d}{dx} \left(\sqrt[n]{g(x)^a} \right) = \frac{ag'(x)}{n \sqrt[n]{g(x)^{n-a}}} \implies \frac{d}{dx} \left(\sqrt{g(x)} \right) = \frac{g'(x)}{2\sqrt{g(x)}}$
3. $\frac{d}{dx} (g(x)^n) = ng(x)^{n-1}g'(x)$
4. $\frac{d}{dx} (\ln(g(x))) = \frac{g'(x)}{g(x)}$
5. $\frac{d}{dx} (\ln |g(x)|) = \frac{g'(x)}{g(x)}$
6. $\frac{d}{dx} \left(e^{g(x)} \right) = g'(x)e^{g(x)}$
7. $\frac{d}{dx} \left(a^{g(x)} \right) = g'(x) \ln(a) a^{g(x)}$
8. $\frac{d}{dx} (f_1 f_2 \dots f_n) = f'_1 f_2 f_3 \dots f_n + f_1 f'_2 f_3 \dots f_n + \dots + f_1 f_2 f_3 \dots f'_n \implies \frac{d}{dx} [f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$

1.5 Integration Rules

1. Definite Integral Definition:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x = \int_a^b f(x) \, dx$$

2. Left-hand Riemann Sum:

$$L_n = \Delta x \sum_{k=0}^{n-1} f(x_k)$$

where n is the number of rectangles.

3. Right-hand Riemann Sum:

$$R_n = \Delta x \sum_{k=1}^n f(x_k)$$

where n is the number of rectangles.

4. Midpoint Riemann Sum:

$$M_n = \Delta x \sum_{k=0}^{n-1} f\left(\frac{x_k + x_{k+1}}{2}\right)$$

where n is the number of rectangles.

5. Integration by “u” Substitution:

$$\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du$$

where $u = g(x)$ and $du = g'(x) \, dx$.

6. Integration by Parts:

$$\int f(x)g'(x) \, dx = f(x)g(x) - \int f'(x)g(x) \, dx$$

7. Mean Value Theorem for Definite Integrals:

If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{\int_a^b f(x) \, dx}{b - a}$$

1.6 Common Integrals

$$1. \int k \, dx = k \int dx = kx + C$$

$$2. \int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$3. \int \frac{1}{x} \, dx = \ln|x| + C$$

$$4. \int \frac{1}{ax+b} \, dx = \frac{\ln|ax+b|}{a} + C$$

$$5. \int \ln(x) \, dx = x \ln(x) - x + C$$

$$6. \int e^x \, dx = e^x + C$$

$$7. \int \cos(x) \, dx = \sin(x) + C$$

$$8. \int \sin(x) \, dx = -\cos(x) + C$$

$$9. \int \sec^2(x) \, dx = \tan(x) + C$$

$$10. \int \sec(x) \tan(x) \, dx = \sec(x) + C$$

$$11. \int \csc(x) \cot(x) \, dx = -\csc(x) + C$$

$$12. \int \csc^2(x) \, dx = -\cot(x) + C$$

$$13. \int \tan(x) \, dx = \ln|\sec(x)| + C$$

$$14. \int \sec(x) \, dx = \ln|\sec(x) + \tan(x)| + C$$

where a and b are arbitrary constants, x is the variable, and C is a constant.

1.7 Common Trigonometric Identities

1. Reciprocal Identities

$$(a) \csc(\theta) = \frac{1}{\sin(\theta)}$$

$$(b) \sec(\theta) = \frac{1}{\cos(\theta)}$$

$$(c) \cot(\theta) = \frac{1}{\tan(\theta)}$$

$$(d) \sin(\theta) = \frac{1}{\csc(\theta)}$$

$$(e) \cos(\theta) = \frac{1}{\sec(\theta)}$$

$$(f) \tan(\theta) = \frac{1}{\cot(\theta)}$$

2. Pythagorean Identities

$$(a) \sin^2(\theta) + \cos^2(\theta) = 1$$

$$(b) 1 + \tan^2(\theta) = \sec^2(\theta)$$

$$(c) 1 + \cot^2(\theta) = \csc^2(\theta)$$

3. Even/Odd Identities

$$(a) \sin(-\theta) = -\sin(\theta)$$

$$(b) \cos(-\theta) = \cos(\theta)$$

$$(c) \tan(-\theta) = -\tan(\theta)$$

$$(d) \csc(-\theta) = -\csc(\theta)$$

$$(e) \sec(-\theta) = \sec(\theta)$$

$$(f) \cot(-\theta) = -\cot(\theta)$$

4. Periodic Identities

$$(a) \sin(\theta + 2\pi n) = \sin(\theta)$$

$$(b) \cos(\theta + 2\pi n) = \cos(\theta)$$

$$(c) \tan(\theta + \pi n) = \tan(\theta)$$

$$(d) \csc(\theta + 2\pi n) = \csc(\theta)$$

$$(e) \sec(\theta + 2\pi n) = \sec(\theta)$$

$$(f) \cot(\theta + \pi n) = \cot(\theta)$$

5. Double Angle Identities

$$(a) \sin(2\theta) = 2\sin(\theta)\cos(\theta)$$

$$(b) \cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

$$(c) \cos(2\theta) = 2\cos^2(\theta) - 1$$

$$(d) \cos(2\theta) = 1 - 2\sin^2(\theta)$$

$$(e) \tan(2\theta) = \frac{2\tan(\theta)}{1 - \tan^2(\theta)}$$

6. Half-Angle Identities

$$(a) \sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos(\theta)}{2}}$$

$$(b) \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos(\theta)}{2}}$$

$$(c) \tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos(\theta)}{1 + \cos(\theta)}}$$

7. Cofunction Identities

$$(a) \sin\left(\frac{\pi}{2} - \theta\right) = \cos(\theta)$$

$$(b) \cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$$

$$(c) \tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta)$$

$$(d) \cot\left(\frac{\pi}{2} - \theta\right) = \tan(\theta)$$

$$(e) \sec\left(\frac{\pi}{2} - \theta\right) = \csc(\theta)$$

$$(f) \csc\left(\frac{\pi}{2} - \theta\right) = \sec(\theta)$$

8. Sum and Difference Identities

$$(a) \sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$$

$$(b) \sin(\alpha - \beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta)$$

$$(c) \cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$

$$(d) \cos(\alpha - \beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$$

$$(e) \tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)}$$

$$(f) \tan(\alpha - \beta) = \frac{\tan(\alpha) - \tan(\beta)}{1 + \tan(\alpha)\tan(\beta)}$$

9. Product-to-Sum Identities

$$(a) \sin(\alpha)\sin(\beta) = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$(b) \cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$(c) \sin(\alpha)\cos(\beta) = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$(d) \cos(\alpha)\sin(\beta) = \frac{1}{2}[\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

10. Sum-to-Product Identities

$$(a) \sin(\alpha) + \sin(\beta) = 2\sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$(b) \sin(\alpha) - \sin(\beta) = 2\cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$(c) \cos(\alpha) + \cos(\beta) = 2\cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$(d) \cos(\alpha) - \cos(\beta) = -2\sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$(e) \tan(\alpha) + \tan(\beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha)\cos(\beta)}$$

$$(f) \tan(\alpha) - \tan(\beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha)\cos(\beta)}$$

11. Additional Identities

$$(a) 1 - \cos(2x) = 2\sin^2(x)$$

$$(b) 1 + \cos(2x) = 2\cos^2(x)$$

$$(c) \sin(x) = \sin(\pi - x)$$

$$(d) \cos(x) = -\cos(\pi - x)$$

$$(e) \tan(x) = -\tan(\pi - x)$$

1.8 Approximations in Calculus (Limits and Derivatives)

Use them with hard question as a shortcut

Binomial Approximation

Use it for very small values of x (most of the time for $|x| < 0.1$). It is most commonly used in limits where x approaches zero.

$$(1+x)^n \approx 1+nx, \quad (\text{First-order binomial approximation})$$

For more accuracy (Second-order) (It's rare to use it):

$$(1+x)^n \approx 1+nx + \frac{n(n-1)}{2}x^2.$$

Example: Approximating $(1+x)^{1/2}$ for small x :

$$(1+x)^{1/2} \approx 1 + \frac{1}{2}x.$$

Taylor Series Expansions for Common Functions

Use them when you face difficult limits or derivatives that need to be evaluated. Additionally, the approximation closely matches the exact value after the first four terms. Therefore, it is recommended to memorize the first four terms of each series and using them as needed.

Exponential Function e^x

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Sine Function $\sin x$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Note: To memorize it, remember that all the powers and factorials are typically the odd numbers and the signs alternates with each term.

Cosine Function $\cos x$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Note: To memorize it, remember that all the powers and factorials are typically the even numbers and the signs alternates with each term.

Tangent Function $\tan x$ For small x :

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$$

Note: You don't want to see the summation notation form (Trust me)

Note: Hi, Hymzam was here, and bruh trust him, you **REALLY** do **NOT** want to see this ~~shit~~ *praying emoji*

1.9 Statistics

1. Arithmetic Mean:

$$\mu = \frac{\sum_{k=1}^n x_k}{n}$$

where μ is the mean, n is the number of values, and x_k is the k th value.

2. Population's Standard Deviation:

$$\sigma_P = \sqrt{\frac{\sum_{k=1}^n (x_k - \mu)^2}{n}}$$

where σ_P is the population's standard deviation.

3. Sample's Standard Deviation c

$$\sigma_S = \sqrt{\frac{\sum_{k=1}^n (x_k - \mu)^2}{n - 1}}$$

where σ_S is the sample's standard deviation.

4. z-score Using μ and σ :

$$z_k = \frac{k - \mu}{\sigma}$$

where z_k is the z-score of k , a value from the data set.

5. The Probability Density Function:

$$P(x < z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{x^2}{2}} dx \approx \frac{1}{\sqrt{2\pi}} \int_{-4}^z e^{-\frac{x^2}{2}} dx$$

where P is the probability of the z-score ($\in \mathbb{R}$).

6. Percentile:

$$P_k = \frac{k}{100}(n + 1)$$

where P_k is the k th percentile.

7. Decile:

$$D_k = \frac{k}{10}(n + 1)$$

where D_k is the k th decile.

8. Quartile:

$$Q_k = \frac{k}{4}(n + 1)$$

where Q_k is the k th quartile.

2 Past Exams Problems

2.1 Midterm, 2016–2017

1. If f is an odd function and $\lim_{x \rightarrow 0} f(x)$ exists, then

$$\lim_{x \rightarrow 0} [2 - 3f(x)]$$

is

- (a) -1
- (b) 0
- (c) 2
- (d) Non-existent

Answer: c

Solution: Since f is an odd function ($f(-x) = -f(x)$) and the limit at 0 exists, $f(0) = 0$ and, hence, $\lim_{x \rightarrow 0} [2 - 3f(x)] = 2 - 0 = 2$.

- 2.

$$\lim_{x \rightarrow 0} \frac{\sin(x) - \tan(x)}{x^2} = \dots\dots$$

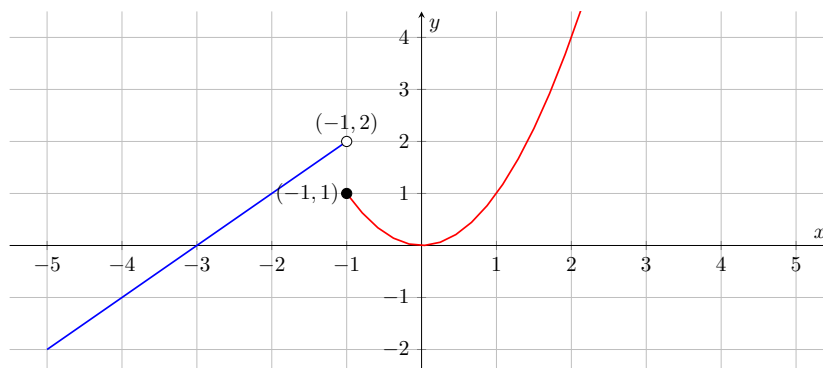
- (a) ∞
- (b) 1
- (c) 0
- (d) Non-existent

Answer: c

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(x) - \tan(x)}{x^2} &= \lim_{x \rightarrow 0} \frac{\sin(x) - \frac{\sin(x)}{\cos(x)}}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{-\sin(x)(1 - \cos(x))}{\cos(x)x^2} = \lim_{x \rightarrow 0} \frac{(1 - \cos(x))}{x^2} \times \left(-\frac{\sin(x)}{\cos(x)} \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{2}(-\tan(x)) = \frac{1}{2} \times (-\tan(0)) = 0 \end{aligned}$$

3. $\lim_{x \rightarrow -1} f(x) = \dots\dots$



- (a) -1
- (b) 1
- (c) 2
- (d) Non-existent

Answer: d

Solution: At $x = -1$, the limits from two sides (right and left) are not equal.

4. Let

$$f(x) = \begin{cases} x^2 - 1 & \text{if } x \neq 1 \\ 4 & \text{if } x = 1 \end{cases}$$

Which of the following statements is (are) true?

- I.* $\lim_{x \rightarrow 1} f(x)$ exists
- II.* $f(1)$ is defined
- III.* f is continuous at $x = 1$

- (a) *I* only
- (b) *II* only
- (c) *I* and *II*
- (d) All of them

Answer: c

Solution: *I* is correct because, at 1, the limits from both sides (right and left) are equal. *II* is right because $f(1)$ is defined in the piecewise function to output 4. *III* is not correct because the outputted value when $x = 1$ is not equal to the limits from both sides (right and left).

5. Given that L and M are distinct roots of the equation $ax^2 + bx + c = 0$, then

$$\lim_{x \rightarrow L} \frac{\sin^2(ax^2 + bx + c)}{(x - L)^2} = \dots\dots$$

- (a) 0
- (b) 1
- (c) $a(L - M)$
- (d) $a^2(L - M)^2$

Answer: d

Solution:

$$\lim_{x \rightarrow L} \frac{\sin^2(ax^2 + bx + c)}{(x - L)^2} = \lim_{x \rightarrow L} \left(\frac{\sin[a(x - L)(x - M)]}{(x - L)} \right)^2 = a^2(L - M)^2$$

6. If f is an even function, and $\lim_{x \rightarrow 0^-} f(x) = 3$, then

- I.* $\lim_{x \rightarrow 0^+} f(x) = 3$
- II.* $\lim_{x \rightarrow 0} f(x) = 3$
- III.* $\lim_{x \rightarrow 0} f(x)$ does not exist

- (a) *I* only
- (b) *II* only
- (c) *I* and *II*
- (d) None of them

Answer: c

Solution: Since f is an even function, $f(-x) = f(x)$, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 3$, and, hence, *I* and *II* are true. *III* is false, as we showed $\lim_{x \rightarrow 0} f(x)$ does exist.

7. If x is measured in degrees, then

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \dots\dots$$

- (a) $\frac{\pi}{180}$
- (b) $\frac{180}{\pi}$
- (c) 1
- (d) 0

Answer: a

Solution:

$$1^\circ = \frac{\pi}{180} \implies \lim_{x \rightarrow 0} \frac{\sin(x^\circ)}{x} = \lim_{x \rightarrow 0} \frac{\sin(\frac{\pi}{180}x)}{x} = \frac{\pi}{180}$$

8. If

$$f(x) = \begin{cases} x^2 - 2x & x > 2 \\ x - 3 & x < 2 \end{cases}$$

and

$$g(x) = \begin{cases} 2x + 1 & x > 2 \\ 2x + 2 & x < 2 \end{cases}$$

then

$$\lim_{x \rightarrow 2} (f(x) + g(x))$$

is

- (a) 2
- (b) 4
- (c) 5
- (d) Non-existent

Answer: c

Solution: We shall take $f(x) + g(x)$'s limit from both sides (right and left):

$$\begin{aligned} \lim_{x \rightarrow 2^+} (f(x) + g(x)) &= [2^2 - 2(2)] + [2(2) + 1] = 5 \\ \lim_{x \rightarrow 2^-} (f(x) + g(x)) &= [2 - 3] + [2(2) + 2] = 5 \end{aligned}$$

Therefore, the answer is 5.

9. The function $f : R \rightarrow R$ is differentiable and $f(-5) = f(9) = 0$. The function $g(x) = \cos(f(x))$ satisfies the condition:

- (a) g' has at least 3 zeros.
- (b) g' has exactly 3 zeros.
- (c) g' has at most one zero.
- (d) g' has no zeros.

Answer: a

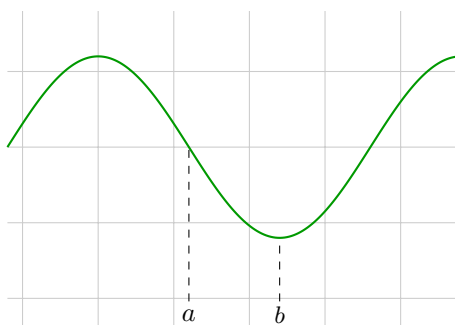
Solution:

$$g'(x) = -\sin(f(x)) f'(x)$$

- Since $f(x)$ has at least two zeros, $\sin(f(x))$ also has at least two zeros.
- By **Rolle's Theorem**, $f'(x)$ has at least one zero over the interval $[-5, 9]$

Then, g' has at least three zeros.

10. Which of the following represents the slope a line drawn between the two points marked in the figure?



- (a) $m = \frac{f(a)+f(b)}{a+b}$
 (b) $m = \frac{a}{b}$
 (c) $m = \frac{f(b)-f(a)}{b-a}$
 (d) $m = \frac{f(a)-f(b)}{b-a}$

Answer: c

Solution:

$$m = \frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

11. Consider the function $f(x) = |x - 2|$. Then

- (a) $f(x)$ is differentiable on R
 (b) $f'(2) = 1$
 (c) $f'(2)$ doesn't exist.
 (d) f is discontinuous at $x = 2$

Answer: c

Solution: $f'(2)$ doesn't exist because:

- Graphically, $f(x)$ has a **corner** at $x = 2$.
- Algebraically,

$$f'(x) = \frac{x-2}{|x-2|}$$

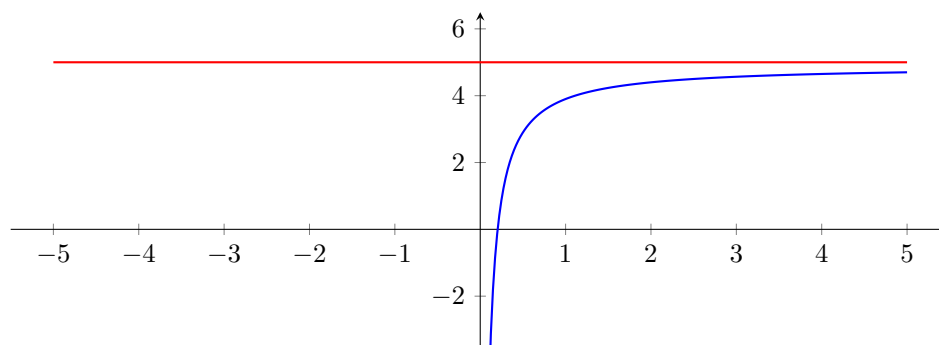
which is undefined at $x = 2$.

12. For $x \geq 0$, the line $y = 5$ is an asymptote for the graph of the function f and $f'(x) \geq 0$. Which of the following statements must be true?

- (a) $f'(5)$ is undefined.
 (b) $\lim_{x \rightarrow \infty} f'(x) = 0$.
 (c) $f''(x) < 0$ for all $x \geq 0$.
 (d) $\lim_{x \rightarrow 5} f(x) = \infty$

Answer: b

Solution: The graph of this function is near horizontal approaching $y = 5$.



13. If the function $f(x) = \begin{cases} bx^2 + 1 & x \leq 2 \\ ax - 3 & x > 2 \end{cases}$ is differentiable at $x = 2$, which of the following is the value of a ?

- (a) 5
- (b) 4
- (c) 3
- (d) 2

Answer: b

Solution:

$$\begin{aligned} \lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^-} f(x) \\ 2a - 3 &= 4b + 1 \\ a &= 2b + 2 \end{aligned} \tag{1}$$

$$\begin{aligned} \lim_{x \rightarrow 2^+} f'(x) &= \lim_{x \rightarrow 2^-} f'(x) \\ \lim_{x \rightarrow 2^+} a &= \lim_{x \rightarrow 2^-} 2bx \\ a &= 4b \end{aligned} \tag{2}$$

From (1) and (2), $a = 4$.

14. If $f(x_0) = 0$ and x_0 is a corner point of $f(x)$. Then the function $g(x) = (x - x_0)f(x) \dots \dots$

- (a) is differentiable at x_0
- (b) is not differentiable at x_0
- (c) is discontinuous at x_0
- (d) has a corner point at x_0

Answer: a

Solution:

$$\begin{aligned} g'(x_0) &= \lim_{h \rightarrow 0} \frac{(x_0 - x_0 + h)f(x_0 + h) - (x_0 - x_0)f(x_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{hf(x_0 + h)}{h} = f(x_0) \end{aligned}$$

Since f is continuous at $x = x_0$, $g'(x_0) = f(x_0)$ is defined and g is differentiable at $x = x_0$.

15. Let $f(x) = x^3$. Find the value c that satisfies the mean value theorem on the closed interval $[1, 3]$.

- (a) 1.414
- (b) 1.732
- (c) 2
- (d) 2.082

Answer: **d**

Solution:

$$f'(c) = \frac{f(3) - f(1)}{3 - 1}$$

$$3c^2 = 13$$

$$c = 2.082$$

2.2 Final Term, 2016–2017

1.

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

is

- (a) 1
- (b) -1
- (c) ± 1
- (d) Non-existent

Answer: d

Solution: The limits from both sides are unequal to each other, and, hence, the limit does not exist; when $x > 0$ (taking limit from the right), $\frac{|x|}{x} = 1$, and when $x < 0$ (taking limit from the left), $\frac{|x|}{x} = -1$.

2.

$$\lim_{x \rightarrow 1} (x - 1) \tan \left(\frac{\pi x}{2} \right)$$

is

- (a) $-\frac{2}{\pi}$
- (b) $\frac{\pi}{2}$
- (c) 1
- (d) Non-existent

Answer: a**Solution:**

$$\begin{aligned}
 \lim_{x \rightarrow 1} (x - 1) \tan \left(\frac{\pi x}{2} \right) &= \lim_{x \rightarrow 1} (x - 1) \cot \left(\frac{\pi}{2} - \frac{\pi x}{2} \right) \\
 &= \lim_{x \rightarrow 1} \frac{(x - 1) \cos \left[\frac{\pi}{2} (1 - x) \right]}{\sin \left[\frac{\pi}{2} (1 - x) \right]} \\
 &= - \lim_{1-x \rightarrow 0} \frac{(1 - x) \cos \left[\frac{\pi}{2} (1 - x) \right]}{\sin \left[\frac{\pi}{2} (1 - x) \right]} \\
 &= - \lim_{1-x \rightarrow 0} \frac{\cos \left[\frac{\pi}{2} (1 - x) \right]}{\frac{\pi}{2}} \\
 &= - \frac{2 \cos \left[\frac{\pi}{2} (0) \right]}{\pi} \\
 &= - \frac{2}{\pi}
 \end{aligned}$$

3.

$$\lim_{x \rightarrow \infty} \frac{x^3 - 2x^2 + 3x - 4}{4x^3 - 3x^2 + 2x - 1} = \dots\dots$$

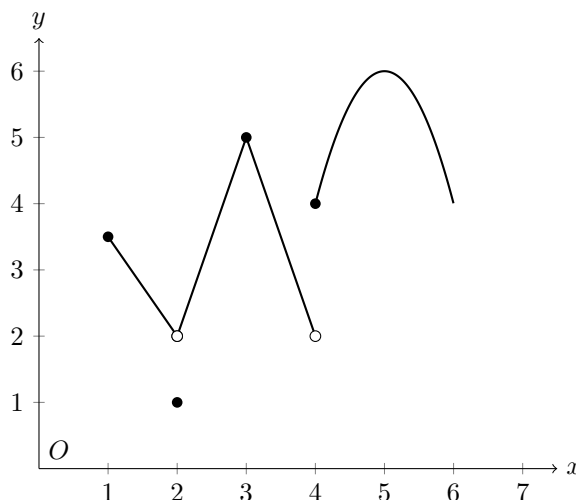
- (a) 4
- (b) 1
- (c) $\frac{1}{4}$
- (d) 0

Answer: c

Solution: Since the rational (with the same-degree numerator and denominator) function's limit is to infinity, the limit will equal the ratio between the leading coefficients:

$$\lim_{x \rightarrow \infty} \frac{x^3 - \cancel{2x^2} + \cancel{3x} - \cancel{4}}{4x^3 - \cancel{3x^2} + \cancel{2x} - \cancel{1}} = \frac{1}{4}$$

4. The graph of the function f is shown below. Which of the following statements is not true?



- (a) $\lim_{x \rightarrow 2} f(x)$ exists
- (b) $\lim_{x \rightarrow 3} f(x)$ exists
- (c) $\lim_{x \rightarrow 4} f(x)$ exists
- (d) The function f is continuous at $x = 3$

Answer: c

Solution: At 4, the limit from right is unequal to the limit from left, and, hence, $\lim_{x \rightarrow 4} f(x)$ does not exist.

5.

$$f(x) = \begin{cases} \frac{2x^2 - 3x - 2}{x - 2} & \text{if } x \neq 2 \\ k & \text{if } x = 2 \end{cases}$$

Let f be the function defined above. For what value of k is f continuous at $x = 2$?

- (a) 0
- (b) 1
- (c) 2
- (d) 5

Answer: d

Solution: For f to be continuous at $x = 2$, the following equation should be valid: $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x) = f(2)$. Knowing that

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(2x + 1)(\cancel{x - 2})}{\cancel{x - 2}} = 2(2) + 1 = 5$$

and that $f(2) = k$, we can deduce that $k = 5$.

6.

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{4x^2} = \dots\dots$$

- (a) $\frac{1}{4}$
- (b) $\frac{1}{2}$
- (c) 1
- (d) 2

*Answer: b**Solution:*

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{4x^2} &= \lim_{x \rightarrow 0} \frac{[\sin^2(x) + \cos^2(x)] - [\cos^2(x) - \sin^2(x)]}{4x^2} \\
&= \lim_{x \rightarrow 0} \frac{2\sin^2(x)}{4x^2} \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sin(x)}{x} \right)^2 \\
&= \frac{1}{2}
\end{aligned}$$

7.

$$f(x) = \begin{cases} cx + d & x \leq 2 \\ x^2 - cx & x > 2 \end{cases}$$

Let f be the function defined above. Where c and d are constants. If f is differentiable at $x = 2$, what is the value of $c + d$?

- (a) -4
- (b) -2
- (c) 0
- (d) 2

*Answer: b**Solution:*

$$\begin{aligned}
\lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^-} f(x) \\
\lim_{x \rightarrow 2^+} x^2 - cx &= \lim_{x \rightarrow 2^-} cx + d \\
4 - 2c &= 2c + d \\
4c &= 4 - d \\
\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} &= \lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} \\
\lim_{h \rightarrow 0^+} \frac{(2+h)^2 - c(2+h) - 2^2 + 2c}{h} &= \lim_{h \rightarrow 0^-} \frac{c(2+h) + d - 2c - d}{h} \\
c &= 2 \\
d &= -4 \\
c + d &= -2
\end{aligned}$$

8. If $f(x) = (x-1)(x-2) \dots (x-2016)(x-2017)$, then $f'(2017) = \dots$

- (a) 0
- (b) 2017
- (c) 2017!
- (d) 2016!

*Answer: d**Solution:* Using the general product rule:

$$\begin{aligned}
\frac{d}{dx}(f_1 f_2 \dots f_n) &= f'_1 f_2 f_3 \dots f_n + f_1 f'_2 f_3 \dots f_n + \dots + f_1 f_2 f_3 \dots f'_n \\
f'(x) &= (x-1)'(x-2) \dots (x-2016)(x-2017) + (x-1)(x-2)'(x-3) \dots (x-2016)(x-2017) \\
&\quad + \dots + (x-1)(x-2) \dots (x-2016)'(x-2017) + (x-1)(x-2) \dots (x-2016)(x-2017)'
\end{aligned}$$

Substituting $x = 2017$ cancels out all terms except the last one.

$$f'(2017) = 2016 \times 2015 \times 2014 \times \dots \times 2 \times 1 = 2016!$$

9. Let f be the function defined by $f(x) = \sqrt{|x-2|}$ for all x . Which of the following is true?

- (a) f is continuous but not differentiable at $x = 2$
- (b) f is not continuous at $x = 2$
- (c) f is differentiable at $x = 2$
- (d) $x = 2$ is vertical asymptote of the graph of f

Answer: a

Solution:

$$\begin{aligned} f(2) &= \sqrt{|2-2|} = 0 \\ \lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^-} f(x) = 0 \\ f(2) &= \lim_{x \rightarrow 2} f(x) = 0 \end{aligned}$$

Then $f(x)$ is continuous at $x = 2$.

The function is not differentiable because:

- Graphically, the graph of $f(x)$ has a **cusp** at $x = 2$.
- Algebraically,

$$f'(x) = \frac{1}{2\sqrt{|x-2|}} \cdot \frac{x-2}{|x-2|}$$

which is undefined at $x = 2$.

10. If $y = |f(x)|$, then $\frac{dy}{dx} = \dots\dots$

- (a) $\frac{f'(x) \cdot f(x)}{|f(x)|}$
- (b) $f'(x)$
- (c) $\frac{1}{|f(x)|}$
- (d) $|f(x)| \cdot f'(x)$

Answer: a

Solution:

$$\begin{aligned} y &= |f(x)| = \left((f(x))^2\right)^{\frac{1}{2}} \\ \frac{dy}{dx} &= \frac{1}{2} \left((f(x))^2\right)^{-\frac{1}{2}} \cdot 2f(x) \cdot f'(x) \\ &= \frac{f(x)}{\sqrt{(f(x))^2}} \cdot f'(x) = \frac{f(x)}{|f(x)|} \cdot f'(x) \end{aligned}$$

Alternative Solution:

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{|f(x+h)| - |f(x)|}{h} = \lim_{h \rightarrow 0} \frac{|f(x+h)| - |f(x)|}{f(x+h) - f(x)} \cdot \frac{f(x+h) - f(x)}{h} \\ &= f'(x) \lim_{h \rightarrow 0} \frac{|f(x+h)| - |f(x)|}{f(x+h) - f(x)} \end{aligned}$$

Let $f(x+h) - f(x) = k$. As $h \rightarrow 0$, $f(x+h) \rightarrow f(x)$ and, consequently, $k \rightarrow 0$. Then, the limit becomes:

$$f'(x) \lim_{k \rightarrow 0} \frac{|f(x) + k| - |f(x)|}{k} = f'(x) \lim_{k \rightarrow 0} \frac{(|f(x) + k| - |f(x)|)(|f(x) + k| + |f(x)|)}{k(|f(x) + k| + |f(x)|)}$$

$$\begin{aligned}
&= f'(x) \lim_{k \rightarrow 0} \frac{(|f(x) + k|)^2 - (|f(x)|)^2}{k(|f(x) + k| + |f(x)|)} = f'(x) \lim_{k \rightarrow 0} \frac{(f(x) + k)^2 - f(x)^2}{k(|f(x) + k| + |f(x)|)} \\
&= f'(x) \lim_{k \rightarrow 0} \frac{2kf(x) + k^2}{k(|f(x) + k| + |f(x)|)} = \frac{f'(x)}{|f(x)|} \cdot f'(x)
\end{aligned}$$

11. Let f and g be differentiable functions such that

$$f(1) = 2, \quad f'(1) = 3, \quad f'(2) = -4$$

$$g(1) = 2, \quad g'(1) = -3, \quad g'(2) = 5$$

If $h(x) = f(g(x))$, then $h'(1) = \dots$

- (a) -9
- (b) -4
- (c) 0
- (d) 12

Answer: d

Solution:

$$h'(x) = f'(g(x)) \cdot g'(x)$$

$$h'(1) = f'(g(1)) \cdot g'(1) = f'(2) \cdot -3 = -4 \times -3 = 12$$

12. Let $f(x) = (2x + 1)^3$ and let g be the inverse function of f . Given that $f(0) = 1$, what is the value of $g'(1)$?

- (a) $-\frac{2}{27}$
- (b) $\frac{1}{54}$
- (c) $\frac{1}{27}$
- (d) $\frac{1}{6}$

Answer: d

Solution:

$$f(0) = 1, \quad g(1) = 0$$

$$f'(x) = 6(2x + 1)^2$$

$$g'(1) = \frac{1}{f'(g(1))} = \frac{1}{f'(0)} = \frac{1}{6}$$

13. Let f be the function given by $f(x) = x^2 - 3x - 2$. Find the number c that satisfies the conclusion of the Mean Value Theorem for f on $[1, 3]$.

- (a) 3
- (b) 2.5
- (c) 2
- (d) 1

Answer: c

Solution:

$$f(3) = 3^2 - 3 \times 3 - 2 = -2$$

$$f(1) = 1^2 - 3 \times 1 - 2 = -4$$

$$f'(x) = 2x - 3$$

$$f'(c) = \frac{f(3) - f(1)}{3 - 1} = 1$$

$$2c - 3 = 1$$

$$c = 2$$

14. Let f be a function that is differentiable on the interval $]1, 10[$. If $f(2) = -5$, $f(5) = 5$, and $f(9) = -5$, which of the following must be true?
- f has at least 2 zeros.
 - The graph of f has at least one horizontal tangent.
- I only
 - II only
 - I and II
 - None of them

Answer: c

Solution:

$$f(2) = -5 < 0, \quad f(5) = 5 > 0$$

By the **intermediate value theorem**, $f(x)$ has at least one zero between $x = 2$ and $x = 5$.

$$f(5) = 5 > 0, \quad f(9) = -5 < 0$$

By the **intermediate value theorem** again, $f(x)$ has at least one zero between $x = 5$ and $x = 9$. Then, $f(x)$ has at least two zeros. $f(x)$ is differentiable over the interval $]1, 10[$. $f(2) = f(9) = -5$. Then, $f(x)$ has at least one horizontal tangent between $x = 2$ and $x = 9$ by **Rolle's Theorem**.

15.

$$\frac{d}{dx} \ln \left| \cos \left(\frac{\pi}{x} \right) \right|$$

- $\frac{-\pi}{x^2 \cos \left(\frac{\pi}{x} \right)}$
- $\frac{\pi}{x} \tan \left(\frac{\pi}{x} \right)$
- $-\tan \left(\frac{\pi}{x} \right)$
- $\frac{\pi}{x^2} \tan \left(\frac{\pi}{x} \right)$

Answer: d

Solution:

$$\begin{aligned} \frac{d}{dx} \ln \left| \cos \left(\frac{\pi}{x} \right) \right| &= \frac{1}{\left| \cos \left(\frac{\pi}{x} \right) \right|} \cdot \frac{\cos \left(\frac{\pi}{x} \right)}{\left| \cos \left(\frac{\pi}{x} \right) \right|} \cdot \left(-\sin \left(\frac{\pi}{x} \right) \right) \cdot \left(\frac{-\pi}{x^2} \right) \\ &= \frac{\pi \sin \left(\frac{\pi}{x} \right) \cos \left(\frac{\pi}{x} \right)}{x^2 \left(\left| \cos \left(\frac{\pi}{x} \right) \right| \right)^2} = \frac{\pi \sin \left(\frac{\pi}{x} \right) \cos \left(\frac{\pi}{x} \right)}{x^2 \cos^2 \left(\frac{\pi}{x} \right)} = \frac{\pi}{x^2} \tan \left(\frac{\pi}{x} \right) \end{aligned}$$

16. If $y = \tan u$, $u = v - \frac{1}{v}$, and $v = \ln x$, what is the value of $\frac{dy}{dx}$ at $x = e$?

- 0
- $\frac{1}{e}$
- 1
- $\frac{2}{e}$

Answer: d

Solution:

$$\begin{aligned} y &= \tan \left(\ln x - \frac{1}{\ln x} \right) \\ y' &= \sec^2 \left(\ln x - \frac{1}{\ln x} \right) \cdot \left(\frac{1}{x} + \frac{1}{x \ln^2 x} \right) \\ \frac{dy}{dx} \Big|_{x=e} &= \sec^2 \left(\ln e - \frac{1}{\ln e} \right) \cdot \left(\frac{1}{e} + \frac{1}{e \ln^2 e} \right) = \frac{2}{e} \end{aligned}$$

Alternative Solution:

$$\begin{aligned}\frac{dy}{du} &= \sec^2 u = \sec^2 \left(\ln x - \frac{1}{\ln x} \right) \\ \frac{du}{dv} &= 1 + \frac{1}{v^2} = 1 + \frac{1}{\ln^2 x} \\ \frac{du}{dx} &= \frac{1}{x} \\ \frac{dy}{dx} \Big|_{x=e} &= \left(\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} \right) \Big|_{x=e} \\ &= \sec^2 \left(\ln e - \frac{1}{\ln e} \right) \cdot \left(1 + \frac{1}{\ln^2 e} \right) \cdot \frac{1}{e} = \frac{2}{e}\end{aligned}$$

17. If $e^{f(x)} = 1 + x^2$, then $f'(x) = \dots\dots$

- (a) $\frac{1}{1+x^2}$
- (b) $\frac{2x}{1+x^2}$
- (c) $2x(1+x^2)$
- (d) $2x(e^{1+x^2})$

Answer: b

Solution:

$$\begin{aligned}\ln(e^{f(x)}) &= \ln(1+x^2) \\ f(x) &= \ln(1+x^2) \\ f'(x) &= \frac{1}{1+x^2} \cdot (2x) = \frac{2x}{1+x^2}\end{aligned}$$

Alternative Solution:

$$\begin{aligned}\frac{d}{dx}(e^{f(x)}) &= \frac{d}{dx}(1+x^2) \\ e^{f(x)} f'(x) &= 2x \\ f'(x) &= \frac{2x}{e^{f(x)}} = \frac{2x}{1+x^2}\end{aligned}$$

18.

$$\lim_{h \rightarrow 0} \frac{\ln(4+h) - \ln(4)}{h} \dots\dots$$

- (a) 0
- (b) $\frac{1}{4}$
- (c) 1
- (d) e

Answer: b

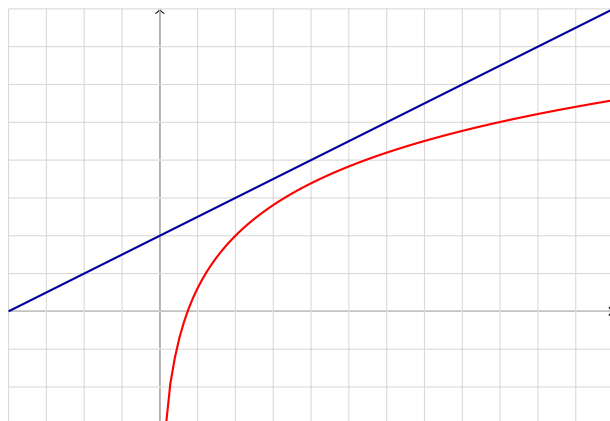
Solution:

$$\begin{aligned}f(x) &= \ln x \\ f'(x) &= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h} = \frac{1}{x} \\ \lim_{h \rightarrow 0} \frac{\ln(4+h) - \ln(4)}{h} &= f'(4) = \frac{1}{4}\end{aligned}$$

Alternative Solution:

$$\lim_{h \rightarrow 0} \frac{\ln(4+h) - \ln(4)}{h} = \lim_{h \rightarrow 0} \frac{\ln\left(\frac{4+h}{4}\right)}{h} = \lim_{h \rightarrow 0} \ln\left(\left(\frac{4+h}{4}\right)^{\frac{1}{h}}\right)$$

$$= \lim_{h \rightarrow 0} \ln \left(\left(1 + \frac{h}{4} \right)^{\frac{1}{h}} \right) = \ln \left(e^{\frac{1}{4}} \right) = \frac{1}{4}$$



19. At which point of the curve $g(x) = \ln x$, the tangent passes through the point of origin?

- (a) $(e, 1)$
- (b) $(\frac{1}{e}, -1)$
- (c) $(2, \ln 2)$
- (d) $(3, \ln 3)$

Answer: a

Solution:

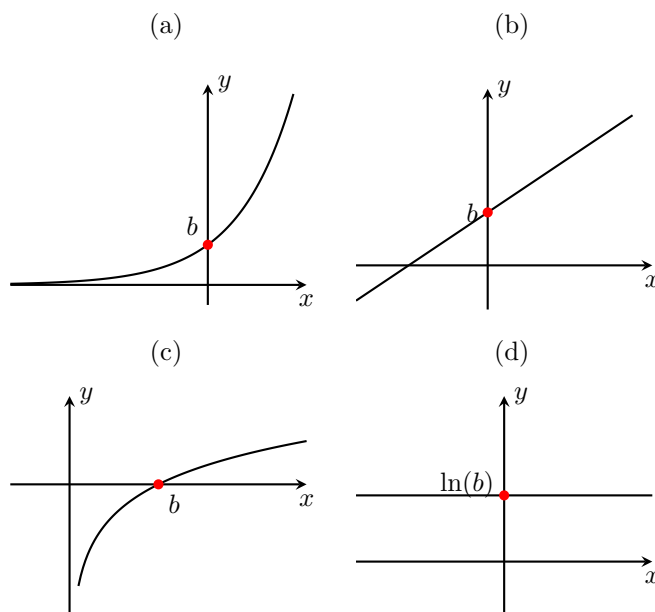
$$g'(x) = \frac{1}{x}$$

Let $(a, g(a))$ be the required point. $g'(a)$ is the slope of the tangent at $x = a$. The y-intercept of the tangent is 0. Then, the equation of the tangent line is $y = \frac{x}{a}$. Since this line passes through the point $(a, g(a))$, the y-coordinate of the point is $y = \frac{a}{a} = 1$.

$$\ln x = 1, \quad x = e$$

Then, the point is $(e, 1)$.

20. Let $f(x) = b^x$ for $b > 1$, and let $g(x) = \frac{f'(x)}{f(x)}$, which of the following graphs could be the graph of $g(x)$:



Answer: d

Solution: $f'(x) = b^x \cdot \ln b$. Substituting in $g(x)$, $g(x) = \frac{b^x \cdot \ln b}{b^x} = \ln b$. Since $\ln b$ is constant, then the graph is a line parallel to the x -axis.

21. Find the area of the region bounded by $y = x^2 - 5x + 4$ and x -axis.

- (a) $\frac{9}{2}$
- (b) $-\frac{9}{2}$
- (c) $\frac{4}{3}$
- (d) $\frac{5}{2}$

Answer: a

Solution: First, we need to find out where $f(x)$ intersects with the x -axis; equating with 0 ($x^2 - 5x + 4 = 0$) and solving for x using the general formula, we get to find two intersections being 1 and 4. Third, we evaluate the definite integral:

$$\begin{aligned} & 0 - \left(\int_1^4 x^2 dx - \int_1^4 5x dx + \int_1^4 4 dx \right) \\ &= - \left(\left[\frac{x^3}{3} \right]_1^4 - 5 \left[\frac{x^2}{2} \right]_1^4 + 4 [x]_1^4 \right) \\ &= - \left(\left[\frac{4^3}{3} - \frac{1}{3} \right] - 5 \left[\frac{4^2}{2} - \frac{1}{2} \right] + 4 [4 - 1] \right) = \frac{9}{2} \end{aligned}$$

22. Estimate the area under the curve for the following function from $x = 0$ to $x = 4$ using a midpoint Riemann sum with $n = 4$ rectangles $f(x) = \frac{1}{4}x^2 + 3$.

- (a) $\frac{57}{16}$
- (b) $\frac{49}{16}$
- (c) $\frac{97}{8}$
- (d) $\frac{69}{4}$

Answer: d

Solution: Having $n = 4$ rectangles from 0 to 4 means that $\Delta x = 1$ and there would be four x_i s that represent the midpoint of each rectangle: $x_1 = 0.5$, $x_2 = 1.5$, $x_3 = 2.5$, and $x_4 = 3.5$. Therefore,

$$\begin{aligned} \text{Area}_{\text{estimation}} &= \Delta x [f(x_1) + f(x_2) + f(x_3) + f(x_4)] \\ &= f(0.5) + f(1.5) + f(2.5) + f(3.5) \\ &= \frac{69}{4} \end{aligned}$$

23. Suppose that f is an integrable function and that $\int_0^1 f(x) dx = 2$, $\int_0^2 f(x) dx = 1$, and $\int_2^4 f(x) dx = 7$. Find $\int_0^4 f(x) dx + \int_1^2 f(x) dx$.

- (a) 2
- (b) 3
- (c) 5
- (d) 7

Answer: d

Solution: Let $\int f(x) dx = g(x) + C$. We can now write $\int_0^4 f(x) dx + \int_1^2 f(x) dx$ as

$$[g(4) - g(0)] + [g(2) - g(1)]$$

We can then write the given equations using $g(x)$ s.

$$\begin{aligned} g(1) - g(0) &= 2 \implies -g(1) = -g(0) - 2 \\ g(2) - g(0) &= 1 \implies -g(0) = -g(2) + 1 \\ g(4) - g(2) &= 7 \end{aligned}$$

Rewriting $[g(4) - g(0)] + [g(2) - g(1)]$ by replacing $-g(1)$ with $-g(0) - 2$ and replacing $-g(0)$ with $-g(2) + 1$, we get

$$[g(4) - g(2)] + 1 + [g(2) - g(0)] - 2 = [7] + 1 + [1] - 2 = 7$$

24. The average temperature for all the days preceding the last for the month of February was 55 degrees Fahrenheit, with a standard deviation of 5 degrees. On the last day, the temperature was 73 degrees F. What is the z-score for the temperature on the last day?

- (a) 2.8
- (b) 3.6
- (c) -3.6
- (d) 1.7

Answer: b

Solution:

$$z = \frac{T - \mu}{\sigma} = \frac{73 - 55}{5} = 3.6$$

25. In a normal curve, $\mu \pm 0.67\sigma$ covers

- (a) 50 % area
- (b) 62.27 % area
- (c) 94.45 % area
- (d) 99.73 % area

Answer: a

Solution: Using the normal distribution table we find the area to be 0.7486 at 0.67σ . To find the area between -0.67σ and 0.67σ , we must subtract the area at -0.67σ , which is $1 - 0.7486 = 0.2514$. Therefore,

$$A_{\text{from } -0.67\sigma \text{ to } 0.67\sigma} = A_{0.67\sigma} - A_{-0.67\sigma} = 0.7486 - 0.2514 \approx 50\%$$

2.3 Midterm, 2017–2018

1. The number of points at which the function $f(x) = \frac{1}{\log|x|}$ is discontinuous is

- (a) 1
(b) 2
(c) 3
(d) 4

Answer: c

Solution: It is discontinuous at 0, -1, and 1.

- 2.

$$\lim_{x \rightarrow 0} \frac{\sqrt{\sec^2(x) - 1}}{x} = \dots\dots$$

- (a) ± 1
(b) 1
(c) 0
(d) Non-existent

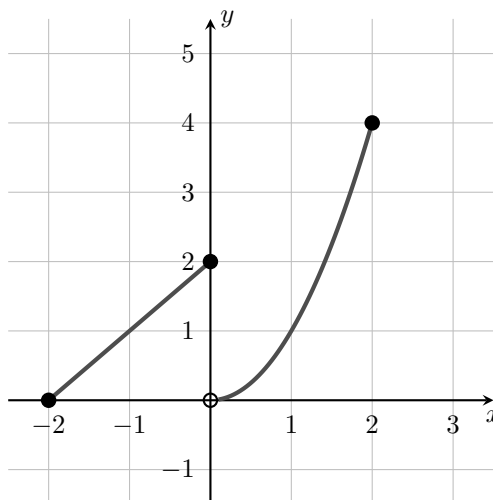
Answer: d

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{\sec^2(x) - 1}}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt{\tan^2(x)}}{x} \\ &= \lim_{x \rightarrow 0} \frac{|\tan(x)|}{x} \end{aligned}$$

which does not exist. This is because $\lim_{x \rightarrow 0^+} \frac{|\tan(x)|}{x} = 1$ and $\lim_{x \rightarrow 0^-} \frac{|\tan(x)|}{x} = -1$, which are unequal, deeming the limit to not exist.

3. $\lim_{x \rightarrow 2} f(x)$ is



- (a) 0
(b) 2
(c) 4
(d) Non-existent

Answer: d

Solution: For the limit to exist at 2, the limit from both right and left must exist and equal each other; in the provided figure, the limit from the right does not exist.

4. Let

$$f(x) = \begin{cases} ax - 3 & \text{if } x < 2 \\ 4 & \text{if } x = 2 \\ 2x + b & \text{if } x > 2 \end{cases}$$

be a continuous function in which $x \in \mathbb{R}$. Then

$$a + b = \dots\dots$$

- (a) $\frac{5}{2}$
- (b) 7
- (c) $\frac{7}{2}$
- (d) -3

Answer: c

Solution:

$$2(2) + b = 4 \implies b = 0$$

$$a(2) - 3 = 4 \implies a = \frac{7}{2}$$

$$a + b = \frac{7}{2} + 0 = \frac{7}{2}$$

5.

$$\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2(x))}{x^2} = \dots\dots$$

- (a) $-\pi$
- (b) π
- (c) $\frac{\pi}{2}$
- (d) 1

Answer: b

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2(x))}{x^2} &= \lim_{x \rightarrow 0} \frac{\frac{d}{dx}[\sin(\pi \cos^2(x))]}{\frac{d}{dx}[x^2]} \\ &= \lim_{x \rightarrow 0} \frac{\cos(\pi \cos^2(x)) \times 2\pi \cos(x) \times (-\sin(x))}{2x} \\ &= \lim_{x \rightarrow 0} -\pi \cos(\pi \cos^2(x)) \cos(x) \\ &= -\pi \cos(\pi \cos^2(0)) \cos(0) \\ &= -\pi \cos(\pi) = \pi \end{aligned}$$

6. If

$$f(x) = \begin{cases} \frac{x - |x|}{x} & \text{if } x \neq 0 \\ 2 & \text{if } x = 0 \end{cases}$$

Then

$$f(0^+) = \dots\dots$$

- (a) 0
- (b) 2
- (c) 4
- (d) Non-existent

Answer: a

Solution: The first function “piece” can be rewritten as $1 - \frac{|x|}{x}$, which would make it clear that it would equal either $1 - 1 = 0$ (from the right) or $1 - -1 = 2$ (from the left). Therefore, $\lim_{x \rightarrow 0^+} f(x)$ will yield 0.

7.

$$\lim_{x \rightarrow -2} \frac{(x^2 - x - 6)^2}{(x + 2)^2} = \dots\dots$$

- (a) 6
- (b) 25
- (c) 9
- (d) 16

*Answer: b**Solution:*

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{(x^2 - x - 6)^2}{(x + 2)^2} &= \lim_{x \rightarrow -2} \left(\frac{(x^2 - x - 6)}{(x + 2)} \right)^2 \\ &= \lim_{x \rightarrow -2} \left(\frac{\cancel{(x+2)}(x-3)}{\cancel{(x+2)}} \right)^2 \\ &= \lim_{x \rightarrow -2} (x - 3)^2 \\ &= (-2 - 3)^2 \\ &= 25 \end{aligned}$$

8.

$$\lim_{x \rightarrow \infty} \frac{(x + 1)^{10} + (x + 2)^{10} + \dots + (x + 100)^{10}}{x^{10} + 10^{10}} = \dots\dots$$

- (a) 100
- (b) 1
- (c) 10
- (d) None of these

*Answer: a**Solution:*

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{(x + 1)^{10} + (x + 2)^{10} + \dots + (x + 100)^{10}}{x^{10} + 10^{10}} \\ &= \lim_{x \rightarrow \infty} \frac{100x^{10} + \text{Lower Order Terms}}{x^{10} + \cancel{10^{10}}} \\ &= \frac{100\cancel{x^{10}}}{\cancel{x^{10}}} = 100 \end{aligned}$$

9. Which of the following functions is continuous and not differentiable at $x = 0$?

- (a) $f(x) = x$
- (b) $f(x) = x^3$
- (c) $f(x) = \frac{1}{x}$
- (d) $f(x) = |x|$

Answer: d

Solution: $|x|$ would be continuous at all points (including 0) as the limits from both sides would always equal each other and the function's output. However, at $x = 0$, there exists a corner that makes it not differentiable; in other words, $\lim_{x \rightarrow 0^-} f'(x) = -1 \neq \lim_{x \rightarrow 0^+} f'(x) = 1$

10. The function $f(x) = x|x|$ is differentiable on

- (a) $\mathbb{R}$
- (b) $\mathbb{R} - \{0\}$
- (c) $\mathbb{R}^+$
- (d) $[0, \infty[$

Answer: a

Solution: In this case, $f(x)$ is equivalent to

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$$

which would be differentiable at all points (including 0) because $\frac{d}{dx}[x^2]\Big|_{x=0} = 0 = \frac{d}{dx}[-x^2]\Big|_{x=0}$.

11. For what value of x , $f(x) = |2x - 7|$ is not differentiable?

- (a) 0
- (b) 7
- (c) $\frac{7}{2}$
- (d) $\frac{2}{7}$

Answer: c

Solution: A function $h(x) = |g(x)|$ is differentiable at all points except the zeros of $g(x)$. $2x - 7$ has one zero at $x = \frac{7}{2}$. Thus, f is not differentiable there.

12. If $f(a) = 2$, $f'(a) = 1$, $g(a) = -1$ and $g'(a) = 2$, find the value of

$$\lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x - a}$$

- (a) -5
- (b) $\frac{1}{5}$
- (c) 5
- (d) None of these

Answer: c

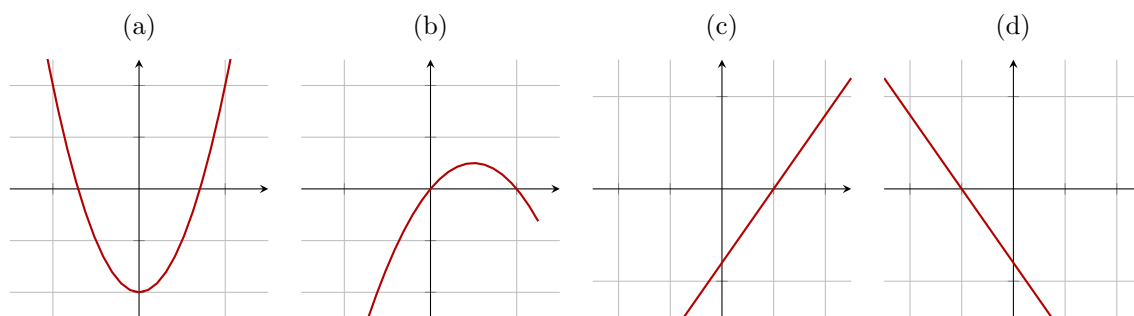
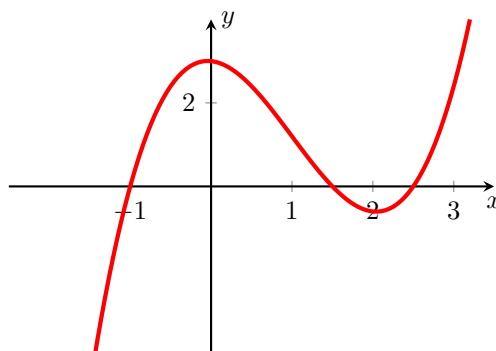
Solution:

$$\begin{aligned} \lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x - a} &= \lim_{x \rightarrow a} \frac{\frac{d}{dx}[2g(x) + f(x)]}{\frac{d}{dx}[x - a]} \\ &= \lim_{x \rightarrow a} \frac{2g'(x) + f'(x)}{1} \\ &= 5 \end{aligned}$$

Alternative Solution:

$$\begin{aligned} \lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x - a} &= \lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(a) + g(a)f(a) - g(a)f(x)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{f(a)[g(x) - g(a)] - g(a)[f(x) - f(a)]}{x - a} \\ &= f(a) \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} - g(a) \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= f(a)g'(a) - g(a)f'(a) \\ &= 5 \end{aligned}$$

13. Which of the following graphs represents the derivative of the function shown below?



Answer: a

Solution: This is a cubic function, so its derivative is a quadratic function. The function has two relative extrema at $x = 0$ and $x = 2$. Those are the zeros of the derivative. For all $x \in \mathbb{R} - [0, 2]$, the function increases and the derivative is positive. For all $x \in [0, 2]$, the function decreases and the derivative is negative.

14. If $f(x + y) = f(x)f(y)$ for all x and y . If $f(5) = 2$, $f'(0) = 3$, find $f'(5)$.

- (a) 5
- (b) 6
- (c) 0
- (d) -10

Answer: b

Solution: Since $f(x + y) = f(x)f(y)$, $f(5) = f(5 + 0) = f(5)f(0)$, which necessitates $f(0)$ to equal 1 in order for $f(5) = f(5)$ to hold. Using the limit definition of the derivative,

$$\begin{aligned}
 f'(5) &= \lim_{h \rightarrow 0} \frac{f(5 + h) - f(5)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(5)f(h) - f(5)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(5)[f(h) - 1]}{h} \\
 &= f(5) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} \\
 &= f(5) \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\
 &= f(5)f'(0) = 2 \times 3 = 6
 \end{aligned}$$

15. Let

$$f(x) = \begin{cases} x^2 & \text{if } x \leq x_0 \\ ax + b & \text{if } x > x_0 \end{cases}$$

The value of the coefficients a and b for which the function f is differentiable at $x = x_0$ are

- (a) $a = x_0, b = -x_0$
- (b) $a = 2x_0, b = -x_0^2$
- (c) $a = x_0^2, b = -x_0$
- (d) $a = x_0, b = -x_0^2$

*Answer: b**Solution:*

$$\begin{aligned} \lim_{x \rightarrow x_0^+} f'(x) &= \lim_{x \rightarrow x_0^-} f'(x) \\ \lim_{x \rightarrow x_0^+} a &= \lim_{x \rightarrow x_0^-} 2x \\ a &= 2x_0 \end{aligned} \tag{1}$$

$$\begin{aligned} \lim_{x \rightarrow x_0^+} f(x) &= \lim_{x \rightarrow x_0^-} f(x) \\ ax_0 + b &= x_0^2 \\ 2x_0^2 + b &= x_0^2 \\ b &= -x_0^2 \end{aligned} \tag{2}$$

16. If $f(x) = x^2g(x)$ and $g(1) = 6, g'(1) = 3$, find the value of $f'(1)$.

- (a) 3
- (b) 6
- (c) 12
- (d) 15

*Answer: d**Solution:* Applying the **Product Rule**:

$$\begin{aligned} f'(x) &= 2xg(x) + x^2g'(x) \\ f'(1) &= 2 \times 6 + 1 \times 3 = 15 \end{aligned}$$

2.4 Final Term, 2017–2018

1. If $0 < a < b$, then $\lim_{n \rightarrow \infty} (b^n + a^n)^{\frac{1}{n}}$ is equal to

- (a) e
- (b) a
- (c) b
- (d) None of these

Answer: c

Solution:

$$\begin{aligned} \lim_{n \rightarrow \infty} (b^n + a^n)^{\frac{1}{n}} &= \lim_{n \rightarrow \infty} \left(b^n \left(1 + \left(\frac{a}{b} \right)^n \right) \right)^{\frac{1}{n}} \\ &= \lim_{n \rightarrow \infty} b^{n \times \frac{1}{n}} \left(1 + \frac{a^n}{b^n} \right)^{\frac{1}{n}} \\ &= \lim_{n \rightarrow \infty} b(1 + (0)^n)^{\frac{1}{n}} \quad \because b^n \gg a^n \\ &= b \end{aligned}$$

2.

$$\lim_{x \rightarrow 0} \frac{(1 - \cos(x))(3 + \cos(x))}{x \tan(x)} = \dots\dots$$

- (a) $\frac{1}{2}$
- (b) 1
- (c) 2
- (d) $-\frac{1}{4}$

Answer: c

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1 - \cos(x))(3 + \cos(x))}{x \tan(x)} &= \lim_{x \rightarrow 0} \frac{\cancel{x}(1 - \cos(x))(3 + \cos(x))}{x^2 \cancel{\tan(x)}} \\ &= \lim_{x \rightarrow 0} \frac{(1 - \cos(x))(3 + \cos(x))}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{3 + \cos(x)}{2} \\ &= \frac{3 + \overbrace{\cos(0)}^1}{2} = 2 \end{aligned}$$

3.

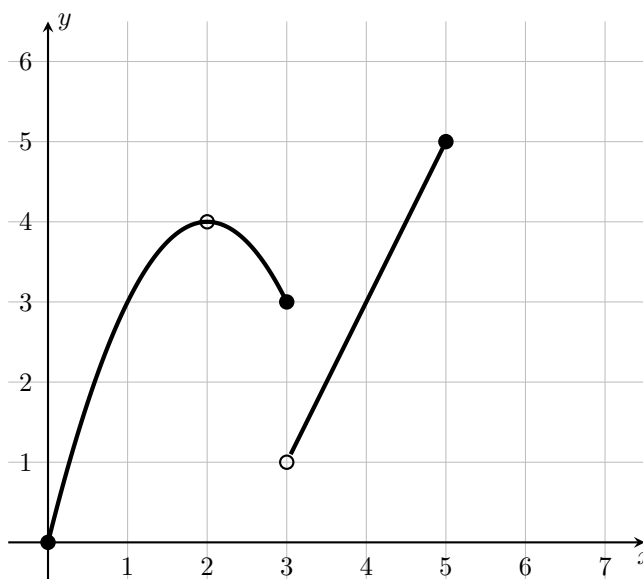
$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin(4x)}{\frac{1}{2} - \sin^2(x)} = \dots\dots$$

- (a) 4
- (b) 2
- (c) 1
- (d) $-\frac{1}{2}$

*Answer: a**Solution:*

$$\begin{aligned}
 \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin(4x)}{\frac{1}{2} - \sin^2(x)} &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin(4x)}{1 - 2 \sin^2(x)} \\
 &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2[2 \sin(2x) \cos(2x)]}{\cos(2x)} \\
 &= \lim_{x \rightarrow \frac{\pi}{4}} 4 \sin(2x) \\
 &= 4 \sin\left(\frac{\pi}{2}\right) \\
 &= 4
 \end{aligned}$$

4. The graph of the function f is shown below. Which of the following statements is not true?



- (a) $\lim_{x \rightarrow 5} f(x) = 5$
 (b) $\lim_{x \rightarrow 3^+} f(x) = 1$
 (c) $\lim_{x \rightarrow 3} f(x)$ does not exist
 (d) $\lim_{x \rightarrow 2} f(x) = 4$

Answer: a

Solution: For a limit to exist, the limits from both sides must exist and equal each other. Looking at the figure at 5, there exists no limit from the right side.

5.

$$f(x) = \begin{cases} cx + 1 & \text{if } x \neq 2 \\ cx^2 - 1 & \text{if } x = 2 \end{cases}$$

For what value of the constant c is the function f continuous over all reals?

- (a) 3
 (b) $\frac{1}{3}$
 (c) 2
 (d) 1

Answer: d

Solution: For f to be continuous, the following equation should be valid for all reals: $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$, where a is any real. Since the two “pieces” of the piecewise are clearly continuous polynomials, there exists only

one point that may cause f to not be continuous: the point $x = 2$. In other words, the two pieces of the piecewise function have to be equal at 2 for f to be continuous. Knowing this fact, we can find c :

$$c(2) + 1 = c(2^2) - 1 \implies c = 1$$

6.

$$f(x) = \begin{cases} x + 2, & x \leq 3 \\ 4x - 7, & x > 3 \end{cases}$$

Let f be the function given above. Which of the following statements are true about f ?

I. $\lim_{x \rightarrow 3} f(x)$ exists.

II. f is continuous at $x = 3$.

III. f is differentiable at $x = 3$.

(a) I only.

(b) II only.

(c) III only.

(d) I and II only.

Answer: d

Solution:

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} x + 2 = 5$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 4x - 7 = 5$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

Then, $\lim_{x \rightarrow 3} f(x)$ exists.

$$f(3) = 3 + 2 = 5 = \lim_{x \rightarrow 3} f(x)$$

Then, f is continuous at $x = 3$.

At $x = 3$, there is a change in the base of the function. So, we should use the limited definition of derivatives. As $h \rightarrow 0^-$, $f(3+h) \rightarrow f(3)^-$. Here we use the base $f(x) = x + 2$. Similarly, As $h \rightarrow 0^+$, $f(3+h) \rightarrow f(3)^+$ and we use the base $f(x) = 4x - 7$.

$$\lim_{h \rightarrow 0^-} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0^-} \frac{5+h-5}{h} = 1$$

$$\lim_{h \rightarrow 0^+} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0^+} \frac{12+4h-12}{h} = 4$$

$$\lim_{h \rightarrow 0^-} \neq \lim_{h \rightarrow 0^+}$$

Then, $f'(3) = \lim_{h \rightarrow 0} \frac{f(3+h)-f(3)}{h}$ does not exist and, consequently, f is not differentiable at $x = 3$.

7.

$$f(x) = \begin{cases} cx + d & x \leq 2 \\ x^2 - cx & x > 2 \end{cases}$$

Let f be the function defined above. Where c and d are constants. If f is differentiable at $x = 2$, what is the value of $c + d$?

(a) -4

(b) -2

(c) 0

(d) 2

Answer: b

Solution:

$$\begin{aligned}
\lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^-} f(x) \\
\lim_{x \rightarrow 2^+} x^2 - cx &= \lim_{x \rightarrow 2^-} cx + d \\
4 - 2c &= 2c + d \\
4c &= 4 - d \\
\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} &= \lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} \\
\lim_{h \rightarrow 0^+} \frac{(2+h)^2 - c(2+h) - 2^2 + 2c}{h} &= \lim_{h \rightarrow 0^-} \frac{c(2+h) + d - 2c - d}{h} \\
c &= 2 \\
d &= -4 \\
c + d &= -2
\end{aligned}$$

8. For inverses where $g(x) = f^{-1}(x)$ and the point (a, b) is on $f(x)$, $g'(b) = \dots\dots$

- (a) $f'(a)$
- (b) $\frac{1}{f'(a)}$
- (c) $f'(b)$
- (d) $\frac{1}{f'(b)}$

*Answer: b**Solution:*

$$\begin{aligned}
f(g(x)) &= f(f^{-1}(x)) = x \\
f'(g(x)) \cdot g'(x) &= 1 \\
g'(x) &= \frac{1}{f'(g(x))} \\
f(a) &= b, \quad g(b) = a \\
g'(b) &= \frac{1}{f'(g(b))} = \frac{1}{f'(a)}
\end{aligned}$$

9. Let f and g be functions that are differentiable for all real numbers, with $g(x) \neq 0$ for $x \neq 0$. If $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} g(x) = 0$ and $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$ exists. Then $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \dots\dots$

- (a) 0
- (b) $\frac{f'(x)}{g'(x)}$
- (c) $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$
- (d) $\frac{f'(x)g(x) - g'(x)f(x)}{(f(x))^2}$

*Answer: c**Solution:*

$$\begin{aligned}
f(0) &= 0, \quad g(0) = 0 \\
\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow 0} \frac{f(x) - 0}{g(x) - 0} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{g(x) - g(0)} \\
&= \lim_{x \rightarrow 0} \frac{\left(\frac{f(x) - f(0)}{x - 0}\right)}{\left(\frac{g(x) - g(0)}{x - 0}\right)} = \frac{\lim_{x \rightarrow 0} \left(\frac{f(x) - f(0)}{x - 0}\right)}{\lim_{x \rightarrow 0} \left(\frac{g(x) - g(0)}{x - 0}\right)} = \frac{f'(0)}{g'(0)} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}
\end{aligned}$$

Note: This is the statement of the special case of **L'Hôpital's rule** for the indeterminate form 0/0.

10. If $f(x) = \frac{x-1}{x+1}$ for all $x \neq -1$, then $f'(1) =$

- (a) -1
- (b) $-\frac{1}{2}$
- (c) 0
- (d) $\frac{1}{2}$

Answer: d

Solution:

$$f'(x) = \frac{(x-1)'(x+1) - (x+1)'(x-1)}{(x+1)^2} = \frac{2}{(x+1)^2}$$

$$f'(1) = \frac{2}{4} = \frac{1}{2}$$

11. Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$, then $\lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{h^3 + 3h} = \dots\dots$

- (a) $\frac{22}{3}$
- (b) $\frac{53}{3}$
- (c) $\frac{26}{3}$
- (d) 12

Answer: b

Solution:

$$\lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{h^3 + 3h} = - \lim_{h \rightarrow 0} \frac{f(1) - f(1-h)}{h(h^2 + 3)} = -f'(1) \lim_{h \rightarrow 0} \frac{1}{h^2 + 3} = \frac{-f'(1)}{3}$$

$$= -\frac{30 - 56 + 30 - 63 + 6}{3} = \frac{53}{3}$$

12. If $f(x) = \sqrt{x^2 + a}$, $f'(2) = \frac{1}{2}$, then $a = \dots\dots$

- (a) 4
- (b) 8
- (c) 12
- (d) 16

Answer: c

Solution:

$$f'(x) = \frac{x}{\sqrt{x^2 + a}}$$

$$f'(2) = \frac{2}{\sqrt{4 + a}} = \frac{1}{2}$$

$$\sqrt{4 + a} = 4, \quad 4 + a = 16$$

$$a = 12$$

13. Let

$$f(x) = \begin{cases} x^a \ln x, & x > 0, \\ 0, & x = 0 \end{cases}$$

Then the value of a for which Rolle's Theorem is applicable in $[0, 1]$ is $\dots\dots$

- (a) $-\frac{2}{3}$
- (b) $-\frac{1}{3}$
- (c) 0

(d) $\frac{1}{2}$ **Answer: d**

Solution: The applicability of **Rolle's Theorem** over the interval $[0, 1]$ requires f to be continuous over $[0, 1]$. This requires:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^a \ln x = f(0)$$

- For $a = 0$,

$$\lim_{x \rightarrow 0^+} x^a \ln x = -\infty \neq f(0)$$

- For $a < 0$,

$$\lim_{x \rightarrow 0^+} x^a \ln x = -\infty \neq f(0)$$

- For $a > 0$,

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-a}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-ax^{(-a-1)}} = \lim_{x \rightarrow 0^+} \frac{-1}{a} x^a = 0 = f(0)$$

Then a must be > 0 to achieve the continuity.

14. If $f(x)$ and $g(x)$ are differentiable functions for all $x \in [0, 1]$ such that $f(0) = g(1) = 2$, $g(0) = 0$ and $f(1) = 6$, then there exist some values α, β of $x \in]0, 1[$ for which

- (a) $f'(\alpha) = g'(\beta)$
 (b) $f'(\alpha) = 4g'(\beta)$
 (c) $f'(\alpha) = 3g'(\beta)$
 (d) $f'(\alpha) = 2g'(\beta)$

Answer: d

Solution: By the **Mean Value Theorem**, there are some values α and β over the interval $]0, 1[$ such that:

$$f'(\alpha) = \frac{f(1) - f(0)}{1 - 0} = \frac{6 - 2}{1} = 4$$

$$g'(\beta) = \frac{g(1) - g(0)}{1 - 0} = \frac{2 - 0}{2 - 0} = 2$$

$$f'(\alpha) = 2g'(\beta)$$

15. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \rightarrow \infty}}}$, then $\frac{dy}{dx} = \dots$

- (a) $\sqrt{\frac{\sin x}{y+1}}$
 (b) $\frac{\sin x}{y+1}$
 (c) $\frac{\cos x}{2y+1}$
 (d) $\frac{\cos x}{2y-1}$

Answer: d**Solution:**

$$y^2 = \left(\sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \rightarrow \infty}}} \right)^2$$

$$y^2 = \sin x + \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \rightarrow \infty}}}$$

$$y^2 = \sin x + y$$

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(\sin x + y)$$

$$2y \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$2y \frac{dy}{dx} - \frac{dy}{dx} = \cos x$$

$$(2y - 1) \frac{dy}{dx} = \cos x$$

$$\frac{dy}{dx} = \frac{\cos x}{2y - 1}$$

16. $y = \ln(x^2 + y^2)$, then the value of $\frac{dy}{dx}$ at the point $(1, 0)$ is

- (a) 0
- (b) $\frac{1}{2}$
- (c) 1
- (d) 2

Answer: d

Solution:

$$\begin{aligned} \frac{d}{dx}(y) &= \frac{d}{dx}(\ln(x^2 + y^2)) \\ \frac{dy}{dx} &= \frac{1}{x^2 + y^2} \cdot \left(2x + 2y \frac{dy}{dx}\right) \end{aligned}$$

Substituting $x = 1$ and $y = 0$,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1^2 + 0^2} \left(2 \times 1 + 2 \times 0 \times \frac{dy}{dx}\right) \\ \frac{dy}{dx} &= 2 \end{aligned}$$

17. $f(x) = \ln(\ln(x))$, then $f'(x) = \dots\dots$

- (a) $\frac{1}{x}$
- (b) $\frac{1}{\ln x}$
- (c) $\frac{\ln x}{x}$
- (d) $\frac{1}{x \ln x}$

Answer: d

Solution:

$$f'(x) = \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \ln x}$$

18. Which of the following families of functions has a constant slope of tangent at $x = 2$?

- (a) $\frac{c}{x}$
- (b) cx^2
- (c) e^{cx}
- (d) $\ln(cx)$

Answer: d

Solution:

$$\begin{aligned} \frac{d}{dx} \left(\frac{c}{x} \right) \Big|_{x=2} &= \left(\frac{-c}{x^2} \right) \Big|_{x=2} = \frac{-c}{4} \\ \frac{d}{dx} (cx^2) \Big|_{x=2} &= (2cx) \Big|_{x=2} = 4c \\ \frac{d}{dx} (e^{cx}) \Big|_{x=2} &= (ce^{cx}) \Big|_{x=2} = ce^{2c} \end{aligned}$$

All those vary with c .

$$\frac{d}{dx} (\ln cx) \Big|_{x=2} = \left(\frac{c}{cx} \right) \Big|_{x=2} = \left(\frac{1}{x} \right) \Big|_{x=2} = \frac{1}{2}$$

This is independent of the value of c . Then, the slope of the tangent is constant whatever the value of c .

19. If $y = \log_{\sin(x)}(\tan(x))$, then, $\frac{dy}{dx}\bigg|_{x=\frac{\pi}{4}}$ is equal to

- (a) $\frac{4}{\ln(2)}$
- (b) $-4 \ln 2$
- (c) $-\frac{4}{\ln 2}$
- (d) None of these

Answer: c

Solution:

$$y = \frac{\ln(\tan x)}{\ln(\sin x)}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{1}{\tan(x)} \cdot \sec^2(x) \cdot \ln(\sin(x)) - \frac{1}{\sin(x)} \cdot \cos(x) \cdot \ln(\tan(x))}{[\ln(\sin(x))]^2} \\ \frac{dy}{dx}\bigg|_{x=\frac{\pi}{4}} &= \frac{\frac{1}{\tan(\frac{\pi}{4})} \cdot \sec^2(\frac{\pi}{4}) \cdot \ln(\sin(\frac{\pi}{4})) - \frac{1}{\sin(\frac{\pi}{4})} \cdot \cos(\frac{\pi}{4}) \cdot \ln(\tan(\frac{\pi}{4}))}{[\ln(\sin(\frac{\pi}{4}))]^2} \\ &= \frac{-4}{\ln 2}\end{aligned}$$

20. Which of the following functions has the property that. “The slope of the tangent at any point = the y -coordinate of the same point.”

- (a) $\ln x$
- (b) $\log x$
- (c) a^x
- (d) e^x

Answer: d

Solution: The required function has the property: $f(x) = f'(x)$.

$$\begin{aligned}\frac{d}{dx}(\ln x) &= \frac{1}{x} \neq \ln x \\ \frac{d}{dx}(\log x) &= \frac{1}{x \ln 10} \neq \log x \\ \frac{d}{dx}(a^x) &= a^x \log a \neq a^x \\ \frac{d}{dx}(e^x) &= e^x \ln e = e^x\end{aligned}$$

21. Find the area of the region bounded by $y = \frac{2}{x+1} - 1$ and the Cartesian axes.

- (a) $\ln(2) + 1$
- (b) $1 - \ln(2)$
- (c) $2 \ln(2) - 1$
- (d) 2

Answer: c

Solution: Before integrating the function, we need to find the upper and lower bound, which will be the points at which the function intersects the y -axis and the x -axis.

The intersection with the y -axis will be $x_1 = 0$.

To find the intersection with the x -axis, we plug $y = 0$:

$$0 = \frac{2}{x+1} - 1 \implies x_2 = 1$$

Therefore, by letting $u = x + 1$ and $du = dx$, we get

$$\begin{aligned}\int_{x_1}^{x_2} y \, dx &= \int_0^1 \frac{2}{x+1} - 1 \, dx \\ &= 2 \int_0^1 \frac{1}{x+1} \, dx - \int_0^1 1 \, dx \\ &= 2 \int_0^1 \frac{1}{u} \, du - \int_0^1 1 \, dx \\ &= 2 [\ln(|x+1|)]_0^1 - [x]_0^1 = 2 \ln(2) - 1\end{aligned}$$

22. Estimate the area under the curve for the following function from $x = 0$ to $x = 4$ using a midpoint Riemann sum with $n = 4$ rectangles $f(x) = \frac{1}{4}x^2 + 3$.

- (a) $\frac{57}{16}$
- (b) $\frac{49}{16}$
- (c) $\frac{97}{8}$
- (d) $\frac{69}{4}$

Answer: d

Solution: Having $n = 4$ rectangles from 0 to 4 means that $\Delta x = 1$ and there would be four x_i s that represent the midpoint of each rectangle: $x_1 = 0.5$, $x_2 = 1.5$, $x_3 = 2.5$, and $x_4 = 3.5$. Therefore,

$$\begin{aligned}\text{Area}_{\text{estimation}} &= \Delta x [f(x_1) + f(x_2) + f(x_3) + f(x_4)] \\ &= f(0.5) + f(1.5) + f(2.5) + f(3.5) \\ &= \frac{69}{4}\end{aligned}$$

23. Suppose that f is an integrable function and that $\int_0^1 f(x) \, dx = 2$, $\int_0^2 f(x) \, dx = 1$, and $\int_2^4 f(x) \, dx = 7$. Find $\int_0^4 f(x) \, dx + \int_1^2 f(x) \, dx$.

- (a) 2
- (b) 3
- (c) 5
- (d) 7

Answer: d

Solution: Let $\int f(x) \, dx = g(x) + C$. We can now write $\int_0^4 f(x) \, dx + \int_1^2 f(x) \, dx$ as

$$[g(4) - g(0)] + [g(2) - g(1)]$$

We can then write the given equations using $g(x)$ s.

$$\begin{aligned}g(1) - g(0) &= 2 \implies -g(1) = -g(0) - 2 \\ g(2) - g(0) &= 1 \implies -g(0) = -g(2) + 1 \\ g(4) - g(2) &= 7\end{aligned}$$

Rewriting $[g(4) - g(0)] + [g(2) - g(1)]$ by replacing $-g(1)$ with $-g(0) - 2$ and replacing $-g(0)$ with $-g(2) + 1$, we get

$$[g(4) - g(2)] + 1 + [g(2) - g(0)] - 2 = [7] + 1 + [1] - 2 = 7$$

24. The average temperature for all the days preceding the last for the month of February was 55 degrees Fahrenheit, with a standard deviation of 5 degrees. On the last day, the temperature was 73 degrees F. What is the z-score for the temperature on the last day?

- (a) 2.8
- (b) 3.6
- (c) -3.6
- (d) 1.7

Answer: b

Solution:

$$z = \frac{T - \mu}{\sigma} = \frac{73 - 55}{5} = 3.6$$

25. Marks on a Chemistry test follow a normal distribution with a mean of 65 and a standard deviation of 12. Approximately what percentage of the students have scores below 50?

- (a) 11 %
- (b) 89 %
- (c) 15 %
- (d) 18 %

Answer: a

Solution:

$$z = \frac{M - \mu}{\sigma} = \frac{50 - 65}{12} = -1.25$$

Using the normal distribution table, we can find 1.25 to have a probability of 0.8944. Therefore, $1 - 0.8944 \approx 11\%$ is the percentage of the students who have scores below 50.

2.5 Midterm, Unknown Year

1.

$$\lim_{x \rightarrow \infty} \frac{2^{-x}}{2^x}$$

is

- (a) -1
- (b) 0
- (c) 1
- (d) ∞

*Answer: b**Solution:*

$$\lim_{x \rightarrow \infty} \frac{2^{-x}}{2^x} = \lim_{x \rightarrow \infty} \frac{1}{(2^x)^2} = \lim_{x \rightarrow \infty} \frac{1}{(2^2)^x} = \lim_{x \rightarrow \infty} \frac{1}{4^x} = 0$$

2.

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

is

- (a) -1
- (b) 1
- (c) ± 1
- (d) Non-existent

Answer: d

Solution: The limits from both sides are unequal to each other, and, hence, the limit does not exist; when $x > 0$ (taking limit from the right), $\frac{|x|}{x} = 1$, and when $x < 0$ (taking limit from the left), $\frac{|x|}{x} = -1$.

3.

$$\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right)$$

is

- (a) -1
- (b) 0
- (c) 1
- (d) Non-existent

*Answer: c**Solution:*

$$\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right) = \lim_{\frac{1}{x} \rightarrow 0} \frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} = 1$$

4. Let

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1} & \text{if } x \neq 1 \\ 4 & \text{if } x = 1 \end{cases}$$

Which of the following statements is (are) true?

- I.* $\lim_{x \rightarrow 1} f(x)$ exists
- II.* $f(1)$ exists
- III.* f is continuous at $x = 1$

- (a) *I* only
- (b) *II* only
- (c) *I* and *II*
- (d) All of them

Answer: c

Solution: *I* is true because, at 1, the limits from both sides are equal. *II* is true because, according to the piecewise function, $f(1)$ is defined to output 4. *III* is not true because, at 1, the limit does not equal the output 4:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(\cancel{x - 1})}{\cancel{x - 1}} = 1 + 1 = 2 \neq 4$$

5. Suppose $\lim_{x \rightarrow -3^-} f(x) = -1$, $\lim_{x \rightarrow -3^+} f(x) = -1$ and $f(-3)$ is not defined. Which of the following statements is (are) true?

- I.* $\lim_{x \rightarrow -3} f(x) = -1$
- II.* f is continuous everywhere except at $x = -3$
- III.* f has a removable discontinuity at $x = -3$

- (a) *I* only
- (b) *III* only
- (c) *I* and *II* only
- (d) *I* and *III* only

Answer: d

Solution: *I* is true because the limits from right and left exist and are equal to -1. *II* is not true because the given information is not enough to deduce that f is continuous everywhere (i.e., being given information about one point is not enough to declare the entirety rest of the function's points to be continuous). *III* is true because a removable discontinuity at a , where a is any constant, means that $\lim_{x \rightarrow a^-} = \lim_{x \rightarrow a^+} \neq f(a)$, which is the case for 3.

6.

$$\lim_{x \rightarrow a} \sin\left(\frac{x - a}{2}\right) \tan\left(\frac{\pi x}{2a}\right) = \dots\dots$$

- (a) $-\frac{a}{\pi}$
- (b) $\frac{a}{\pi}$
- (c) $-\frac{\pi}{a}$
- (d) $\frac{\pi}{a}$

Answer: a

Solution:

$$\begin{aligned}
 \lim_{x \rightarrow a} \sin\left(\frac{x-a}{2}\right) \tan\left(\frac{\pi x}{2a}\right) &= \lim_{x \rightarrow a} \sin\left(\frac{x-a}{2}\right) \cot\left(\frac{\pi}{2} - \frac{\pi x}{2a}\right) \\
 &= \lim_{x \rightarrow a} \frac{\sin\left(\frac{x-a}{2}\right) \cos\left[\frac{\pi}{2}\left(1 - \frac{x}{a}\right)\right]}{\sin\left[\frac{\pi}{2}\left(1 - \frac{x}{a}\right)\right]} \\
 &= \lim_{x \rightarrow a} \frac{\sin\left(\frac{x-a}{2}\right)}{\sin\left[-\frac{\pi}{2}\left(\frac{x-a}{a}\right)\right]} \\
 &= \lim_{\frac{x-a}{2} \rightarrow 0} \frac{\sin\left(\frac{x-a}{2}\right)}{\sin\left[-\frac{\pi}{a}\left(\frac{x-a}{2}\right)\right]} \\
 &= \frac{1}{-\frac{\pi}{a}} \\
 &= \frac{a}{-\pi}
 \end{aligned}$$

7.

$$\lim_{x \rightarrow 0} \frac{1}{|x|}$$

is

- (a) 0
- (b) 1
- (c) ∞
- (d) Non-existent

Answer: c

Solution: Remember the $\frac{1}{x}$ graph; now multiply every negative output with -1 , which will give you the graph of $\frac{1}{|x|}$, the graph of an even function that outputs positive numbers only. When x approaches 0, $\frac{1}{|x|}$ approaches ∞ . Hence, $\lim_{x \rightarrow 0} \frac{1}{|x|} = \infty$

8.

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{\sqrt{x+1} - \sqrt{1-x}}$$

is

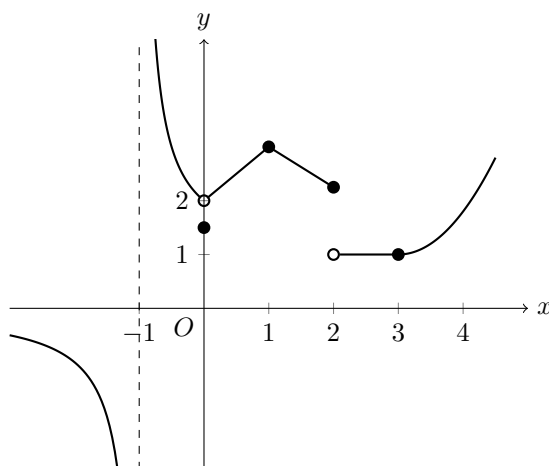
- (a) -1
- (b) 0
- (c) 1
- (d) 2

Answer: c

Solution:

Multiplying the expression by the **conjugate** of the function's denominator, we get

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\sin(x)}{\sqrt{x+1} - \sqrt{1-x}} &= \lim_{x \rightarrow 0} \frac{\sin(x)}{\sqrt{x+1} - \sqrt{1-x}} \times \frac{\sqrt{x+1} + \sqrt{1-x}}{\sqrt{x+1} + \sqrt{1-x}} \\
 &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+1} + \sqrt{1-x}) \cancel{\sin x}}{2\cancel{x}} \quad (\text{rationalizing the denominator}) \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} (\sqrt{x+1} + \sqrt{1-x}) \\
 &= \frac{1}{2} \times 2 \\
 &= 1
 \end{aligned}$$



9. The graph of a function f is shown above. If $\lim_{x \rightarrow b} f(x)$ exists, and f is not continuous at b . Then, $b = \dots\dots$

- (a) -1
- (b) 0
- (c) 1
- (d) 2

Answer: b

Solution: The given description is for a removable discontinuity, which is resembled at $x = 0$.

10. When $x \rightarrow \infty$, the function f has the asymptote $y = x + 1$, while the function g has the asymptote $y = 2x$. Then:

- (a) $\lim_{x \rightarrow \infty} (f - g) = -\infty$
- (b) $\lim_{x \rightarrow \infty} \frac{f-g}{x} = 0$
- (c) $\lim_{x \rightarrow \infty} \frac{f-g}{x} = \infty$
- (d) $\lim_{x \rightarrow \infty} (f + g) = -\infty$

Answer: a

Solution: Since we know the behaviour of f and g at ∞ to be $x + 1$ and $2x$, we can simply replace $\lim_{x \rightarrow \infty} f(x)$ with $\lim_{x \rightarrow \infty} x + 1$ and replace $\lim_{x \rightarrow \infty} g(x)$ with $\lim_{x \rightarrow \infty} 2x$. Implementing this method, we find option **a** to be the only correct option:

$$\lim_{x \rightarrow \infty} x + 1 - 2x = \lim_{x \rightarrow \infty} -x + 1 = -\infty$$

11. If $\frac{d}{dx} f(x) = g(x)$ and if $h(x) = x^2$, then $\frac{d}{dx} f(h(x)) = \dots\dots$

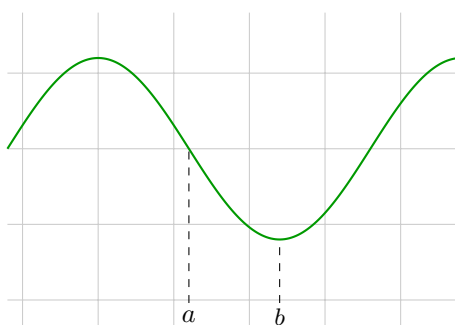
- (a) $g(x^2)$
- (b) $2xg(x)$
- (c) $g'(x)$
- (d) $2xg(x^2)$

Answer: d

Solution:

$$\frac{d}{dx} f(h(x)) = \frac{d}{dx} f(x^2) = \frac{d}{dx} f(x^2) \cdot \frac{d}{dx} x^2 = 2xg(x^2)$$

12. Which of the following represents the slope of a line drawn between the two points marked in the figure below?



(a) $m = \frac{f(a)+f(b)}{a+b}$

(b) $m = \frac{a}{b}$

(c) $m = \frac{f(b)-f(a)}{b-a}$

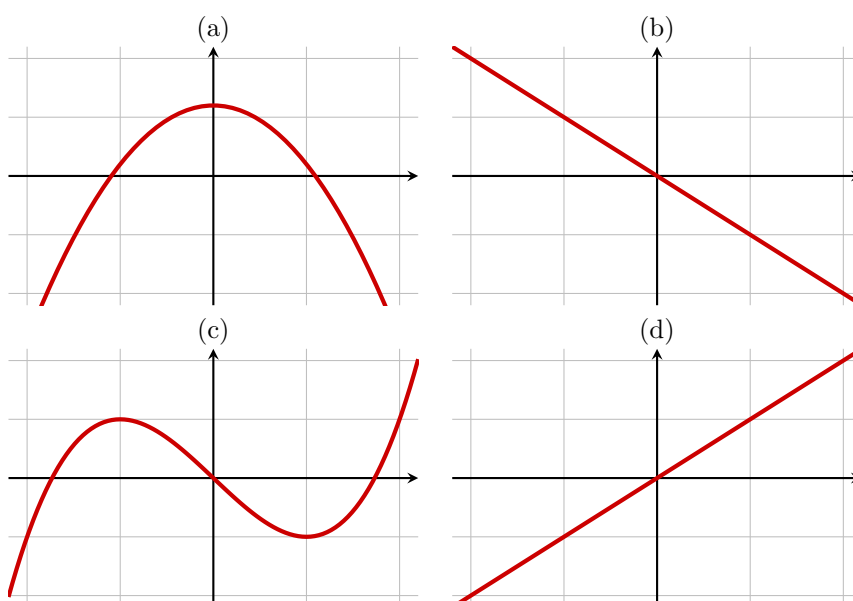
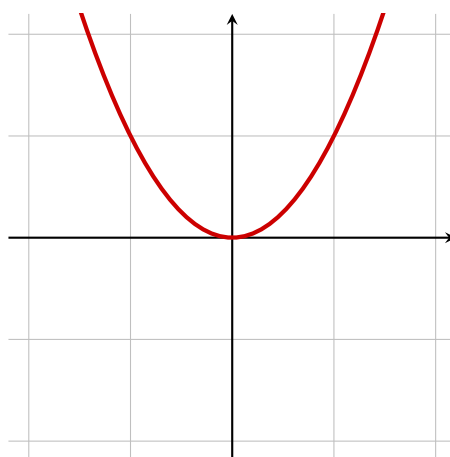
(d) $m = \frac{f(a)-f(b)}{b-a}$

Answer: c

Solution

$$m = \frac{\Delta x}{\Delta y} = \frac{f(b) - f(a)}{b - a}$$

13. Which of the following graphs is the graph of the derivative of the function shown in the figure?



Answer: d**Solution:** The derivative of a positive quadratic function is a positive linear function.14. If $f(x) = \pi^2$, then what is $f'(x)$?

- (a) 2π
- (b) π^2
- (c) 0
- (d) 2

Answer: c**Solution:** Since $f(x) = \pi^2$ is a constant function, $f'(x) = 0$ 15. If $f(3x) = x^2 + 4x$, then $f'(3) = \dots\dots$

- (a) 10
- (b) 6
- (c) 2
- (d) 1

Answer: c**Solution:**

$$f(x) = \left(\frac{x}{3}\right)^2 + \frac{4x}{3}$$

$$f'(x) = \frac{2}{3}\left(\frac{x}{3}\right) + \frac{4}{3}$$

$$f'(3) = 2$$

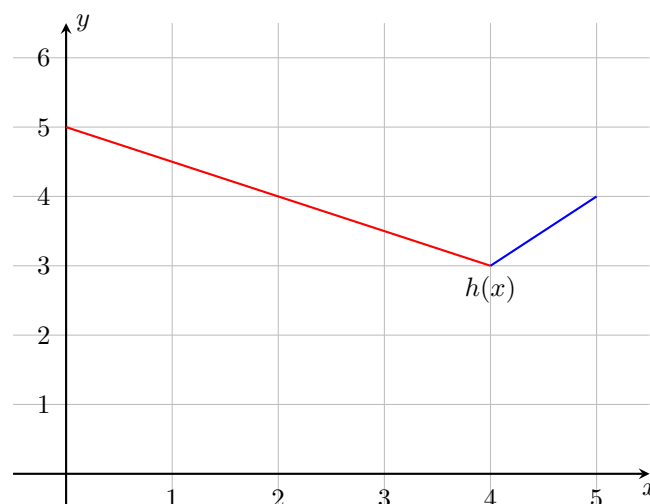
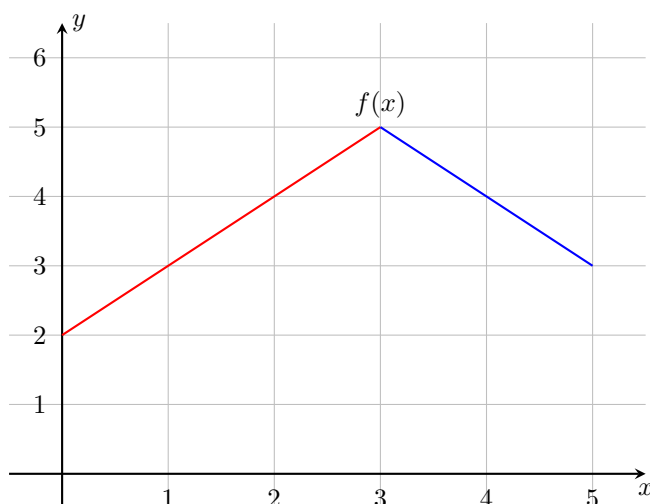
16. If $f(3) = 2$, $f'(3) = 4$, $g(3) = 1$, $g'(3) = 3$, and $h(x) = f(x)g(x)$, then what is $h'(3)$?

- (a) 2
- (b) 10
- (c) 11
- (d) 12

Answer: b**Solution:**

$$h'(x) = f'(x)g(x) + g'(x)f(x)$$

$$h'(3) = 4 \times 1 + 3 \times 2 = 10$$

Use the following information to answer questions 17, 18. The functions $f(x)$ and $h(x)$ are plotted below:

17. If the function $g = 2f \cdot h$, what is $g'(2)$?

- (a) -1
- (b) 2
- (c) 4
- (d) 32

Answer: c

Solution: Using the multiplication rule,

$$g'(2) = 2f'(2) \cdot h(2) + 2f(2) \cdot h'(2)$$

Since $f(x)$ is a linear function, then its slope at a point is the derivative at the same point. In this way, the red part of the function where point $(2, 4)$ is located has a slope of 1. Then:

$$f'(2) = 1$$

Applying the same to $h(x)$:

$$h'(x) = -\frac{1}{2}$$

$$g'(2) = ((2 \cdot 1) \cdot 4) + \left((2 \cdot 4) \cdot -\frac{1}{2} \right) = 4$$

18. If the function $g(x) = f(h(x))$, what is $g'(2)$?

- (a) $\frac{1}{2}$
- (b) 1
- (c) 3
- (d) 4

Answer: a

Solution: Using chain rule:

$$g'(2) = f'(h(2)) \cdot h'(2)$$

Since $f(x)$ is a linear function, then its slope at a point is the derivative at the same point. In this way, the blue part of the function where point $(4, 4)$ is located has a slope of -1. Also, refer to Question 17 where $h'(2) = -\frac{1}{2}$. Then:

$$f'(h(2)) = f'(4) = -1$$

$$g'(x) = -1 \cdot -\frac{1}{2} = \frac{1}{2}$$

19. If the function $f : R \rightarrow R$ is differentiable, $f(-1) = 0$ and $f(6) = 3$, then its derivative $f'(x)$ in the interval $] -1, 6[$ =

- (a) Assumes the value $\frac{7}{3}$ in at (least one) point
- (b) Assumes the value $\frac{3}{7}$ in at (least one) point
- (c) Vanishes in (at least) one point
- (d) Is never zero

Answer: b

Solution: By the **Mean Value Theorem**, there is at least one value c in the interval $] -1, 6[$ such that

$$f'(c) = \frac{f(6) - f(-1)}{6 - (-1)} = \frac{3}{7}$$

20. If $f(x)$ is differentiable at x_0 , then the function $g(x) = (x - x_0)f(x)$

- (a) is differentiable at x_0
- (b) is not differentiable at x_0

(c) is discontinuous at x_0

(d) None of these

Answer: a

Solution:

$$\begin{aligned} g'(x_0) &= \lim_{h \rightarrow 0} \frac{(x_0 - x_0 + h) f(x_0 + h) - (x_0 - x_0) f(x_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h f(x_0 + h)}{h} \\ &= f(x_0) \end{aligned}$$

Since f is continuous at $x = x_0$, $g'(x_0) = f(x_0)$ is defined and g is differentiable at $x = x_0$.

2.6 Midterm, 2020–2021

1. Which of the following functions is continuous for all real numbers x ?

I. $y = x^{\frac{3}{2}}$

II. $y = e^x$

III. $y = \ln(x)$

- (a) I only
 (b) II only
 (c) I and II only
 (d) I, II and III

Answer: b

Solution: The domain of the function $x^{\frac{3}{2}}$ is the nonnegative number, so it's not continuous. $\ln 0$ is undefined, so $\ln x$ is not continuous. So the only continuous function is e^x .

2. Three students were given a function f and asked to estimate $\lim_{x \rightarrow 2} f(x)$. Each student created a table as shown below, which one is the best evaluating the limit?

Table 1: [I]

x	-1	0	1	2	3	4
$f(x)$	0.2	1.3	2.9	3.25	2.9	1.3

Table 2: [II]

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	3.32	3.264	3.251	3.249	3.242	3.31

Table 3: [III]

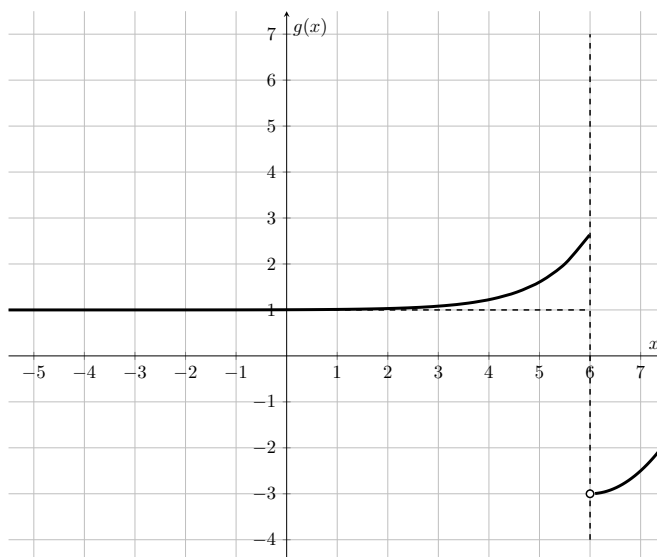
x	2.001	2.01	2.1	2.25	2.5	3
$f(x)$	3.249	3.242	3.31	3.2	3.05	2.9

- (a) (I)
 (b) (II)
 (c) (III)
 (d) None

Answer: b

Solution: The first table studies values that are too far from 2, so it cannot demonstrate the limit clearly. The third table studies values that are larger than 2 and does not study any value below 2 so it studies just the right-handed limit. The second table is the best because it studies values near 2 from both sides.

3. What appears to the value of $\lim_{x \rightarrow 6^+} ?$



- (a) -3
- (b) Zero
- (c) 6
- (d) Undefined

Answer: a

Solution: From the right, as the value of x gets near 6, the value of the function gets to -3 .

4. If $\lim_{x \rightarrow 3} f(x) = 7$, which of the following must be satisfied:

- I. f is continuous at $x = 3$
- II. $f(3) = 7$

- (a) I only
- (b) II only
- (c) I and II
- (d) None

Answer: d

Solution: None of the two conditions is necessary for the limit to be equal to 7.

5. Let f be a function defined by the following:

$$f(x) = \begin{cases} \sin(x), & \text{if } x < 0 \\ x^2, & \text{if } 0 \leq x < 1 \\ 2 - x, & \text{if } 1 \leq x < 2 \\ x - 3, & \text{if } x \geq 2 \end{cases}$$

for what value of x f is NOT continuous?

- (a) 0 only
- (b) 1 only
- (c) 2 only
- (d) 0,1 and 2

Answer: c

Solution: We check 0:

$$\sin 0 = 0, 0^2 = 0$$

The function is continuous for 0. We check 1

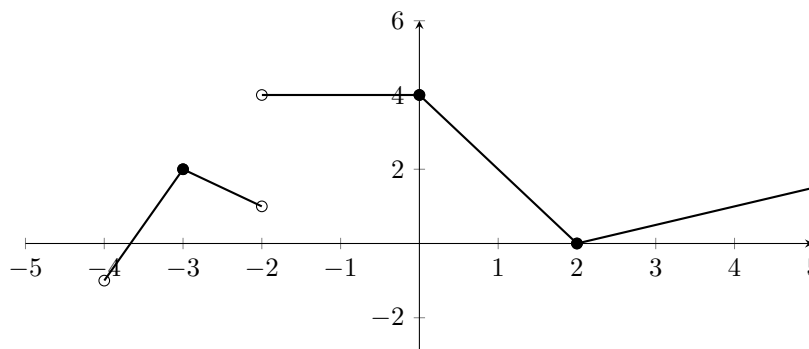
$$1^2 = 1, 2 - 1 = 1$$

The function is continuous for 1. We check 2

$$2 - 2 = 0, 2 - 3 = -1$$

The function is not continuous for 2, so the answer is 2.

6. Find $\lim_{x \rightarrow -2} (g(x) + h(x))$



- (a) limit doesn't exist
- (b) 8
- (c) 2
- (d) 5

Answer: d

Solution: We cannot divide the limit to be $\lim_{x \rightarrow -2} g(x)$ and $\lim_{x \rightarrow -2} h(x)$ because both of the two limits do not exist (The right-handed limit is different from the left-handed one as shown in the graphs). We have to get the $\lim_{x \rightarrow -2+} (g(x) + h(x))$ and $\lim_{x \rightarrow -2-} (g(x) + h(x))$ and check if they are equal:

$$\lim_{x \rightarrow -2+} (g(x) + h(x)) = \lim_{x \rightarrow -2+} (g(x)) + \lim_{x \rightarrow -2+} (h(x)) = 4 + 1 = 5$$

$$\lim_{x \rightarrow -2-} (g(x) + h(x)) = \lim_{x \rightarrow -2-} (g(x)) + \lim_{x \rightarrow -2-} (h(x)) = 1 + 4 = 5$$

As the right-handed limit and the left-handed one are equal, the limit exists and is equal to 5.

7. Let f be a function such that $\frac{f(x) - f(a)}{x - a} = 5$, then $x = a$ is

- (a) defined and discontinuous
- (b) undefined and discontinuous
- (c) defined and continuous
- (d) undefined and continuous

Answer: c

Solution: $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ is another definition of derivative of $f(x)$ at point a . As the limit exists, the function is differentiable at a . as a result, it's defined and continuous.

8. If $f(x) = \frac{x}{x^2 - x}$, then determine the point(s), at which function f is discontinuous, as jump, removable, infinite, or other.

- (a) infinite discontinuity at $x = 0$ and removable discontinuity at $x = 1$.
- (b) removable discontinuity at $x = 0$ and jump discontinuity at $x = 1$.
- (c) removable discontinuity at $x = 0$ and infinite discontinuity at $x = 1$.

(d) infinite discontinuity at $x = 0$ and jump discontinuity at $x = 1$.

Answer: c

Solution: $f(0)$ is undefined in its current shape. However, we can make:

$$f(x) = \frac{x}{x^2 - x} = \frac{x}{x(x-1)} = \frac{1}{x-1}$$

This enables us to get $f(0) = -1$. Hence, the function has a removable discontinuity at $x = 0$. There is a vertical asymptote at $x = 1$. As a result, there is an infinite discontinuity at $x = 1$.

9. Find the value of k that makes the function f continuous on its domain:

$$f(\theta) = \begin{cases} \cos(\theta + k\pi), & \text{if } -\frac{\pi}{2} \leq \theta \leq 0 \\ \frac{\sin \theta}{\theta}, & \text{if } 0 < \theta \leq \frac{\pi}{2} \end{cases}$$

- (a) $2n$ where $n \in \mathbb{Z}$
- (b) $2n + 1$ where $n \in \mathbb{Z}$
- (c) n where $n \in \mathbb{Z}$
- (d) $2n - 1$ where $n \in \mathbb{Z}$

Answer: (a)

Solution: Since $\cos(0 + k\pi) = 1$, the value of $\cos(k\pi)$ must be 1. This happens when $k = 2n$, where $n \in \mathbb{Z}$.

10. If

$$f(x) = \begin{cases} \frac{x^2 - 1}{|1 - x|}, & \text{if } x \neq 1 \\ 4, & \text{if } x = 1 \end{cases}$$

Find $\lim_{x \rightarrow 1^-} f(x)$.

- (a) 4
- (b) does not exist
- (c) 2
- (d) -2

Answer: (d)

Solution:

We simplify the expression:

$$f(x) = \frac{x^2 - 1}{|1 - x|} = \frac{(x - 1)(x + 1)}{|1 - x|}$$

Note that: For $x < 1$, we have $|1 - x| = 1 - x$, so for $x \rightarrow 1^-$:

$$f(x) = \frac{(x - 1)(x + 1)}{1 - x} = \frac{-(1 - x)(x + 1)}{1 - x} = -(x + 1)$$

Now take the limit:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} -(x + 1) = -2$$

So, the final answer is **(d)** -2.

11. Find $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 5x} - \sqrt{x^2 + 2x})$.

- (a) ∞
- (b) $\frac{-3}{2}$
- (c) 0
- (d) $\frac{3}{2}$

Answer: d

Solution: by using the conjugate method the limit will be $\lim_{x \rightarrow \infty} \frac{x^2 + 5x - (x^2 + 2x)}{\sqrt{x^2 + 5x} + \sqrt{x^2 + x}}$ by dividing by x the value will of the limit will be $\frac{3}{2}$

12. Find the limit

$$\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x^2 + 1} - \sqrt{x^2 - 1}}{\sqrt{x + 1} + \sqrt{x - 1}} \right)$$

- (a) ∞
- (b) 2
- (c) 0
- (d) $-\infty$

Answer: c

Solution:

$$\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x^2 + 1} - \sqrt{x^2 - 1}}{\sqrt{x + 1} + \sqrt{x - 1}} \times \frac{(\sqrt{x^2 + 1} + \sqrt{x^2 - 1})(\sqrt{x + 1} - \sqrt{x - 1})}{(\sqrt{x^2 + 1} + \sqrt{x^2 - 1})(\sqrt{x + 1} - \sqrt{x - 1})} \right) = \lim_{x \rightarrow \infty} \frac{\sqrt{x + 1} - \sqrt{x - 1}}{\sqrt{x^2 + 1} + \sqrt{x^2 - 1}} = 0$$

13. The position of a particle moving in the xy -plane is given by the parametric equations:

$$x(t) = \frac{6t}{t + 1}, \quad y(t) = \frac{-8t}{t^2 + 1}$$

What is the limit of y as $x \rightarrow 6$?

- (a) ∞
- (b) 0
- (c) -8
- (d) $-\infty$

Answer: b

Solution: As $t \rightarrow \infty$, we have:

$$x(t) = \frac{6t}{t + 1} \rightarrow 6, \quad \text{so } x \rightarrow 6^-$$

$$y(t) = \frac{-8t}{t^2 + 1} \rightarrow 0$$

Therefore, $\lim_{x \rightarrow 6} y = 0$.

14. $\lim_{x \rightarrow \infty} \frac{\sin(x)}{x + 1} = \dots\dots$

- (a) 1
- (b) -1
- (c) 0
- (d) doesn't exist

Answer: c

Squeeze Theorem

$$\text{Solution: } \lim_{x \rightarrow \infty} \frac{\overbrace{\left[\frac{\sin(x)}{x} \right]}^{\text{Squeeze Theorem}}}{\frac{x+1}{x}} = \frac{0}{1} = 0$$

15. If

$$f(x) = \begin{cases} e^{kx}, & \text{if } 0 \leq x \leq 2 \\ x + 3, & \text{if } 2 < x \leq 8 \end{cases}$$

Is it possible to define f at $x = 2$ in such a way that f becomes continuous at $x = 2$? If so, then the value of k is:

- (a) 5

(b) $\ln \sqrt{5}$

(c) $\sqrt{\ln 5}$

(d) $\ln 5$

Answer: **b**

Solution: to make the function continuous then $\lim_{x \rightarrow 2} e^{kx} = \lim_{x \rightarrow 2} x + 3$ so $e^{2k} = 5$ so $k = \ln(\sqrt{5})$

2.7 Midterm, 2021–2022

1. If
- $f(x) = x^3 - 7x + 5$
- , then

$$\lim_{x \rightarrow -2} \frac{f(x) - f(-2)}{x + 2} = \dots\dots$$

- (a) 0
 (b) 5
 (c) 1
 (d) 11

*Answer: b**Solution:*

$$\begin{aligned} f'(x) &= 3x^2 - 7 \\ \lim_{x \rightarrow -2} \frac{f(x) - f(-2)}{x + 2} &= f'(-2) \\ &= 3 \times (-2)^2 - 7 \\ &= 5 \end{aligned}$$

Alternative Solution:

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{f(x) - f(-2)}{x + 2} &= \lim_{x \rightarrow -2} \frac{x^3 - 7x + 6}{x + 2} \\ &= \lim_{x \rightarrow -2} \frac{\frac{d}{dx} [x^3 - 7x + 6]}{\frac{d}{dx} [x + 2]} \\ &= \lim_{x \rightarrow -2} \frac{3x^2 - 7}{1} \\ &= \lim_{x \rightarrow -2} 3x^2 - 7 \\ &= 3(4) - 7 \\ &= 5 \end{aligned}$$

- 2.

$$\lim_{x \rightarrow 0} \frac{\sin(e^x - 1)}{e^{2x} - 1} = \dots\dots$$

- (a) zero
 (b) 3
 (c) $\frac{1}{2}$
 (d) $\frac{1}{4}$

*Answer: c**Solution:*

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(e^x - 1)}{e^{2x} - 1} &= \lim_{x \rightarrow 0} \frac{\sin(e^x - 1)}{(e^x - 1)(e^x + 1)} \\ &= \lim_{x \rightarrow 0} \left[\frac{\sin(e^x - 1)}{e^x - 1} \right] \cdot \lim_{x \rightarrow 0} \left[\frac{1}{e^x + 1} \right] \\ &= \frac{1}{2} \end{aligned}$$

- 3.

$$\lim_{x \rightarrow \infty} [\ln(x) - x] = \dots\dots$$

- (a) zero
 (b) 1
 (c) ∞
 (d) $-\infty$

Answer: d

Solution: As $x \rightarrow \infty$, the decline of the function $y = -x$ to $-\infty$ is faster than the growth of the function $y = \ln x$ to ∞ . Thus,

$$\lim_{x \rightarrow \infty} [\ln x - x] = -\infty$$

Alternative Solution:

$$\begin{aligned} \lim_{x \rightarrow \infty} [\ln(x) - x] &= \lim_{x \rightarrow \infty} x \left(\frac{\ln(x) - x}{x} \right) \\ &= \lim_{x \rightarrow \infty} x \cdot \lim_{x \rightarrow \infty} \frac{\frac{d}{dx} [\ln x - x]}{\frac{d}{dx} [x]} \\ &= \lim_{x \rightarrow \infty} x \cdot \lim_{x \rightarrow \infty} \frac{1}{x} - 1 \\ &= -\infty \end{aligned}$$

4.

$$\lim_{x \rightarrow \infty} \frac{e^{x+1} + \pi^{x-1}}{e^{x-1} + \pi^{x+1}} = \dots\dots$$

- (a) π^2
- (b) e^2
- (c) $\frac{1}{\pi^2}$
- (d) $\frac{1}{e^2}$

Answer: c

Solution: Since $\pi > e$, exponential functions with base π grow faster to ∞ than those with base e . Thus, we can neglect exponential functions with base e as $x \rightarrow \infty$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{e^{x+1} + \pi^{x-1}}{e^{x-1} + \pi^{x+1}} &= \lim_{x \rightarrow \infty} \frac{\pi^{x-1}}{\pi^{x+1}} \\ &= \frac{1}{\pi^2} \end{aligned}$$

Alternative Solution: Since $-1 < \frac{e}{\pi} < 1$, $\lim_{x \rightarrow \infty} \left(\frac{e}{\pi}\right)^x = 0$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{e^{x+1} + \pi^{x-1}}{e^{x-1} + \pi^{x+1}} &= \lim_{x \rightarrow \infty} \frac{\pi^{x+1} \left(\frac{e^{x+1}}{\pi^{x+1}} + \pi^{-2} \right)}{\pi^{x-1} \left(\frac{e^{x-1}}{\pi^{x-1}} + \pi^2 \right)} \\ &= \lim_{x \rightarrow \infty} \frac{\pi^2 \left(\left(\frac{e}{\pi}\right)^{x+1} + \pi^{-2} \right)}{\left(\frac{e}{\pi}\right)^{x-1} + \pi^2} \\ &= \frac{1}{\pi^2} \end{aligned}$$

5. If the function f where

$$f(x) = \begin{cases} x+6 & \text{if } x \leq a \\ x^2 & \text{if } x > a \end{cases}$$

is continuous on $\mathbb{R}$, then $a = \dots\dots$

- (a) 3 or 2
- (b) -3 or -2
- (c) 3 or -2
- (d) 2 or -3

*Answer: c**Solution:*

$$\begin{aligned}
 \lim_{x \rightarrow a^+} f(x) &= \lim_{x \rightarrow a^-} f(x) \\
 \lim_{x \rightarrow a^+} x^2 &= \lim_{x \rightarrow a^-} x + 6 \\
 a^2 &= a + 6 \\
 a &= 3 \text{ or } a = -2
 \end{aligned}$$

6. If $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ and $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$, then

$$\lim_{x \rightarrow a} (f(x) \times g(x)) = \dots\dots$$

- (a) $\frac{4}{3}$
- (b) $\frac{-3}{4}$
- (c) $\frac{3}{4}$
- (d) 2

*Answer: c**Solution:* Assuming f and g being continuous functions,

$$f(a) + g(a) = 2 \quad (1)$$

$$f(a) - g(a) = 1 \quad (2)$$

Solving (1) and (2),

$$\begin{aligned}
 \lim_{x \rightarrow a} (f(x) \times g(x)) &= f(a) \times g(a) \\
 &= \frac{3}{2} \times \frac{1}{2} = \frac{3}{4}
 \end{aligned}$$

7. If the function

$$f : f(x) = \begin{cases} (ax - 1)^3 & , x < 2 \\ a^2x^2 - 1 & , x \geq 2 \end{cases}$$

is continuous on $\mathbb{R}$, then:I. $a = 0$ II. $a = \frac{1}{2}$ III. $a = \frac{3}{2}$

- (a) I only
- (b) II and III only
- (c) II only
- (d) I, II, and III

Answer: d

Solution:

$$\begin{aligned}
 \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^+} f(x) \\
 \lim_{x \rightarrow 2^-} (ax - 1)^3 &= \lim_{x \rightarrow 2^+} a^2 x^2 - 1 \\
 (2a - 1)^3 &= 4a^2 - 1 \\
 8a^3 - 12a^2 + 6a - 1 &= 4a^2 - 1 \\
 8a^3 - 16a^2 + 6a &= 0 \\
 a(4a^3 - 8a^2 + 3) &= 0 \\
 4a \left(a - \frac{1}{2}\right) \left(a - \frac{3}{2}\right) &= 0 \\
 a = 0, \quad a = \frac{1}{2}, \quad a = \frac{3}{2}
 \end{aligned}$$

8. Consider the function

$$f : f(x) = \begin{cases} \ln x & , 0 < x < 1 \\ ax^2 + b & , x \geq 1 \end{cases}$$

If $f(2) = 3$, determine the values of a and b for which $f(x)$ is continuous over its domain.

(a) $a = 1, b = -1$

(b) $a = -1, b = 1$

(c) $a = -1, b = -1$

(d) $a = 1, b = 1$

Answer: a

Solution:

$$\begin{aligned}
 f(2) &= 3 \\
 4a + b &= 3 \tag{1} \\
 \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^-} f(x) \\
 \lim_{x \rightarrow 1^+} ax^2 + b &= \lim_{x \rightarrow 1^-} \ln x \\
 a + b &= 0 \tag{2}
 \end{aligned}$$

From (1) and (2), $a = 1$ and $b = -1$.

9. If

$$f : f(x) = \frac{x^5 - x^4 - 2x^3}{x^4 - 3x^3 - x^2 + 3x}$$

For what values of $x = a$ does f have a removable discontinuity at a ?

(a) 1 and 3

(b) 0 and -1

(c) 1 and -1

(d) 0, 1, -1, and 3

Answer: b

Solution:

$$f(x) = \frac{x^3(x+1)(x-2)}{x(x-1)(x+1)(x-3)}$$

A rational function has a removable discontinuity at $x = a$ if both the numerator and the denominator have the factor $(x - a)$. Thus, f has removable discontinuities at $x = 0$ and $x = -1$.

- 10 Determine whether the **Mean Value Theorem** can be applied to the function $f(x) = 3 + \sqrt{x}$ over $[0, 4]$. If so, find all possible values of c .
- (a) No, it cannot be applied.
 (b) Yes, at $c = 1$.
 (c) Yes, at $c = 2$.
 (d) Yes, at $c = 0$.

Answer: b

Solution: Since f is continuous over $[0, 4]$ and differentiable over $(0, 4)$, the **Mean Value Theorem** is applicable over the given interval.

$$f'(c) = \frac{f(4) - f(0)}{4 - 0}$$

$$\frac{1}{2\sqrt{c}} = \frac{1}{2}$$

$$c = 1$$

10. Determine whether the **Mean Value Theorem** can be applied to the function $f(x) = \sin(2x)$ over $[0, \pi]$. If so, find all possible values of c .
- (a) No, it cannot be applied.
 (b) Yes, at $c = \frac{\pi}{4}$ and $c = \frac{\pi}{2}$.
 (c) Yes, at $c = \frac{\pi}{4}$ and $c = \frac{3\pi}{4}$.
 (d) Yes, at $c = \frac{3\pi}{2}$ and $\frac{\pi}{2}$.

Answer: c

Solution: Since f is continuous over $[0, \pi]$ and differentiable over $(0, \pi)$, the **Mean Value Theorem** is applicable over the given interval.

$$f'(c) = \frac{f(\pi) - f(0)}{\pi - 0}$$

$$2 \cos(2c) = 0$$

$$c = \frac{(2n + 1)\pi}{4}$$

for any $n \in \mathbb{Z}$.

11. Suppose we know that: f is continuous and differentiable over $[-7, 0]$, $f(-7) = -3$, and $f'(x) \leq 2$. What is the greatest possible value of $f(0)$?
- (a) 11
 (b) 7
 (c) 2
 (d) zero

Answer: a

Solution: Since f is continuous and differentiable over $[-7, 0]$, the **Mean Value Theorem** is applicable over this interval.

$$f'(c) = \frac{f(0) - f(-7)}{0 - (-7)}$$

$$f'(c)_{max} = \frac{f(0)_{max} - f(-7)}{0 - (-7)}$$

$$2 = \frac{f(0)_{max} + 3}{7}$$

$$f(0)_{max} = 11$$

The maximum possible value of $f'(c)$ is 2.

12.

$$\frac{d}{dx} \left[\ln \sqrt{\frac{x^2+1}{x+3}} \right] = \dots$$

- (a) $\frac{x}{x^2+1} + \frac{1}{2x+6}$
 (b) $\frac{x}{x^2+1} - \frac{1}{2x+6}$
 (c) $\frac{2x}{x^2+1} + \frac{1}{x+3}$
 (d) $\frac{2x}{x^2+1} - \frac{1}{x+3}$

Answer: b**Solution:**

$$\begin{aligned} \frac{d}{dx} \left[\ln \sqrt{\frac{x^2+1}{x+3}} \right] &= \frac{d}{dx} \left[\ln \left(\frac{x^2+1}{x+3} \right)^{\frac{1}{2}} \right] \\ &= \frac{d}{dx} \left[\frac{1}{2} \ln \left(\frac{x^2+1}{x+3} \right) \right] \\ &= \frac{1}{2} \cdot \frac{d}{dx} [\ln(x^2+1) - \ln(x+3)] \\ &= \frac{1}{2} \left(\frac{2x}{x^2+1} - \frac{1}{x+3} \right) \\ &= \frac{x}{x^2+1} - \frac{1}{2x+6} \end{aligned}$$

13. If $f(x) = x^{100}e^x$, then $f^{(100)}(0) = \dots$

- (a) 0
 (b) 99!
 (c) 100!
 (d) 101!

Answer: c

Solution: Using $\overbrace{\text{General Leibniz Rule}}^{\text{Link to Info}}$, a generalization of the $\overbrace{\text{Product Rule}}^{\text{Link to Info}}$,

$$\begin{aligned} (fg)^{(n)} &= \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)} \\ (x^{100}e^x)^{(100)} &= \sum_{k=0}^{100} \binom{100}{k} (x^{100})^{(k)} (e^x)^{(100-k)} \\ &= \sum_{k=0}^{100} \binom{100}{k} \frac{(100)!}{(100-k)!} x^{100-k} e^x \end{aligned}$$

When substituting with $x = 0$ to calculate $f^{(100)}(0)$, we cancel all terms with the factor x out. The expansion of this summation produces 101 terms with decreasing non-negative power of x . Thus, the only free-of- x term is when $k = 100$.

$$\begin{aligned} f^{(100)}(0) &= \frac{100!}{0!} x^0 e^0 \\ &= 100! \end{aligned}$$

Link to Info

Alternative Solution: Using the **Power Series Expansion** of e^x ,

$$\begin{aligned}
 f(x) &= x^{100} e^x \\
 &= x^{100} \sum_{n=0}^{\infty} \frac{x^n}{n!} \\
 &= \sum_{n=0}^{\infty} \frac{x^{n+100}}{n!} \\
 f^k(x) &= \sum_{n=0}^{\infty} \frac{(n+100)!}{(n+100-k)!} \frac{x^{n+100-k}}{n!} \\
 f^{(100)}(x) &= \sum_{n=0}^{\infty} \frac{(n+100)!}{n!} \frac{x^n}{n!}
 \end{aligned}$$

When substituting with $x = 0$ to calculate $f^{(100)}(0)$, we cancel all terms with the factor x out. The expansion of this summation produces an infinite number of terms with increasing non-negative power of x . Thus, the only free-of- x term is when $n = 0$.

$$\begin{aligned}
 f^{(100)}(0) &= \frac{100!}{0!} \frac{x^0}{0!} \\
 &= 100!
 \end{aligned}$$

14. If $f(x^3 - 7) = 11x^2 + 16x + 5$, then $f'(1) = \dots\dots$

- (a) 8
- (b) 7
- (c) 6
- (d) 5

Answer: **d**

Solution:

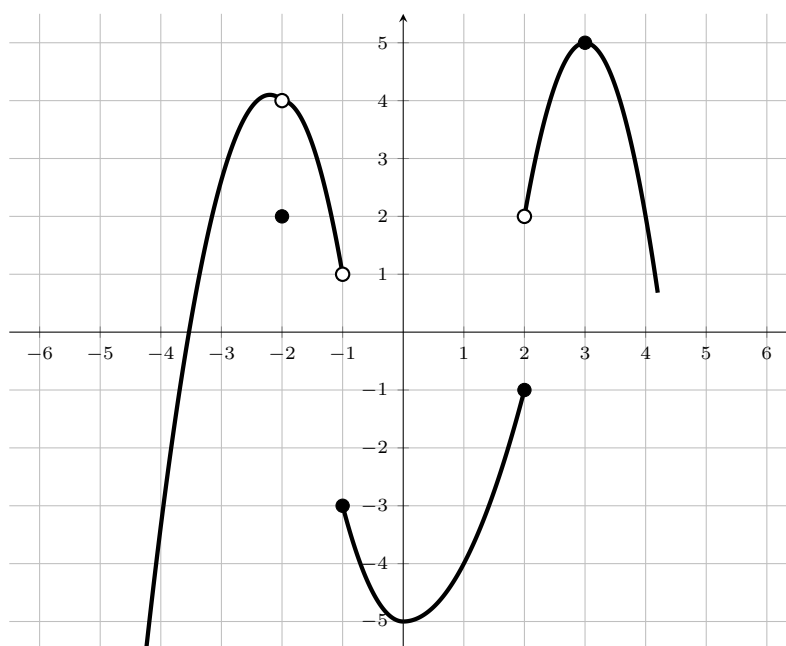
$$\begin{aligned}
 \frac{d}{dx} [f(x^3 - 7)] &= \frac{d}{dx} [11x^2 + 16x + 5] \\
 3x^2 \cdot f'(x^3 - 7) &= 22x + 16 \\
 f'(x^3 - 7) &= \frac{22x + 16}{3x^2}
 \end{aligned}$$

To find $f'(1)$, we substitute with $x = 2$.

$$f'(1) = \frac{22 \times 2 + 16}{3 \times 4} = 5$$

2.8 Final Term, 2021–2022

1. The figure below shows the graph of a function $f(x)$. Which of the following is **NOT** true about f ?



- (a) $\lim_{x \rightarrow 3} f(x) = 5$
 (b) f is discontinuous at $x = 2$
 (c) $f(-2) = 2$
 (d) $\lim_{x \rightarrow -2} f(x)$ does not exist.

Answer: d

Solution: As shown in the graph, The $\lim_{x \rightarrow 2} f(x)$ exists:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

So the wrong statement is *d* here.

2. If $f(x) = x^4 + 5x - 3$, then,

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \dots\dots$$

- (a) zero
 (b) 1
 (c) 3
 (d) 9

Answer: d

Solution:

$$\begin{aligned} f'(x) &= 4x^3 + 5 \\ \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} &= f'(1) \\ &= 4(1)^3 + 5 \\ &= 9 \end{aligned}$$

- 3.

$$\lim_{x \rightarrow 2} \frac{\pi^x - \pi^2}{e^x - e^2} = \dots\dots$$

- (a) $\frac{\pi^2}{e^2}$

- (b) $\frac{\pi}{e} \ln \pi$
 (c) $\frac{\pi^2}{e^2} \ln \pi$
 (d) None of them

Answer: c

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{\pi^x - \pi^2}{e^x - e^2} &= \lim_{x \rightarrow 2} \frac{\frac{d}{dx} [\pi^x - \pi^2]}{\frac{d}{dx} [e^x - e^2]} \\ &= \lim_{x \rightarrow 2} \frac{\pi^x \ln \pi}{e^x} \\ &= \frac{\pi^2}{e^2} \ln \pi\end{aligned}$$

4. If f is a continuous function at $x = 2$, where:

$$f(x) = \begin{cases} \frac{ax^2 + bx + c}{(x-2)^2} & , x \neq 2 \\ 5 & , x = 2 \end{cases}$$

then $(a, b, c) = \dots\dots$

- (a) $(1, -4, 4)$
 (b) $(1, 4, -4)$
 (c) $(5, -20, 20)$
 (d) $(5, 20, 20)$

Answer: c

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{ax^2 + bx + c}{(x-2)^2} &= 5 \\ ax^2 + bx + c &= 5(x-2)^2 \\ ax^2 + bx + c &= 5x^2 - 20x + 20 \\ a = 5, \quad b = -20, \quad c &= 20\end{aligned}$$

5. If f is a function where

$$f(x) = \begin{cases} 8x - 8 & , x < -2 \\ \frac{2}{3}x^3 & , x \geq -2 \end{cases}$$

then f is $\dots\dots$

- (a) differentiable at $x = 2$
 (b) continuous at $x = 2$
 (c) continuous and not differentiable at $x = -2$
 (d) not differentiable at $x = -2$

Answer: d

Solution:

$$\begin{aligned}\lim_{x \rightarrow -2^-} (8x - 8) &= 8 \times (-2) - 8 = -24 \\ \lim_{x \rightarrow -2^+} \left(\frac{2}{3}x^3 \right) &= \frac{2}{3} \times (-2)^3 = \frac{-16}{3} \\ \therefore \lim_{x \rightarrow -2^-} f(x) &\neq \lim_{x \rightarrow -2^+} f(x)\end{aligned}$$

Then, f is not continuous and, consequently, not differentiable at $x = -2$.

6. If $y = \ln \sqrt{(x^2 + 1)(x + 3)}$, then $\frac{dy}{dx} = \dots\dots$

- (a) $\frac{x}{x^2+1} + \frac{1}{2x+6}$
- (b) $\frac{x}{x^2+1} - \frac{1}{2x+6}$
- (c) $\frac{2x}{x^2+1} + \frac{1}{x+3}$
- (d) $\frac{2x}{x^2+1} - \frac{1}{x+3}$

Answer: a

Solution:

$$\begin{aligned}
 y &= \ln \sqrt{(x^2 + 1)(x + 3)} \\
 &= \ln [(x^2 + 1)(x + 3)]^{\frac{1}{2}} \\
 &= \frac{1}{2} \ln (x^2 + 1)(x + 3) \\
 &= \frac{1}{2} (\ln (x^2 + 1) + \ln (x + 3)) \\
 \frac{dy}{dx} &= \frac{1}{2} \left(\frac{2x}{x^2 + 1} + \frac{1}{x + 3} \right) \\
 &= \frac{x}{x^2 + 1} + \frac{1}{2x + 6}
 \end{aligned}$$

Alternative Solution:

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{\sqrt{(x^2 + 1)(x + 3)}} \cdot \frac{2x(x + 3) + (x^2 + 1)}{2\sqrt{(x^2 + 1)(x + 3)}} \\
 &= \frac{2x(x + 3) + (x^2 + 1)}{2(x^2 + 1)(x + 3)} \\
 &= \frac{x}{x^2 + 1} + \frac{1}{2x + 6}
 \end{aligned}$$

7. Given that: $f(-3) = 12$, $f'(-3) = 9$, $g(-3) = -4$, and $g'(-3) = 7$, then $\left(\frac{g}{f}\right)'(-3) = \dots\dots$

- (a) $-\frac{15}{2}$
- (b) $\frac{5}{6}$
- (c) $-\frac{7}{9}$
- (d) $\frac{9}{7}$

Answer: b

Solution:

$$\begin{aligned}
 \left(\frac{g}{f}\right)'(-3) &= \frac{g'(-3)f(-3) - g(-3)f'(-3)}{(f(-3))^2} \\
 &= \frac{7 \times 12 - (-4) \times 9}{12^2} = \frac{5}{6}
 \end{aligned}$$

8. If $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots} \rightarrow \infty}}$, then $\frac{dy}{dx} = \dots$

- (a) $\frac{\sin x}{1-2y}$
- (b) $\frac{\sin x}{2y-1}$
- (c) $\frac{\cos x}{1-2y}$
- (d) $\frac{\cos x}{2y-1}$

*Answer: a**Solution:*

$$\begin{aligned}
 y &= \sqrt{\cos x + y} \\
 y^2 &= \cos x + y \\
 \frac{d}{dx} [y^2] &= \frac{d}{dx} [\cos x + y] \\
 2y \frac{dy}{dx} &= -\sin x + \frac{dy}{dx} \\
 2y \frac{dy}{dx} - \frac{dy}{dx} &= -\sin x \\
 \frac{dy}{dx} \cdot (2y - 1) &= -\sin x \\
 \frac{dy}{dx} &= \frac{-\sin x}{2y - 1} \\
 &= \frac{\sin x}{1 - 2y}
 \end{aligned}$$

9. If $\frac{x}{y} - \frac{y}{x} = 5$, then $\frac{dy}{dx} = \dots\dots$

- (a) $\frac{2x+5y}{5x+2y}$
- (b) $\frac{2x+5y}{5x-2y}$
- (c) $\frac{2y-5x}{5y+2x}$
- (d) $\frac{2x-5y}{5x+2y}$

*Answer: d**Solution:*

$$\begin{aligned}
 \frac{x^2 - y^2}{xy} &= 5 \\
 x^2 - y^2 &= 5xy \\
 \frac{d}{dx} [x^2 - y^2] &= \frac{d}{dx} [5xy] \\
 2x - 2y \frac{dy}{dx} &= 5y + 5x \frac{dy}{dx} \\
 5x \frac{dy}{dx} + 2y \frac{dy}{dx} &= 2x - 5y \\
 \frac{dy}{dx} \cdot (5x + 2y) &= 2x - 5y \\
 \frac{dy}{dx} &= \frac{2x - 5y}{5x + 2y}
 \end{aligned}$$

10. If $f(x^3 - 1) = x^6 - 5x^3 + 3$, then $f'(-2) = \dots\dots$

- (a) 7
- (b) -7
- (c) 21
- (d) -21

*Answer: b**Solution:*

$$\begin{aligned}
 \frac{d}{dx} [f(x^3 - 1)] &= \frac{d}{dx} [x^6 - 5x^3 + 3] \\
 3x^2 f'(x^3 - 1) &= 6x^5 - 15x^2 \\
 f'(x^3 - 1) &= 2x^3 - 5
 \end{aligned}$$

To find $f'(-2)$, we substitute with $x = -1$.

$$f'(-2) = 2(-1) - 5 = -7$$

11. If $f(x) = \sqrt{x} g(x)$, $g(4) = 8$, and $g'(4) = 7$, then $f'(4) = \dots\dots$

- (a) 16
- (b) 28
- (c) $\frac{7}{4}$
- (d) $\frac{4}{7}$

Answer: a

Solution:

$$\begin{aligned} f'(x) &= \frac{g(x)}{2\sqrt{x}} + \sqrt{x} g'(x) \\ f'(4) &= \frac{g(4)}{2\sqrt{4}} + \sqrt{4} g'(4) \\ &= 16 \end{aligned}$$

12. Let f be a differentiable function at $x = 1$, $f(1) = 0$, and $\lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$, then $f'(1) = \dots\dots$

- (a) 6
- (b) 5
- (c) 1
- (d) zero

Answer: b

Solution:

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(1+h) - 0}{h} \\ &= 5 \end{aligned}$$

13. The average rate of change of $f(x) = 4x + 5$ on the interval $[a, b]$ is $\dots\dots$

- (a) 5
- (b) 4
- (c) 1
- (d) None of them

Answer: b

Solution: Since f is a linear function, its average rate of change over any interval is equal to its derivative. $f'(x) = 4$.

14. The function f where $f(x) = ax^3 + bx^2 + 11x - 6$ satisfies the conditions of **Rolle's Theorem** over $[1, 3]$ at $x = 2 + \frac{1}{\sqrt{3}}$, then the values of a and b respectively are $\dots\dots$

- (a) -3, 2
- (b) 1, -6
- (c) 1, 3
- (d) None of them

*Answer: b**Solution:*

$$\begin{aligned}
 f(1) &= f(3) \\
 a + b + 5 &= 27a + 9b + 27 \\
 26a + 8b &= -22
 \end{aligned} \tag{1}$$

$$\begin{aligned}
 f'(x) &= 3ax^2 + 2bx + 11 \\
 f'\left(2 + \frac{1}{\sqrt{3}}\right) &= 0 \\
 (13 + 4\sqrt{3})a + \frac{12 + 2\sqrt{3}}{3}b &= -11
 \end{aligned} \tag{2}$$

From (1) and (2), $a = 1$ and $b = -6$.15. If $y = 2^{\sin x}$, then $\frac{dy}{dx} = \dots\dots$

- (a) $2^{\sin x} \cos x$
- (b) $-2^{\sin x} \cos x$
- (c) $2^{\sin x} \ln 2^{\cos x}$
- (d) $-2^{\sin x} \cos x \ln 2$

*Answer: c**Solution:*

$$\begin{aligned}
 \frac{dy}{dx} &= 2^{\sin x} \ln(2) \cdot \cos x \\
 &= 2^{\sin x} \ln 2^{\cos x}
 \end{aligned}$$

16.

$$\int e^{(1+\ln x)} dx = \dots\dots + C$$

- (a) $e^{(1+\ln x)}$
- (b) $xe^{(1+\ln x)}$
- (c) $ex + \frac{1}{2}x^2$
- (d) $\frac{1}{2}ex^2$

*Answer: d**Solution:*

$$\begin{aligned}
 \int e^{(1+\ln x)} dx &= \int e^1 \cdot e^{\ln x} dx \\
 &= \int ex dx \\
 &= \frac{1}{2}ex^2 + C
 \end{aligned}$$

17.

$$\int (e^x + 2^x) dx = \dots\dots$$

- (a) $e^x + 2^x + C$
- (b) $e^x + 2^x \ln 2 + C$
- (c) $e^x + \frac{2^x}{\ln 2} + C$
- (d) $\ln x + \ln 2 + C$

Answer: c

18. Find the specific antiderivative of $f'(x) = 2 - \cos x$, $f\left(\frac{\pi}{2}\right) = 2$

- (a) $2x + \sin x + (1 + \pi)$
- (b) $2x - \sin x + (2 - \pi)$
- (c) $2x + \sin x + (2 - \pi)$
- (d) $2x - \sin x + (3 - \pi)$

Answer: d

Solution:

$$\begin{aligned}
 f(x) &= \int (2 - \cos x) \, dx \\
 &= 2x - \sin x + C \\
 f\left(\frac{\pi}{2}\right) &= 2 \\
 2\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{2}\right) + C &= 2 \\
 \pi - 1 + C &= 2 \\
 C &= 3 - \pi \\
 f(x) &= 2x - \sin x + (3 - \pi)
 \end{aligned}$$

19.

$$\int \frac{\sin x}{\cos^2 x} \, dx = \dots\dots + C$$

- (a) $\sec x \tan x$
- (b) $\sec x$
- (c) $\sec^2 x$
- (d) $\ln |\cos^2 x|$

Answer: b

Solution:

$$\begin{aligned}
 \int \frac{\sin x}{\cos^2 x} \, dx &= \int \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} \, dx \\
 &= \int \sec x \tan x \, dx \\
 &= \sec x + C
 \end{aligned}$$

20. The value of c that satisfies the

for which $f(x) = |x|$ in the interval $[-1, 1]$ is

- (a) $-\frac{1}{2}$
- (b) zero
- (c) $\frac{1}{2}$
- (d) None existent in the interval.

Answer: d

Solution: For the to be applicable over $[-1, 1]$, the function f must be differentiable over $(-1, 1)$. $f(x) = |x|$ is not differentiable at $x = 0$ because it has a corner there. Thus, the value of c is non-existent in the interval.

21. If x is a normal distribution variable its mean is μ , its standard deviation is σ , then $p(x \leq \mu) = \dots$

- (a) -0.5
- (b) 0
- (c) 0.5

(d) 1

*Answer: c**Solution:* It's known that the probability of some normal distribution variable being less than the mean is 0.5, and it's 0.5 being more than the mean.

22. In normal distribution, the mean is 50. A score of 70 corresponds to a standard score of 2. The standard deviation of the distribution is:

(a) 6
 (b) 8
 (c) 10
 (d) 12

*Answer: c**Solution:* We know that:

$$Z = \frac{x - \mu}{\sigma} \Rightarrow 2 = \frac{70 - 50}{\sigma} \Rightarrow \sigma = \frac{20}{2} = 10$$

23. Medical tests indicate that the amount of an antibiotic needed to destroy a bacterial infection in a patient is normally distributed with a mean of 120 mg and a standard deviation of 15 mg. The percentage of patients who would require more than 135 mg to clear the infection is:

(a) 16%
 (b) 2.5%
 (c) 5%
 (d) 95%

*Answer: A**Solution:* We know that:

$$Z = \frac{x - \mu}{\sigma} \Rightarrow Z = \frac{135 - 120}{15} = 1$$

and we want the values more than 135, then we should find the probability of $Z > 1$. From the table, we will find that the probability of $Z > 1$ after removing the left half is 0.3413. Then the answer is $(0.5 - 0.3413) \cdot 100 = 15.87\%$.

24. A fishing boat catches a load of fish and finds the mass of each fish. The masses of the fish are normally distributed with a mean of 800g and a standard deviation of 75g. If a fish is chosen from the catch, its mass will almost certainly lie between:

(a) 575g and 1075g
 (b) 725g and 875g
 (c) 650g and 950g
 (d) 800g and 1025g

*Answer: B**Solution:* We know that the standard deviation is a measure of the amount of variation of a random variable about its mean. From this information, the range will be from $[(800-75) , (800+75)]$ which equals answer B.

25. The table below shows the mean and the standard deviation of times in the 100 m by the same group of athletes on four different days. It also shows Matt's time on each of these days.

On what day did Matt give his best performance?

Day	Mean	Standard deviation	Matt's time
8Jan.	10.21	0.15	10.12
15Jan.	10.22	0.12	10.11
22Jan.	10.14	0.09	10.05
29Jan.	10.48	0.28	10.30

(a) 8 Jan.
 (b) 15 Jan.

(c) 22 Jan.

(d) 29 Jan.

Answer: **A**

Solution: we will begin with finding the value Z for all days.

$$Z_8 = \frac{10.12 - 10.21}{0.15} = -0.6$$

$$Z_{15} = \frac{10.11 - 10.22}{0.12} = -0.916$$

$$Z_{22} = \frac{10.05 - 10.14}{0.09} = -1$$

$$Z_{29} = \frac{10.30 - 10.48}{0.28} = -0.643$$

From the results, it appears that his best performance was achieved on 8 Jan as the highest Z value.

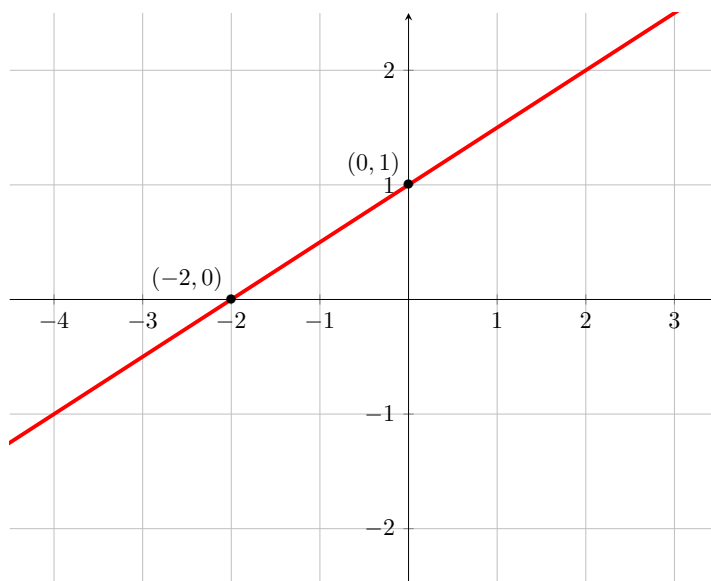
2.9 Midterm, 2022–2023

1. If $y = f(x)$, substituting $x = 1.9, 1.99, 1.999, \dots$ to find

- (a) $\lim_{x \rightarrow 1} f(x)$
- (b) $\lim_{x \rightarrow 2^+} f(x)$
- (c) $\lim_{x \rightarrow 2^-} f(x)$
- (d) $f(2)$

Answer: **c**

2. Below the graph of a linear function $y = f(x)$, then $\lim_{x \rightarrow 10} f(x) = \dots$



- (a) $\frac{1}{4}$
- (b) $\frac{1}{2}$
- (c) **6**
- (d) Doesn't exist

Answer: **c**

Solution:

$$m = \frac{\Delta y}{\Delta x} = \frac{1}{2}$$

$$y = mx + c = \frac{1}{2}x + 1$$

Since f is a continuous (linear) function,

$$\begin{aligned} \lim_{x \rightarrow 10} f(x) &= f(10) \\ &= 6 \end{aligned}$$

3. The value of

$$\lim_{x \rightarrow a} \frac{\sqrt{x^2 + 3a^2} - 2a}{x - a} = \dots$$

where $a > 0$

- (a) $\frac{1}{2a}$
- (b) $\frac{1}{a}$
- (c) **$\frac{1}{2}$**

(d) a *Answer: c**Solution:*

$$\begin{aligned}
\lim_{x \rightarrow a} \frac{\sqrt{x^2 + 3a^2} - 2a}{x - a} &= \lim_{x \rightarrow a} \frac{\sqrt{x^2 + 3a^2} - 2a}{x - a} \cdot \frac{\sqrt{x^2 + 3a^2} + 2a}{\sqrt{x^2 + 3a^2} + 2a} \\
&= \lim_{x \rightarrow a} \frac{(x^2 + 3a^2) - (2a)^2}{(x - a)(\sqrt{x^2 + 3a^2} + 2a)} \\
&= \lim_{x \rightarrow a} \frac{x^2 - a^2}{(x - a)(\sqrt{x^2 + 3a^2} + 2a)} \\
&= \lim_{x \rightarrow a} \frac{\cancel{(x - a)}(x + a)}{\cancel{(x - a)}(\sqrt{x^2 + 3a^2} + 2a)} \\
&= \frac{1}{2}
\end{aligned}$$

4. If

$$\lim_{x \rightarrow \infty} (a\sqrt{2x^2 + x + 1} - bx) = 2$$

where $a, b \in \mathbb{R}$, then the value of $a^2 + b^2 = \dots$

(a) 32

(b) 64

(c) 48

(d) 96

*Answer: d**Solution:*

$$\lim_{x \rightarrow \infty} \left(x \left(a\sqrt{2 + \frac{1}{x} + \frac{1}{x^2}} - b \right) \right) = 2$$

Since $\lim_{x \rightarrow \infty} x = \infty$, $\lim_{x \rightarrow \infty} \left(x \left(a\sqrt{2 + \frac{1}{x} + \frac{1}{x^2}} - b \right) \right)$ is finite if and only if $\lim_{x \rightarrow \infty} \left(a\sqrt{2 + \frac{1}{x} + \frac{1}{x^2}} - b \right) = 0$. Then,

$$\sqrt{2}a = b \quad (1)$$

$$\lim_{x \rightarrow \infty} (a\sqrt{2x^2 + x + 1} - \sqrt{2}ax) = 2$$

$$a \lim_{x \rightarrow \infty} (\sqrt{2x^2 + x + 1} - \sqrt{2}x) = 2$$

$$a \lim_{x \rightarrow \infty} \frac{(2x^2 + x + 1) - 2x^2}{(\sqrt{2x^2 + x + 1} + \sqrt{2}x)} = 2$$

$$a \lim_{x \rightarrow \infty} \frac{x + 1}{(\sqrt{2x^2 + x + 1} + \sqrt{2}x)} = 2$$

$$a \lim_{x \rightarrow \infty} \frac{\cancel{x} \left(1 + \frac{1}{x} \right)}{\cancel{x} \left(\sqrt{2 + \frac{1}{x} + \frac{1}{x^2}} + \sqrt{2} \right)} = 2$$

$$\frac{a}{2\sqrt{2}} = 2$$

$$a = 4\sqrt{2} \quad (2)$$

From (1) and (2), $b = 8$ and $a^2 + b^2 = (4\sqrt{2})^2 + 8^2 = 96$.

Alternative Way of Obtaining (1):

$$\begin{aligned}\lim_{x \rightarrow \infty} (a\sqrt{2x^2 + x + 1} - bx) &= \lim_{x \rightarrow \infty} 2 \\ \lim_{x \rightarrow \infty} \left(\frac{a\sqrt{2x^2 + x + 1} - bx}{x} \right) &= \lim_{x \rightarrow \infty} \frac{2}{x} \\ \lim_{x \rightarrow \infty} \left(a\sqrt{2 + \frac{1}{x} + \frac{1}{x^2}} - b \right) &= 0 \\ \sqrt{2}a &= b\end{aligned}$$

5. If

$$\lim_{x \rightarrow 0} \frac{\sin^2(3x^3)}{x^n} = 9$$

then $n = \dots$

(a) 6

(b) 5

(c) 3

(d) 2

Answer: a

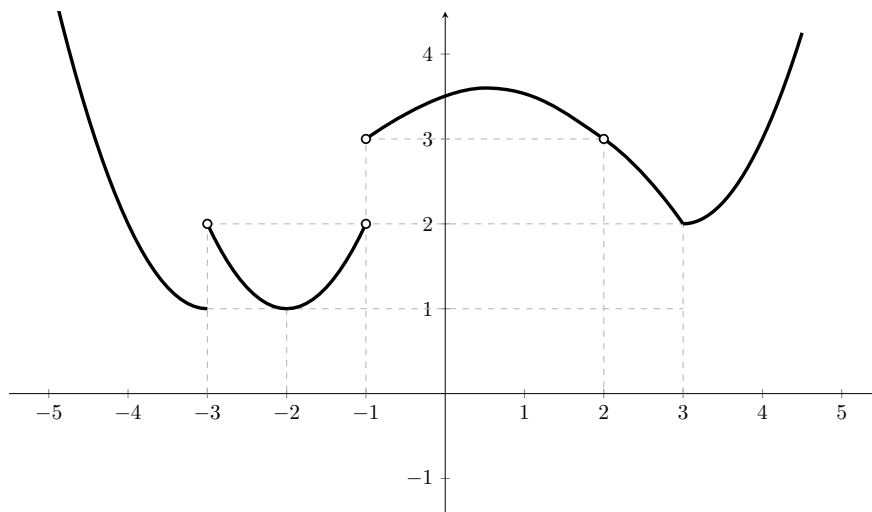
Solution:

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{\sin(3x^3)}{x^{\frac{n}{2}}} \right)^2 &= 3^2 \\ \Rightarrow \lim_{x \rightarrow 0} \frac{\sin(3x^3)}{x^{\frac{n}{2}}} &= 3 \\ \Rightarrow \frac{n}{2} &= 3 \\ \Rightarrow n &= 6\end{aligned}$$

Alternative Solution:

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{\sin(3x^3)}{x^{\frac{n}{2}}} \right)^2 &= 9 \\ \lim_{x \rightarrow 0} \left(\frac{3}{x^{\frac{n}{2}-3}} \cdot \frac{\sin(3x^3)}{3x^3} \right)^2 &= 9 \\ \lim_{x \rightarrow 0} \left(\frac{3}{x^{\frac{n}{2}-3}} \right)^2 &= 9 \\ \lim_{x \rightarrow 0} \frac{\cancel{3}}{x^{n-6}} &= \cancel{9} \\ \lim_{x \rightarrow 0} x^{6-n} &= 1 \\ 6-n &= 0 \\ n &= 6\end{aligned}$$

6. The opposite figure of the function $y = f(x)$. The function can be redefined at $x = \dots$ to be continuous.



- (a) -3
 (b) -1
 (c) 2
 (d) Each of -3, -1, and 2

Answer: c

Solution: If a function g has a removable discontinuity at $x = a$,

$$\lim_{x \rightarrow a^+} g(x) = \lim_{x \rightarrow a^-} f(x) \neq g(a)$$

It is the only type of discontinuities that can be overcome by redefining the function at the point of discontinuity. f has a removable discontinuity at only $x = 2$.

7. To find $\lim_{x \rightarrow 0} \sqrt{1 - \sin x}$ we use which of the following properties of limits

- (a) $\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = \lim_{x \rightarrow 0} f(x) \cdot \lim_{x \rightarrow 0} g(x)$
 (b) $\lim_{x \rightarrow 0} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow 0} f(x)}{\lim_{x \rightarrow 0} g(x)}$, where $\lim_{x \rightarrow 0} g(x) \neq 0$
 (c) $\lim_{x \rightarrow 0} (kf(x)) = k \lim_{x \rightarrow 0} f(x)$
 (d) $\lim_{x \rightarrow 0} f(g(x)) = f(\lim_{x \rightarrow 0} g(x))$

Answer: d

8. If the function

$$f(x) = \begin{cases} \frac{x \tan 2x}{1 - \cos x} & x \neq 0 \\ k & x = 0 \end{cases}$$

is continuous at $x = 0$, then $k = \dots$

- (a) $\frac{1}{4}$
 (b) $\frac{1}{2}$
 (c) 2
 (d) 4

*Answer: d**Solution:*

$$\begin{aligned}
f(0) &= \lim_{x \rightarrow 0} f(x) \\
k &= \lim_{x \rightarrow 0} \frac{x \tan 2x}{1 - \cos x} \\
&= \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos x} \cdot \frac{\tan 2x}{2x} \cdot 2 \\
&= 2 \cdot \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \right)^{-1} \cdot \lim_{2x \rightarrow 0} \frac{\tan 2x}{2x} \\
&= 2 \times 2 \times 1 \\
&= 4
\end{aligned}$$

9. If

$$f(x) = \begin{cases} x + a & , x \leq 0 \\ |x - 4| & , x > 0 \end{cases}, \quad g(x) = \begin{cases} x + 1 & , x < 0 \\ (x - 4)^2 + b & , x \geq 0 \end{cases}$$

are continuous on $\mathbb{R}$, then $(g \circ f)(2) + (f \circ g)(-2) = \dots$ **(a) -8**

(b) -10

(c) 8

(d) 10

*Answer: a**Solution:*

$$\begin{aligned}
f(0) &= \lim_{x \rightarrow 0^+} f(x) \\
a &= 4 \\
g(0) &= \lim_{x \rightarrow 0^-} g(x) \\
16 + b &= 1 \\
b &= -15 \\
(g \circ f)(2) + (f \circ g)(-2) &= g(2) + f(-1) \\
&= -11 + 3 \\
&= -8
\end{aligned}$$

10. Let f be identified as follows:

$$f(x) = \begin{cases} \sin x & \text{if } x < \pi \\ mx + n & \text{if } x \geq \pi \end{cases}$$

where m and n are constants. Determine m and n such that f is differentiable on $\mathbb{R}$.**(a) $m = -1, n = \pi$** (b) $m = 1, n = \pi$ (c) $m = -1, n = -\pi$ (d) $m = 1, n = -\pi$ *Answer: a*

Solution:

$$\lim_{h \rightarrow 0^+} \frac{m(x+h) + n - mx - n}{h} = \lim_{h \rightarrow 0^-} \frac{\sin(x+h) - \sin x}{h}$$

$$m = -1 \quad (1)$$

$$f(\pi) = \lim_{x \rightarrow \pi^-}$$

$$m\pi + n = 0$$

$$n = \pi \quad (2)$$

11. If Rolle's Theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$ where $x \in [1, 2]$ with $f'(\frac{4}{3}) = 0$ then the order pair (a, b) equal to

- (a) $(-5, 8)$
 (b) $(5, -8)$
 (c) $(5, 8)$
 (d) $(-5, -8)$

*Answer: c**Solution:*

$$f(1) = f(2)$$

$$1 - a + b - 4 = 8 - 4a + 2b - 4$$

$$3a - b = 7 \quad (1)$$

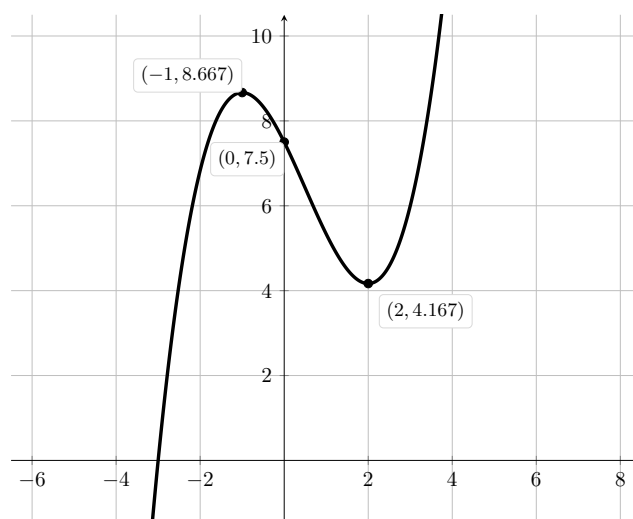
$$f'(x) = 3x^2 - 2ax + b$$

$$f'(\frac{4}{3}) = 0$$

$$b - \frac{8}{3}a = -\frac{16}{3} \quad (2)$$

The solution of (1) and (2) is $a = 5$ and $b = 8$.

12. The opposite figure shows the graph of function $y = f(x)$. The curve of $f'(x)$ intersects x -axis at $x = \dots$



- (a) -1 and 2
 (b) -1 and 3
 (c) 2
 (d) -3

Answer: a

Solution: The graph of $f'(x)$ intersects the x -axis at $(c, 0)$ where $f'(c) = 0$. **The First Derivative Theorem for Local Extreme values** states that if a differentiable function f has a local extreme value at an interior point c of its domain, $f'(c) = 0$. Then, the two local extrema of f , at $x = -1$ and $x = 2$, are the x -intercepts of $f'(x)$.

13. Suppose f is a function, $f(x) = \frac{1}{2}(x^2 + 4x + 1)$, then $f'(1) = \dots$

(a) 6

(b) 3

(c) 2

(d) 1

Answer: b

Solution:

$$f'(x) = \frac{1}{2}(2x + 4)$$

$$f'(1) = 3$$

14. For the function $f : f(x) = x + \frac{1}{x}$, $x \in [1, 3]$, the value of c in the mean value theorem

(a) zero

(b) $\sqrt{2}$

(c) 1

(d) $\sqrt{3}$

Answer: d

Solution:

$$f'(x) = 1 - \frac{1}{x^2}$$

$$f'(c) = \frac{f(3) - f(1)}{3 - 1}$$

$$1 - \frac{1}{c^2} = \frac{4}{6}$$

$$c = \sqrt{3}$$

15. Rolle's Theorem is not applicable to the function $f : f(x) = |x|$, on $[-1, 1]$ because

(a) $f(-1) \neq f(1)$

(b) $f(-1) = f(1) \neq 0$

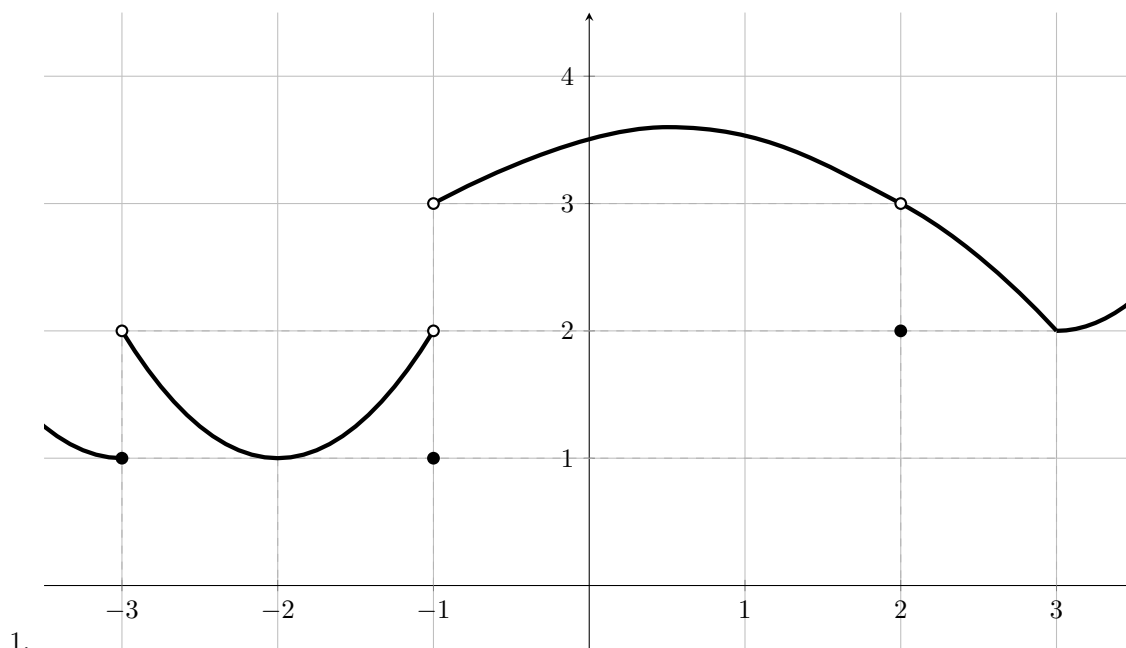
(c) f is not continuous on $[-1, 1]$

(d) f is not differentiable on $[-1, 1]$

Answer: d

Solution: Rolle's Theorem requires the function to be differentiable over $(-1, 1)$. This is not the case here since f has a corner at $x = 0$ and, consequently, $f'(0)$ is not defined.

2.10 Final, 2022–2023



1.

At which value of $x = a$ the function f satisfies:

$f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, $f(a) \neq \lim_{x \rightarrow a} f(x)$

- (a) -3
- (b) -1
- (c) 2
- (d) 3

Answer: c

Solution: In points -1 and -3 , the limit doesn't exist because right-handed and left-handed limits are different. The condition $f(a) \neq \lim_{x \rightarrow a} f(x)$ is satisfied in 3, so the answer is 2.

2. If the function $f(x) = \begin{cases} 2x + a^2, & \text{if } x \geq m \\ x^2 - 3a^2 + 1, & \text{if } x < m \end{cases}$ is continuous at $x = m$ where $a > 0$, then ...

- (a) $m = 2a + 1$
- (b) $m = 2a$
- (c) $m = a^2$
- (d) $m = 2a^2 + 1$

Answer: a

Solution: As the function is continuous at $x = m$, then:

$$2m + a^2 = m^2 - 3a^2 + 1$$

$$4a^2 = m^2 - 2m + 1 = (m - 1)^2$$

$$2a = m - 1 \Rightarrow m = 2a + 1$$

3. If $f(x) = \begin{cases} x^2 + x, & x < 1 \\ 3x + 1, & x \geq 1 \end{cases}$, $g(x) = \begin{cases} 2x - 5, & x < 0 \\ x - 1, & 0 \leq x < 4 \\ x^2 + 1, & x \geq 4 \end{cases}$

Then $\lim_{x \rightarrow 2^-} (f(x - 1) \cdot g(x + 2)) = \dots$

- (a) -6
- (b) 5
- (c) 6
- (d) 14

*Answer: c**Solution:*

$$\begin{aligned}\lim_{x \rightarrow 2^-} (f(x-1) \cdot g(x+2)) &= \lim_{x \rightarrow 1^-} f(x) \times \lim_{x \rightarrow 4^-} g(x) \\ \lim_{x \rightarrow 1^-} x^2 + x \times \lim_{x \rightarrow 4^-} x - 1 &= 2 \times 3 = 6\end{aligned}$$

4. $\lim_{x \rightarrow 2} \frac{k(x-2)}{x^2 - 4x + 4} = 0$, then $k = \dots$

- (a) 4
- (b) 2
- (c) 1
- (d) 0

*Answer: d**Solution:*

$$\lim_{x \rightarrow 2} \frac{k(x-2)}{x^2 - 4x + 4} = \lim_{x \rightarrow 2} \frac{k(x-2)}{(x-2)^2} = \lim_{x \rightarrow 2} \frac{k}{(x-2)} = 0$$

The limit of the function is equal to ∞ or $-\infty$ in every value of k instead of 0, so the answer is 0.

5. $\lim_{x \rightarrow \infty} \frac{(2x-1)(2x-2) \cdots (2x-n)}{(x-1)(x-2) \cdots (x-n)} = 32$, then $n = \dots$

- (a) 5
- (b) 4
- (c) 2
- (d) 32

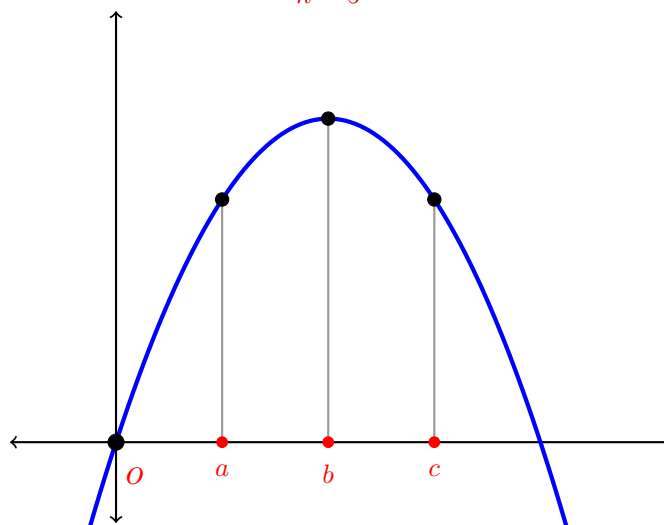
*Answer: a**Solution:*

$$\lim_{x \rightarrow \infty} \frac{(2x-1)(2x-2) \cdots (2x-n)}{(x-1)(x-2) \cdots (x-n)} = \lim_{x \rightarrow \infty} \frac{2x-1}{x-1} \times \lim_{x \rightarrow \infty} \frac{2x-2}{x-2} \cdots \lim_{x \rightarrow \infty} \frac{2x-n}{x-n}$$

We know that we can neglect added constants in limits that approach infinity, so:

$$\lim_{x \rightarrow \infty} \frac{2x-1}{x-1} \times \lim_{x \rightarrow \infty} \frac{2x-2}{x-2} \cdots \lim_{x \rightarrow \infty} \frac{2x-n}{x-n} = \lim_{x \rightarrow \infty} \frac{2x}{x} \times \lim_{x \rightarrow \infty} \frac{2x}{x} \cdots \lim_{x \rightarrow \infty} \frac{2x}{x} = 2^n = 32$$

$n = 5$



6. The figure above shows the function $f(x)$, then $\dots$

- (a) $f'(a) < f'(b) < f'(c)$
- (b) $f'(a) < f'(c) < f'(b)$

(c) $f'(c) < f'(b) < f'(a)$

(d) $f'(b) < f'(a) < f'(c)$

Answer: c**Solution:** From the figure, the tangent slope on a is positive, on b is 0 and negative on c .

7. If $g(x) = \begin{cases} x^2, & x < 1 \\ 2x, & x \geq 1 \end{cases}$, then $f'(1^-) = \dots$

(a) Doesn't exist

(b) 2

(c) 1

(d) 0

Answer: b**Solution:** Taking the left-side derivative, we will work only on the first function x^2 as we want the left derivative which is less than 1, the function will equal $2x$, then subsisting by 1 equals 2.

8. If $f(x) = (x-1)(x-2)\cdots(x-6)$, then $f'(3) = \dots$

(a) 12

(b) -12

(c) -4

(d) 4

Answer: b**Solution:** When applying the multiplication rule for more than 2 functions, we take the derivative of the first function and multiply it by the rest of the functions, then do this for all functions and add the terms.

$$f'(x) = (x-1)'(x-2)\cdots(x-6) + (x-1)(x-2)'\cdots(x-6) + \cdots (x-1)(x-2)\cdots(x-6)'$$

We want to get $f'(3)$. Every term in the last expression is equal to 0 except the third one. So we are left with:

$$(3-1)(3-2)(3-4)(3-5)(3-6) = 2 \times 1 \times -3 = -12$$

9. If f and g are differentiable functions in $]0, 2[$ and $f(2) = g(2) = -1$, $f(0) = 5$, $g(0) = 1$, then for some $c \in]0, 2[$:

(a) $f'(c) = -3g'(c)$

(b) $f'(c) = 3g'(c)$

(c) $2f'(c) = 3g'(c)$

(d) $3f'(c) = g'(c)$

Answer: bAccording to Rolle's theorem, if the function is continuous and differentiable through $[a, b]$, there exists a value c that satisfies $f'(c) = \frac{f(b)-f(a)}{b-a}$. For function f , $f'(c) = \frac{-1-5}{2-0} = -3$. Also, for g , $g'(c) = \frac{-1-1}{2-0} = -1$. Thus, $f'(c) = 3g'(c)$.

10. A weight hanging from a spring stretched 6 units beyond its rest position ($s = 0$) and released at time $t = 0$ to bob up and down. Its position at any time t is $s = 6 \cos t$, then its velocity and acceleration at $t = \frac{\pi}{4}$ are $\dots, \dots$ respectively

(a) $3\sqrt{2}, 3\sqrt{2}$

(b) $3\sqrt{2}, -3\sqrt{2}$

(c) $-3\sqrt{2}, 3\sqrt{2}$

(d) $-3\sqrt{2}, -3\sqrt{2}$

Answer: d**Solution:** Velocity is the first derivative of time and acceleration is the second derivative. Thus, $v = -6 \sin t$ and $a = -6 \cos t$. The velocity is $-3\sqrt{2}$ and the acceleration is $-3\sqrt{2}$.

11. If $f(x) = e^{\frac{1}{x}}$, $f'(x) = \dots$

- (a) $e^{\frac{1}{x}}$
- (b) $-\frac{1}{x}e^{\frac{1}{x}}$
- (c) $-\frac{1}{x^2}e^{\frac{1}{x}}$
- (d) $\frac{1}{x^2}e^{\frac{1}{x}}$

Answer: c

Solution: Derivative of $a^{f(x)}$ is $a^{f(x)} \cdot f'(x) \cdot \ln a$. Thus, $f'(x) = -\frac{1}{x^2}e^{\frac{1}{x}}$.

12. Apply l'Hopital's rule to find the limit $\lim_{x \rightarrow 0} \frac{x - \sin x}{1 - \cos x}$

- (a) 0
- (b) 1
- (c) -1
- (d) Doesn't exist

Answer: a

Solution:

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(x - \sin x)}{\frac{d}{dx}(1 - \cos x)} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{\sin x}{\cos x} = 0$$

13. If $e^{f(x)} = 1 + x^2$, then $f'(x) = \dots$

- (a) $\frac{1}{1+x^2}$
- (b) $\frac{2x}{1+x^2}$
- (c) $2x(1+x^2)$
- (d) $2x \ln 1 + x^2$

Answer: b

Solution:

$$e^{f(x)} = 1 + x^2 \Rightarrow f(x) \ln e = \ln(1 + x^2) \Rightarrow f(x) = \ln(1 + x^2)$$

$$f'(x) = \frac{1}{1+x^2} \times 2x = \frac{2x}{1+x^2}$$

14. If $y = 2^{\sin x}$, then $\frac{dy}{dx} = \dots$

- (a) $2^{\sin x} \cos x$
- (b) $-2^{\sin x} \cos x$
- (c) $2^{\sin x} \ln(\cos x)$
- (d) $2^{\sin x} \cos x \ln 2$

Answer: d

Solution: We know that:

$$\frac{d}{dx} a^{f(x)} = a^{f(x)} f'(x) \ln a$$

Thus:

$$\frac{dy}{dx} = 2^{\sin x} \cos x \ln 2$$

15. If $f(x_0) = 0$ and x_0 is a corner point of $f(x)$. Then the function $g(x) = (x - x_0)f(x)$

- (a) Is differentiable at x_0
- (b) Is not differentiable at x_0
- (c) Is discontinuous at x_0
- (d) Has a corner point at x_0

*Answer: a**Solution:*

$$\begin{aligned}
 g'(x_0) &= \lim_{h \rightarrow 0} \frac{(x_0 - x_0 + h) f(x_0 + h) - (x_0 - x_0) f(x_0)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{h} f(x_0 + h)}{\cancel{h}} \\
 &= f(x_0)
 \end{aligned}$$

Since f is continuous at $x = x_0$, $g'(x_0) = f(x_0)$ is defined and g is differentiable at $x = x_0$.

16. Let $f(x) = x^3$. Find the value c that satisfies the mean value theorem on the closed interval $[1, 3]$.

- (a) 1.414
- (b) 1.732
- (c) 2
- (d) 2.082

*Answer: d**Solution:*

$$\begin{aligned}
 f'(c) &= \frac{f(3) - f(1)}{3 - 1} \\
 3c^2 &= 13 \\
 c &= 2.082
 \end{aligned}$$

17. $\int \frac{e^{3x} + 1}{e^x + 1} dx = f(x) + c$ where $f(x)$ has no term free of x , then $f(0) = \dots$

- (a) $-\frac{1}{2}$
- (b) $\frac{1}{2}$
- (c) 1
- (d) 0

*Answer: a**Solution:*

$$\begin{aligned}
 \int \frac{e^{3x} + 1}{e^x + 1} dx &= \int \frac{(e^x + 1)(e^{2x} - e^x + 1)}{e^x + 1} dx = \int e^{2x} - e^x + 1 dx = \frac{1}{2}e^{2x} - e^x + x + C \\
 f(0) &= \frac{1}{2} - 1 + 0 = -\frac{1}{2}
 \end{aligned}$$

18. If $-\frac{3}{4}\pi < x < \frac{\pi}{4}$, then $\int \sqrt{1 - \sin 2x} dx = a \sin x + b \cos x + c$, then $a + b = \dots$

- (a) 0
- (b) 1
- (c) 2
- (d) $1 + \sqrt{3}$

*Answer: c**Solution:*

$$\begin{aligned}
 \int \sqrt{1 - \sin 2x} dx &= \int \sqrt{\sin^2 x + \cos^2 x - \sin 2x} dx = \int \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cos x} dx \\
 &= \int \sqrt{(\cos x - \sin x)^2} dx = \int \cos x - \sin x dx = \sin x + \cos x + C
 \end{aligned}$$

Then $a = b = 1$, $a + b = 2$

19. If $\int \frac{1}{x \ln x} dx = f(x) + C$ where $f(x)$ has no term free of x , then $f(e) = \dots$

- (a) 1
- (b) e
- (c) $1 + e$
- (d) 0

Answer: d

Solution:

$$\int \frac{1}{x \ln x} dx = \int \frac{1}{x} (\ln x)^{-1} dx$$

We know that $\frac{d}{dx} \ln x = \frac{1}{x}$. Thus, assuming $\ln x = u$, we can rewrite the problem:

$$\int \frac{1}{x} (\ln x)^{-1} dx = \int (u)^{-1} du = \ln u + C = \ln (\ln x) + C$$

$$f(e) = \ln (\ln e) = \ln 1 = 0$$

20. $\int e^{ax} [af(x) + f'(x)] dx = \dots + C$

- (a) ae^{ax}
- (b) $e^{ax} f'(x)$
- (c) $e^{ax} f(x)$
- (d) $af(x)$

Answer: c

Solution: We notice that:

$$\frac{d}{dx} e^{ax} f(x) = ae^{ax} f(x) + e^{ax} f'(x) = e^{ax} [af(x) + f'(x)]$$



2.11 Midterm, 2023–2024

1. Suppose f , g and h are functions with limit $\lim_{x \rightarrow 2} f(x) = 7$, $\lim_{x \rightarrow 2} g(x) = -4$, $\lim_{x \rightarrow 2} h(x) = \frac{3}{5}$ find the limit:

$$\lim_{x \rightarrow 2} \frac{f(x)g(x)h(x)}{f(x) + g(x) - 5h(x)}$$

- (a) $-\frac{84}{5}$
 (b) zero
 (c) $\frac{84}{5}$
 (d) Does not exist.

Answer: d

Solution substituting the values in the limit:

$$\lim_{x \rightarrow 2} \frac{f(x)g(x)h(x)}{f(x) + g(x) - 5h(x)} = \frac{\lim_{x \rightarrow 2} f(x) \times \lim_{x \rightarrow 2} g(x) \times \lim_{x \rightarrow 2} h(x)}{\lim_{x \rightarrow 2} f(x) + \lim_{x \rightarrow 2} g(x) - 5 \lim_{x \rightarrow 2} h(x)} = \frac{7 \times -4 \times \frac{3}{5}}{7 - 4 - 3}$$

As the expression involves a 0 in the denominator, the limit does not exist.

2.

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)}{1-x+x^2-x^3} = \dots$$

- (a) $\frac{1}{4}$
 (b) $-\frac{1}{2}$
 (c) $\frac{1}{2}$
 (d) Doesn't exist

Answer: b

Solution Using L'Hopital's rule:

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)}{1-x+x^2-x^3} = \lim_{x \rightarrow 1} \frac{\frac{d}{dx} \sin(x-1)}{\frac{d}{dx} 1-x+x^2-x^3} = \lim_{x \rightarrow 1} \frac{\cos(x-1)}{-1+2x-3x^2} = \frac{1}{-2}$$

3. If $2^{k^2-k-2} = 1$ where $k > 0$, then:

$$\lim_{x \rightarrow k} (x^2 + x - 2) = \dots$$

- (a) 4
 (b) 3
 (c) 2
 (d) 1

Answer: a

Solution

$$2^{k^2-k-2} = 1 \Rightarrow (k^2 - k - 2) \ln 2 = \ln 1 = 0$$

$$k^2 - k - 2 = 0 \Rightarrow (k-2)(k+1) = 0$$

k is equal to 2 or -1 . As $k > 0$, $k = 2$.

$$\lim_{x \rightarrow k} (x^2 + x - 2) = \lim_{x \rightarrow 2} (x^2 + x - 2) = 4 + 2 - 2 = 4$$

4.

$$\lim_{x \rightarrow 2^+} \frac{4-x^2}{x|2-x|}$$

- (a) 2
 (b) -2
 (c) 0

(d) Doesn't exist

Answer: b

Solution

$$\lim_{x \rightarrow 2^+} \frac{4 - x^2}{x|2 - x|} = \lim_{x \rightarrow 2^+} \frac{(2 - x)(2 + x)}{x|2 - x|}$$

From the right, $|2 - x|$ will reach some small positive value while $2 - x$ will approach the same value with a negative sign, so we can cancel them out and put -1 :

$$\lim_{x \rightarrow 2^+} \frac{(2 - x)(2 + x)}{x|2 - x|} = \lim_{x \rightarrow 2^+} \frac{-(2 + x)}{x} = -2$$

5. Find the value of

$$\lim_{x \rightarrow \infty} \frac{ax^2 + bx + c}{nx^3 + mx^2 + lx + d}$$

where a, b, c, m, n, l and d are non-zero real constants

(a) $\frac{a}{n}$

(b) $\frac{a}{m}$

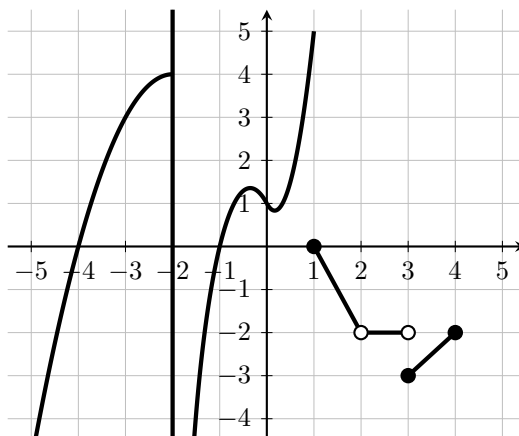
(c) 0

(d) ∞

Answer: c

Solution The power of the denominator is larger than the power of the nominator. As the limit approaches ∞ , the limit approaches 0 regardless of the value of the constants.

6. The opposite figure shows the curve of the function f , then $\lim_{x \rightarrow 2} ((2 - 3x) \times f(x))$



(a) -3

(b) -1

(c) 8

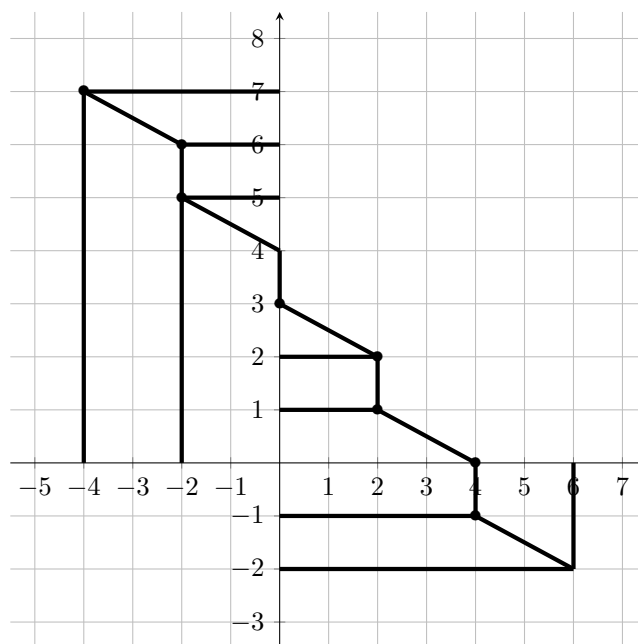
(d) Doesn't exist

Answer: c

Solution:

$$\lim_{x \rightarrow 2} ((2 - 3x) \times f(x)) = \lim_{x \rightarrow 2} 2 - 3x \times \lim_{x \rightarrow 2} f(x) = (2 - 3 \times 2) \times -2 = 8$$

7. Use the opposite graph of the function f to find $\sum_{r=-2}^2 (\lim_{x \rightarrow r^+} f(x))$



(a) 16

(b) 12

(c) 10

(d) 8

Answer: a*Solution:*

$$\sum_{r=-2}^2 (\lim_{x \rightarrow r^+} f(x)) = \lim_{x \rightarrow -2^+} f(x) + \lim_{x \rightarrow -1^+} f(x) \cdots \lim_{x \rightarrow 2^+} f(x) = 5 + 4.5 + 3 + 2.5 + 1 = 16$$

8. If $f(x) = \sqrt{4 - x^2}$ which one of the following statements is correct

(a) $\lim_{x \rightarrow 2^+} \sqrt{4 - x^2} = 0$ (b) $\lim_{x \rightarrow 2^-} \sqrt{4 - x^2} = 0$ (c) $\lim_{x \rightarrow 2} \sqrt{4 - x^2} = 0$ (d) $\lim_{x \rightarrow -2^+} \sqrt{4 - x^2}$ Doesn't exist -**Answer: b**

Solution The value under the root cannot be negative. Thus, $\lim_{x \rightarrow 2^+} \sqrt{4 - x^2}$ does not exist, as any value above 2 is not in the domain of the function. As $\lim_{x \rightarrow 2^+} \sqrt{4 - x^2}$ D.N.E, $\lim_{x \rightarrow 2} \sqrt{4 - x^2}$ D.N.E also. $\lim_{x \rightarrow 2^-} \sqrt{4 - x^2}$ exists as it gives a positive value under the root that approaches 0, so the only correct answer is b.

9. If the function $f(x) = \begin{cases} ax - 3, & x > 2 \\ 2x + 5, & x \leq 2 \end{cases}$ is continuous at $x = 2$, then $a = \dots$

(a) 3

(b) 4

(c) 6

(d) 8

Answer: c*Solution* As the function is continuous:

$$2a - 3 = 2 \times 2 + 5$$

$$2a = 12 \Rightarrow a = 6$$

10. If $f(x) = x^2 - 3x + 2, g(x) = \sqrt{x}$ which of the following is differentiable at $x = -1$

- (a) $f(x) + g(x)$
- (b) $f(x) \times g(x)$
- (c) $(f \circ g)(x)$
- (d) $(g \circ f)(x)$

Answer: d*Solution:*

$$f(x) + g(x) = x^2 - 3x + 2 + \sqrt{x}$$

We know that $\sqrt{-1}$ is not defined, the function is undefined at -1 . Thus, the function $f(x) + g(x)$ is not differentiable at -1 . The same thing is true for the function $f(x) \times g(x)$.

$$(f \circ g)(x) = x - 3\sqrt{x} + 2$$

We have $\sqrt{-1}$ in the function, it's neither defined nor differentiable.

$$(g \circ f) = \sqrt{x^2 - 3x + 2}$$

Blogging -1 in the function, it will give us 6, so it's defined and differentiable (I'm too lazy to take the derivative but try to take it, it works).

11. If $f(x) = \sqrt[3]{g(x)}$, then $\frac{g'(x)}{f'(x)} = \dots$

- (a) $\sqrt[3]{g(x)}$
- (b) $\sqrt[3]{f(x)}$
- (c) $\frac{1}{3\sqrt[3]{(g(x))^2}}$
- (d) $3(f(x))^2$

Answer: d*Solution*

$$f'(x) = \frac{d}{dx} \sqrt[3]{g(x)} = \frac{1}{3}(g(x))^{-\frac{2}{3}} \times g'(x) = \frac{g'(x)}{3(\sqrt[3]{g(x)})^2}$$

$$\frac{g'(x)}{f'(x)} = g'(x) \div \frac{g'(x)}{3(\sqrt[3]{g(x)})^2} = 3(\sqrt[3]{g(x)})^2 = 3(f(x))^2$$

12. If the function $f(x) = \begin{cases} 4 - ax^2, & x > 2 \\ x + b, & x \leq 2 \end{cases}$ is differentiable at $x = 2$, then $b - 4a = \dots$

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Answer: b

Solution: As the function is differentiable at $x = 2$, we can take the derivative of either function and they should be equal for 2.

$$\frac{d}{dx}(4 - ax^2) = -2ax, \quad \frac{d}{dx}(x + b) = 1$$

$$-4a = 1 \Rightarrow a = -\frac{1}{4}$$

As the function is differentiable at 2, it's continuous at 2, so:

$$4 - 4a = 2 + b \Rightarrow 2 + b = 5 \Rightarrow b = 3$$

$$b - 4a = 3 + 1 = 4$$

13. If $f(3x - 5) = 2x^2 + x - 1$, then $f'(1) + f(1) = \dots$

- (a) 10
(b) 12
 (c) 14
 (d) 16

Answer: b

Solution: Let $3x - 5 = u$. $\frac{1}{3}(u + 5)$. Thus:

$$f(u) = \frac{2}{9}(u + 5)^2 + \frac{1}{3}(u + 5) - 1$$

$$f'(u) = \frac{4}{9}(u + 5) + \frac{1}{3}$$

Hence:

$$f'(1) + f(1) = \frac{2}{9}(1 + 5)^2 + \frac{1}{3}(1 + 5) - 1 + \frac{4}{9}(1 + 5) + \frac{1}{3} = 12$$

14. If $f(x) = 2x^2 + 3$, then $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \dots$

- (a) 0
 (b) 3
(c) 4
 (d) undefined

Answer: c

Solution $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ is the definition of the derivative of $f(x)$. Thus, $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$ is $f'(1)$.

$$f'(x) = 4x$$

$$f'(1) = 4 \times 1 = 4$$

15. The value of C in the lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$ where $x \in [0, 1]$ is:

- (a) $\frac{1}{3}(4 - \sqrt{7})$
 (b) $\frac{1}{3}(4 - \sqrt{5})$
 (c) $\frac{1}{3}$
 (d) $\frac{1}{4}$

Answer: a

Solution: Mean value theorem tells us that if $f(x)$ is continuous at $[a, b]$ and differentiable at (a, b) , there is a value c that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$f(1) = 1 - 4 + 8 + 11 = 16$, $f(0) = 0 - 0 + 0 + 11 = 11$. Hence:

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = 5$$

$$f'(x) = 3x^2 - 4x + 8 \Rightarrow f'(c) = 3c^2 - 4c + 8 = 5$$

$$3c^2 - 4c + 3 = 0 \Rightarrow c = \frac{1}{3}(4 \pm \sqrt{7})$$

2.12 Final term, 2023–2024

1. Compute the limit

$$\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2}$$

- a. $\frac{1}{4}$
- b. $\frac{1}{2}$
- c. 1
- d. 2

Answer: b

Solution:

$$\lim_{x \rightarrow 0} \frac{(1 - \cos(x))(1 + \cos(x))}{x^2(1 + \cos(x))} = \lim_{x \rightarrow 0} \frac{1 - \cos^2(x)}{(1 + \cos^2(x)) \times x^2} = \frac{\sin^2(x)}{x^2} \times \frac{1}{1 + \cos^2(x)} = 1 \times \frac{1}{2} = \frac{1}{2}$$

2. Compute the limit

$$\lim_{x \rightarrow \infty} \sqrt{x^2 + x - 1} - x$$

- a. 0
- b. ∞
- c. $\frac{1}{2}$
- d. $-\infty$

Answer: c

Solution:

$$\begin{aligned} \lim_{x \rightarrow \infty} \sqrt{x^2 + x - 1} - x &= \lim_{x \rightarrow \infty} (\sqrt{x^2 + x - 1} - x) \times \frac{\sqrt{x^2 + x - 1} + x}{\sqrt{x^2 + x - 1} + x} = \lim_{x \rightarrow \infty} \frac{x^2 + x - 1 - x^2}{\sqrt{x^2 + x - 1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{(x - 1) \div x}{(\sqrt{x^2 + x - 1} + x) \div x} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{x}}{\sqrt{1 + \frac{1}{x} - \frac{1}{x^2}} + 1} = \frac{1}{2} \end{aligned}$$

3. Consider the function $f(x) = |x|$ in the interval $-1 \leq x \leq 1$. At the point $x = 0$, then $f(x)$ is $\dots$

- a. Continuous and Differentiable
- b. Discontinuous and Differentiable
- c. Continuous and Non-Differentiable
- d. Neither Continuous nor Differentiable

Answer: c

solution: by the definition of continuous and differentiable functions, we get that $f(x) = |x|$ is continuous but not differentiable at the point $x = 0$

4. If $f(x)$ is continuous at $x = 0$, then the value of K is:

$$f(x) = \begin{cases} \frac{\sin(5x)}{x^2 + 2x}, & x \neq 0 \\ K + \frac{1}{2}, & x = 0 \end{cases}$$

- a. 2
- b. 1
- c. -2
- d. $\frac{1}{2}$

answer:a

solutions:

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} \times \frac{1}{2 + x} = \lim_{x \rightarrow 0} \frac{\sin(5x)}{x} \times \lim_{x \rightarrow 0} \frac{1}{2 + x} = \frac{5}{2}$$

to get the value of K , we know that $\frac{5}{2} = K + \frac{1}{2} \rightarrow K = 2$

5. At how many points is the function $f(x) = [x]$ discontinuous:

- a. 1
- b. 2
- c. 3
- d. ∞

Answer: d

Solution: since the function is floor function it's not continuous for every integer.

6. If $x^m + y^n = 1$, such that $\frac{dy}{dx} = -\frac{x}{y}$, then what should be the value of m :

- a. 0
- b. 1
- c. 2
- d. 3

answer:c

solution: by differentiating with respect to x we get that

$$mx^{m-1} + ny^{n-1} \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{m x^{m-1}}{n y^{n-1}} \implies m = n$$

$$m - 1 = 1 \rightarrow m = 2$$

7. If $f(x) = e^{|x|}$. Then which one of the following is correct?

- a. $f'(0) = 1$
- b. $f'(0) = -1$
- c. $f'(0) = 0$
- d. $f'(0)$ doesn't exist

answer:d

solution: by using the definition of derivative we get that

$$f'(0) = \lim_{h \rightarrow 0} \frac{e^{|1+h|} - 1}{h}$$

we know that the value of $e^{|1+h|}$ is positive for all real values of h , so $\lim_{x \rightarrow 0^+} \neq \lim_{x \rightarrow 0^-}$ so, the function is not differentiable at $x = 0$

8. Find the first derivative of $f(x) = \sin^2(2x + 5)$ with respect to x :

- a. $4 \sin(2x + 5)$
- b. $4 \sin(4x + 10)$
- c. $2 \sin(2x + 5)$
- d. $2 \sin(4x + 10)$

Answer:d

Solution: using the chain rule

$$y' = 2 \times 2 \sin(2x + 5) \cos(2x + 5) = 2 \sin(4x + 10) \text{ (double angle formula)}$$

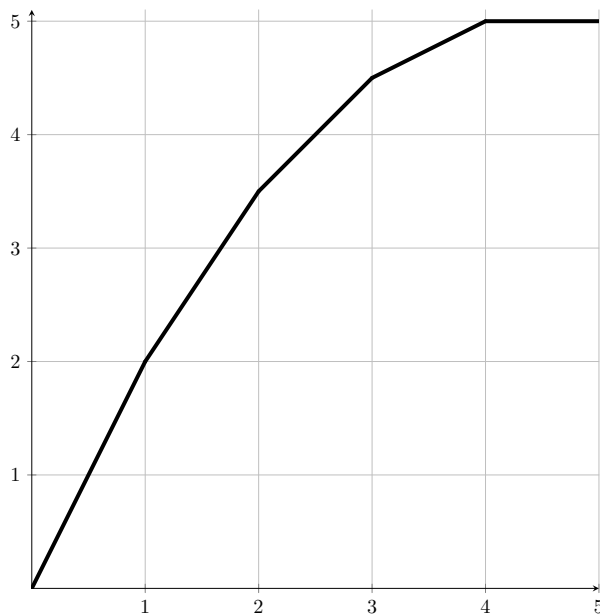
9. If $f(0) = 4$ and $f'(x) = \frac{3}{x^2+2}$, then the lower bound of $f(2)$ estimated by mean value theorem is $\dots$

- a. 0
- b. 5
- c. 7
- d. 12

Answer:b*Solution:* We know that from the mean value theorem

$$\frac{f(2) - 4}{2} = \frac{3}{x^2 + 2} \implies f(2) = \frac{6}{x^2 + 2} + 4 \rightarrow f(2) = \frac{6}{2^2 + 2} + 4 = 1 + 4 = 5$$

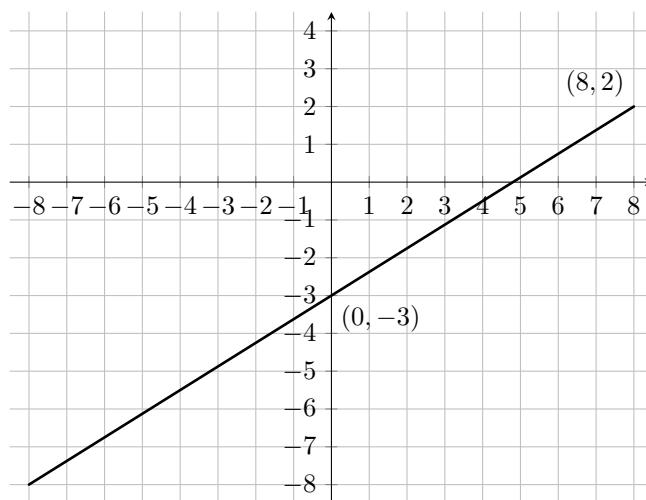
10. The graph below shows the distance in (m) miles and time in (h) hours, which interval has the greatest rate of change:



- a. hour 0 to hour 1
- b. hour 1 to hour 2
- c. hour 2 to hour 3
- d. hour 3 to hour 4

Answer: a*Solution:* from the graph we can see that the higher slope is in the interval [0,1]

11. Which is the average rate of change for the line graphed in the opposite figure?



- a. $\frac{3}{5}$
- b. $\frac{5}{8}$
- c. $\frac{5}{12}$
- d. $\frac{3}{15}$

Answer:b*Solution:* the rate of change = sloop = $\frac{2-(-8)}{8-(-8)} = \frac{10}{16} = \frac{5}{8}$

12. If the rate of change of the radius of a circle is 6cm/s, then find the rate of change of area of the circle when $r = 2$ cm. ($\pi = 3.14$):

- a. 74.36 cm^2/s
- b. 75.36 cm^2/s
- c. 15.36 cm^2/s
- d. 65.36 cm^2/s

Answer : b*Solution:* The area of the circle $A = 2\pi r^2$. Thus, the rate of change of the area of the circle with radius is

$$\frac{dA}{dt} = \frac{dr}{dt} 2\pi r$$

so the change of area = $2 \times 6 \times 2 \times \pi \approx 75.36$

13. If $y = x^{\sqrt{x}}$ then $\frac{dy}{dx} = \dots$

- a. $\frac{\ln(x)}{2\sqrt{2}}$
- b. $\frac{x^{\sqrt{x}}}{\sqrt{x}}$
- c. $\frac{y \ln(x)}{2\sqrt{x}}$
- d. $\frac{y(\ln(x)+2)}{2\sqrt{x}}$

Answer:d*Solution:* Take the natural logarithm of both sides:

$$\ln y = \sqrt{x} \cdot \ln x$$

Differentiate both sides implicitly with respect to x :

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{d}{dx} (\sqrt{x} \cdot \ln x) = \frac{\ln x}{2\sqrt{x}} + \frac{1}{\sqrt{x}}$$

Multiply both sides by $y = x^{\sqrt{x}}$:

$$\frac{dy}{dx} = x^{\sqrt{x}} \left(\frac{\ln x + 2}{2\sqrt{x}} \right)$$

Final answer:

$$\frac{dy}{dx} = \frac{y(\ln x + 2)}{2\sqrt{x}}$$

14. A-25-year-old student is offered an opportunity to invest some money in a retirement account that pays 5% annual interest compound continuously. How much the student need to invest today to have 1\$ million when she retires at age 65
- a. 135 335.28
 - b. 125 335.28
 - c. 115 335.28
 - d. 105 335.28

Answer:a*Solution:* we know that

$$1 \times 10^6 = P e^{\frac{5}{100} \times 40} \rightarrow P = 135\,335.28$$

15. If $y = \log_y x$, then $\frac{dy}{dx} = \dots$

- a. $\frac{1}{x+\log(y)}$
- b. $\frac{1}{\log(x)(1+y)}$
- c. $\frac{1}{x(1+\ln(y))}$
- d. $\frac{1}{y+\log(x)}$

Answer:c

Solution:

$$y = \frac{\ln x}{\ln y}$$

$$y \ln y = \ln x$$

$$\frac{dy}{dx} \cdot \ln y + y \cdot \frac{1}{y} \frac{dy}{dx} = \frac{1}{x} = 2 \ln y \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{x(1 + \ln(y))}$$

16.

$$\int \sin^3(x) \cos^2(x) dx = \dots$$

- a. $\frac{-1}{3} \cos^3(x) + \frac{1}{5} \sin^5(x) + C$
- b. $\frac{1}{3} \sin^3(x) + \frac{1}{5} \cos^5(x) + C$
- c. $\frac{-1}{3} \cos^3(x) + \frac{1}{5} \cos^5(x) + C$
- d. none of the above

Answer:c

Solution:

$$\int \sin(x)(\sin^2(x))(\cos^2(x))dx = \int \sin(x)(1 - \cos^2(x))(\cos^2(x))dx =$$

$$\int \sin(x) \cos^2(x) dx - \int \sin(x) \cos^4(x) dx$$

$$\text{set } \cos(x) = u$$

$$-\sin(x) dx = du$$

$$-\int u^2 du + \int u^4 du = \frac{-1}{3} u^3 + \frac{1}{5} u^5$$

so

$$\int \sin^3(x) \cos^2(x) dx = \frac{-1}{3} \cos^3(x) + \frac{1}{5} \cos^5(x) + c$$

17. Evaluate:

$$\int 6x(x^2 + x) dx$$

- a. $\frac{3x^4}{2} + 18x^2 + C$
- b. $\frac{3x^4}{2} + 18x + C$
- c. $\frac{3x^4}{2} + 18x^2 + C$
- d. $\frac{3x^4}{2} + 2x^3 + C$

Answer:d

Solution:

$$\int 6x^3 + 6x^2 dx = \frac{3x^4}{2} + 2x^3 + C$$

18. Evaluate:

$$\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$$

- a. $\cos(\sqrt{x}) + c$
- b. $-2 \csc(\sqrt{x}) + c$

c. $-2\cos(\sqrt{x}) + c$

d. $\sec(\sqrt{x}) + c$

Answer:c*Solution:*

$$\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx = 2 \int \sin(\sqrt{x}) \times \frac{1}{2\sqrt{x}} dx = 2 \int \sin(u) du = -2\cos(\sqrt{x}) + c$$

19. Evaluate:

$$\int e^{5\ln(x)} dx$$

a. $6\ln(x) + c$

b. $e^6 + c$

c. $\frac{x^6}{6} + c$

d. $\frac{x^5}{5} + e^x + c$

Answer:*Solutions:* it's known that $e^{x\ln(y)} = y^x$, so

$$\int e^{5\ln(x)} dx = \int x^5 dx = \frac{x^6}{6} + c$$

20. Evaluate:

$$\int \frac{\cos(\ln(x))}{x} dx$$

a. $\sin(\ln(x)) + c$

b. $\cos(\ln(x)) + c$

c. $\frac{-\cos(x)}{x^2} + c$

d. $\frac{-\sin(x)}{x^2} + c$

answer:a*solution:*

$$\int \cos(\ln(x)) \times \frac{1}{x} dx$$

set $\ln(x) = u$

$$\int \cos(\ln(x)) \times \frac{1}{x} dx = \int \cos(u) du = \sin(u) = \sin(\ln(x)) + C$$

21. Find the first derivative of $f(x) = 4x^{e^x}$ with respect to x :

a. $x^{e^x} e^{-x} (x \log(x) + 1)$

b. $-4x^{e^x-1} e^x (x \log(x) + 1)$

c. $4x^{e^x} e^x (x \log(x) + 1)$

d. $4x^{e^x-1} e^x (x \ln(x) + 1)$

Answer:d*Solution:*

$$\frac{d4x^{e^x}}{dt} = 4 \frac{d}{dt} x^{e^x}$$

we know that $x^{e^x} = e^{e^x \ln(x)}$

$$4 \frac{d}{dt} x^{e^x} = 4 \frac{d}{dt} e^{e^x \ln(x)} = 4 \left(\frac{e^x}{x} + e^x \ln(x) \right) e^{e^x \ln(x)} = 4e^{e^x \ln(x)+x} \left(\frac{1}{x} + \ln(x) \right) = 4x^{e^x-1} e^x (x \ln(x) + 1)$$

22. Which of the following is true about Percentile, Quartile, and Median?

a. $P50 > Q2 > Median$

b. $P50 = Q2 > Median$

- c. $P50 < Q2 < Median$
- d. $P50 = Q2 = Median$

Answer:d

Solution: it's known that $P50 = Q2 = Median$

23. Which of the following is true about standardization?

- a. It changes the units of measurement and the shape of the distribution
- b. The resulting distribution has a mean of 0 and a standard deviation of 1
- c. It transforms raw distributions of scores to distribution with any desired mean and standard deviation
- d. In a normal distribution, the standard deviation varies depending on the raw scores

Answer:b

Solution: it's known that when standardizing a normal distribution the mean becomes 0 and the standard deviation becomes 1

24. Normal distribution is symmetric about:

- a. Standard deviation
- b. Mean
- c. Covariance
- d. Variance

Answer: b

Solution: The normal distribution is typically symmetric about its mean which equals 0 in the case of normal distributions.

25. Which of the following is false about normal distribution?

- (a) Normal distribution is applied for discrete random variable
- (b) The shape of the normal curve is bell shaped
- (c) The area under a standard normal curve is 1
- (d) The standard normal curve is symmetric about the value 0

Answer: a

Solution: Distribution is applied for discrete Random Distribution. This statement is false because the Normal distribution is typically applied to continuous random variables, not discrete ones. For discrete random variables, other probability distributions like the Poisson distribution or the Binomial distribution are more appropriate.

2.13 Mid term, 2024–2025

1. If

$$f(x) = \begin{cases} \frac{x^2 - m^2}{x - m}, & x \neq m \\ 0, & x = m \end{cases}$$

then $\lim_{x \rightarrow m} f(x) =$ a. $2m$ b. m c. 0 d. $2m - 1$ *Answer: a**Solution:* Since the limit is not affected by the actual value of the function, hence

$$\lim_{x \rightarrow m} \frac{x^2 - m^2}{x - m} = \lim_{x \rightarrow m} \frac{(x - m)(x + m)}{x - m} = \lim_{x \rightarrow m} (x + m) = m + m = 2m$$

2. If

$$f(x) = \begin{cases} \frac{x^7 - 64x}{3x - 6}, & x \neq 2 \\ k, & x = 2 \end{cases}$$

is continuous at $x = 2$ then $k = ?$

a. 64

b. 128

c. 32

d. 256

*Answer: b**Solution:*

Since the function is continuous, then

$$\lim_{x \rightarrow 2} f(x) = f(2)$$

Which implies

$$\lim_{x \rightarrow 2} \frac{x^7 - 64x}{3x - 6} = k$$

We have

$$\lim_{x \rightarrow 2} \frac{x(x^6 - 64)}{3(x - 2)} = \lim_{x \rightarrow 2} \left[\left(\frac{x}{3} \right) \left(\frac{x^6 - 64}{x - 2} \right) \right]$$

Using product property limits :

$$\lim_{x \rightarrow 2} \left[\left(\frac{x}{3} \right) \left(\frac{x^6 - 64}{x - 2} \right) \right] = \lim_{x \rightarrow 2} \left(\frac{x}{3} \right) \times \lim_{x \rightarrow 2} \left(\frac{x^6 - 64}{x - 2} \right)$$

$$\lim_{x \rightarrow 2} \left(\frac{x}{3} \right) = \frac{2}{3}$$

Using the limit identity

$$\lim_{x \rightarrow a} \frac{x^b - a^b}{x^c - a^c} = \frac{b}{c} \times a^{b-c}$$

We have

$$\lim_{x \rightarrow 2} \left(\frac{x^6 - 64}{x - 2} \right) = 6 \times 2^5$$

So

$$\lim_{x \rightarrow 2} \left[\left(\frac{x}{3} \right) \left(\frac{x^6 - 64}{x - 2} \right) \right] = \left(\frac{2}{3} \right) (6 \times 2^5) = 128 = k$$

3. 3. If

$$f(x) = x(\sqrt{x} + \sqrt{x+1})$$

then $f(x) =$

- a. Continuous on $\mathbb{R}$, but not differentiable at $x = 0$
- b. Is differentiable at $x = 0$
- c. **Is not differentiable at $x = 0$**
- d. None of the above

Answer (According to the original source): b

The correct answer: c

The question seems to be written incorrectly on the original website, but that doesn't really affect the final answer, because the function is not differentiable at $x = 0$ either way.

Doubtnut marked the answer as (b), but in their video solution they actually solved it as (c), which is simply a piece of stupidity. Testbook also chose (b), but both are wrong. The correct answer is (c), since the function isn't differentiable at $x = 0$.

Solution:

The function $f(x) = x(\sqrt{x} + \sqrt{x+1})$ is only defined for $x \geq 0$ since $\sqrt{x}$ is undefined for negative x in the real numbers. At $x = 0$, the function is continuous, but the left-hand derivative does not exist because the function is not defined for $x < 0$. Therefore, $f(x)$ is **not differentiable at $x = 0$** .

4. If

$$\lim_{x \rightarrow 1} \frac{x^2 - mx + n}{x - 1} = 5$$

then $m =$

- a. 2
- b. -2
- c. 3
- d. **-3**

Answer: d

Solution:

We are given the limit:

$$\lim_{x \rightarrow 1} \frac{x^2 - mx + n}{x - 1} = 5$$

As $x \rightarrow 1$, the denominator becomes zero. For the limit to exist, the numerator must also approach zero at $x = 1$. So we require:

$$x^2 - mx + n = 0 \quad \text{when } x = 1$$

This suggests that $(x - 1)$ is a factor of the numerator. So, we try factoring the numerator as:

$$x^2 - mx + n = (x - 1)(x - a)$$

$$\lim_{x \rightarrow 1} \frac{(x - 1)(x - a)}{x - 1} = \lim_{x \rightarrow 1} (x - a) = 1 - a = 5 \implies a = -4$$

So we have

$$x^2 - mx + n = (x - 1)(x + 4) = x^2 + 3x - 4$$

So

$$-m = 3 \implies m = -3$$

5. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + x^2}} - \sqrt{2}}{x^2}$$

- a. 0
- b. $\frac{1}{4\sqrt{2}}$
- c. $\frac{1}{2\sqrt{2}}$
- d. Do not exist

Answer: **b**

Solution:

We can solve this using binomial approximation

$$\sqrt{1+u} \approx 1 + \frac{1}{2}u \quad |u| < 1$$

$$\text{Let } f(x) = \frac{\sqrt{1 + \sqrt{1 + x^2}} - \sqrt{2}}{x^2}$$

$$\sqrt{1+x^2} \approx 1 + \frac{1}{2}x^2$$

$$\sqrt{1 + \sqrt{1 + x^2}} \approx \sqrt{2 + \frac{1}{2}x^2} = \sqrt{2} \sqrt{1 + \frac{1}{4}x^2} \approx \sqrt{2} \left(1 + \frac{1}{8}x^2\right)$$

$$f(x) \approx \frac{\sqrt{2} + \frac{\sqrt{2}}{8}x^2 - \sqrt{2}}{x^2} = \frac{\frac{\sqrt{2}}{8}x^2}{x^2} = \frac{\sqrt{2}}{8} = \frac{1}{4\sqrt{2}}$$

or, use the conjugate method:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + x^2}} - \sqrt{2}}{x^2} &= \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + x^2}} - \sqrt{2}}{x^2} \cdot \frac{\sqrt{1 + \sqrt{1 + x^2}} + \sqrt{2}}{\sqrt{1 + \sqrt{1 + x^2}} + \sqrt{2}} \\ &= \lim_{x \rightarrow 0} \frac{1 + \sqrt{1 + x^2} - 2}{x^2 (\sqrt{1 + \sqrt{1 + x^2}} + \sqrt{2})} \\ &= \lim_{x \rightarrow 0} \frac{\sqrt{1 + x^2} - 1}{x^2 (\sqrt{1 + \sqrt{1 + x^2}} + \sqrt{2})} \\ &= \lim_{x \rightarrow 0} \frac{1}{(\sqrt{1 + x^2} + 1) (\sqrt{1 + \sqrt{1 + x^2}} + \sqrt{2})} \\ &= \frac{1}{(1 + 1)(\sqrt{2} + \sqrt{2})} = \frac{1}{4\sqrt{2}} \end{aligned}$$

6. If

$$f(x) = [x]$$

then $f(x)$

- a. is continuous along interval $[1, 2]$
- b. is not continuous along interval $[1, 2]$
- c. $\lim_{x \rightarrow 2^-} f(x) = 2$
- d. $\lim_{x \rightarrow 1^+} f(x) = 2$

Answer: **b**

Solution:

Let $f(x) = [x]$ be the greatest integer (floor) function, which is discontinuous at all integers because it has jump discontinuities. In the interval $[1, 2]$, the function is discontinuous at $x = 2$.

Therefore, $f(x)$ is not continuous on $[1, 2]$.

$$\lim_{x \rightarrow 2^-} f(x) = 1 \quad \text{but} \quad f(2) = 2 \Rightarrow \text{Discontinuity at } x = 2$$

$$\lim_{x \rightarrow 1^+} f(x) = 1 \quad \text{not } 2$$

Hence, the correct option is b

7. If

$$f(x) = \begin{cases} 2x + 1, & x < 2 \\ k, & x = 2 \\ 3x - 1, & x > 2 \end{cases}$$

is continuous at $x = 2$, then $k =$

a. 5

b. 3

c. 6

d. 2

Answer: a

Solution:

To make $f(x)$ continuous at $x = 2$, we must have:

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2) = k$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (2x + 1) = 2(2) + 1 = 5$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3x - 1) = 3(2) - 1 = 5$$

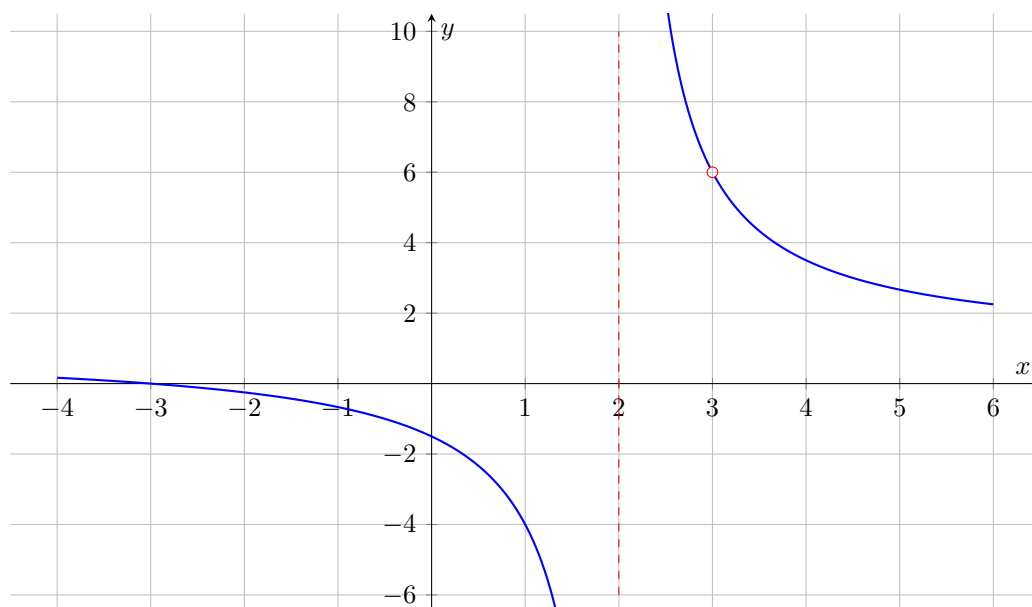
$$\text{So, } \lim_{x \rightarrow 2} f(x) = 5 \Rightarrow k = 5$$

$$k = 5$$

8. The function

$$f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$$

is not continuous at:



- a. $\{2\}$
- b. $\{3\}$
- c. $\{2, 3\}$
- d. $\{-3, 2, 3\}$

Answer: **c**

Solution: The function

$$f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$$

is not continuous at the values of x that make the denominator zero, since division by zero is undefined.

$$\begin{aligned} x^2 - 5x + 6 &= 0 \\ (x - 2)(x - 3) &= 0 \end{aligned}$$

Thus, the function is undefined (and hence not continuous) at:

$$x = 2 \quad \text{and} \quad x = 3$$

Answer: $\{2, 3\}$

9. Evaluate

$$\lim_{x \rightarrow \infty} \sqrt{\frac{x - \sin x}{x + \cos^2 x}}$$

- a. 1
- b. 0
- c. -1
- d. Does not exist

Answer: **a**

Solution:

We divide the numerator and denominator inside the square root by x :

$$= \sqrt{\frac{x \left(1 - \frac{\sin x}{x}\right)}{x \left(1 + \frac{\cos^2 x}{x}\right)}} = \sqrt{\frac{1 - \frac{\sin x}{x}}{1 + \frac{\cos^2 x}{x}}}$$

Since $-1 \leq \sin x \leq 1$, it follows that $-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$, and since $0 \leq \cos^2 x \leq 1$, we have $0 \leq \frac{\cos^2 x}{x} \leq \frac{1}{x}$.

Thus, the expression inside the square root is squeezed as follows:

$$\sqrt{\frac{1 - \frac{1}{x}}{1 + \frac{1}{x}}} \leq \sqrt{\frac{1 - \frac{\sin x}{x}}{1 + \frac{\cos^2 x}{x}}} \leq \sqrt{1 + \frac{1}{x}}$$

Taking the limit as $x \rightarrow \infty$ of all parts:

$$\lim_{x \rightarrow \infty} \sqrt{\frac{1 - \frac{1}{x}}{1 + \frac{1}{x}}} = \sqrt{1} = 1, \quad \lim_{x \rightarrow \infty} \sqrt{1 + \frac{1}{x}} = \sqrt{1} = 1$$

By the Squeeze Theorem, we conclude:

1

10. For the function $f : f(x) = x + \frac{1}{x}$, the value of c in the Lagrange's mean value theorem for the function f , where $x \in [1, 3]$

- a. $\sqrt{5}$
- b. $\sqrt{3}$
- c. $\sqrt{2}$
- d. 1

Answer: **b**

Solution:

$$f(1) = 1 + \frac{1}{1} = 2$$

$$f(3) = 3 + \frac{1}{3} = \frac{10}{3}$$

$$\frac{f(3) - f(1)}{3 - 1} = \frac{\frac{10}{3} - 2}{2} = \frac{\frac{4}{3}}{2} = \frac{2}{3}$$

$$f'(x) = 1 - \frac{1}{x^2}$$

$$1 - \frac{1}{c^2} = \frac{2}{3} \implies \frac{1}{c^2} = \frac{1}{3} \implies c^2 = 3 \implies c = \sqrt{3}$$

11. For the function $f : f(x) = \sqrt{\sin(3x + 1)}$, the derivative $f'(x) =$

- a. $3 \cos(3x + 1)$
- b. $\frac{1}{2\sqrt{\sin(3x + 1)}}$
- c. $\frac{\cos(3x + 1)}{2\sqrt{\sin(3x + 1)}}$
- d. $\frac{3 \cos(3x + 1)}{2\sqrt{\sin(3x + 1)}}$

Answer: **d**

Solution: By using the Chain Rule

$$f'(x) = \frac{d}{dx} (\sin(3x + 1))^{1/2} = \frac{1}{2\sqrt{\sin(3x + 1)}} \cdot \frac{d}{dx} (\sin(3x + 1))$$

$$= \frac{1}{2\sqrt{\sin(3x + 1)}} \cdot 3 \cos(3x + 1)$$

$$= \frac{3 \cos(3x + 1)}{2\sqrt{\sin(3x + 1)}}$$

12. If $f : f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \frac{x^{98}}{98} \dots + \frac{x^2}{2} + x + 1$, then $f'(1) =$

- a. 0
- b. 1
- c. $f'(0)$
- d. $100f'(0)$

Answer: d

Solution:

$$f'(x) = \frac{100x^{99}}{100} + \frac{99x^{98}}{99} + \dots + \frac{2x}{2} + 1 = x^{99} + x^{98} + \dots + x + 1$$

$$f'(1) = \underbrace{1 + 1 + \dots + 1}_{100 \text{ terms}} = 100$$

$$f'(0) = 0 + 0 + \dots + 0 + 1 = 1$$

Thus, $f'(1) = 100f'(0)$.

13. If $f : f(x) = f(x) = x \sin x$, then $f'(\frac{\pi}{2}) =$

- a. 1
- b. 0
- c. $\frac{1}{2}$
- d. -1

Answer: a

Solution:

$$f'(x) = 1 \cdot \sin x + x \cdot \cos x = \sin x + x \cos x$$

$$f'(\frac{\pi}{2}) = \sin(\frac{\pi}{2}) + \frac{\pi}{2} \cos(\frac{\pi}{2}) = 1 + \frac{\pi}{2} \cdot 0 = 1$$

14. Suppose that the functions f , g and their derivatives f' , g' with respect to x have the given values in the table, then at $x = 0$ the value of $\frac{d}{dx}[f(g(x))]$ =

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	5	2
1	3	-4	4	-1

- a. 3
- b. 4
- c. 7
- d. 8

Answer: d

Solution:

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

From the table:

Since $g(0) = 1$, we need $f'(1)$. From the table:

$$\frac{d}{dx}[f(g(0))] = f'(g(0)) \cdot g'(0) = f'(1) \cdot 2 = 4 \cdot 2 = 8$$

15. Using the table in question NO.14, then at $x = 1$ the value of $\frac{d}{dx} \frac{f(x)}{g(x)} =$

- a. 2
- b. $-\frac{16}{17}$

c. $-\frac{13}{16}$

d. None of them

*Answer: c**Solution:*

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

$$f(1) = 3, \quad g(1) = -4, \quad f'(1) = 4, \quad g'(1) = -1$$

$$\left. \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] \right|_{x=1} = \frac{4 \cdot (-4) - 3 \cdot (-1)}{(-4)^2} = \frac{-16 + 3}{16} = \frac{-13}{16}$$

وَفَقَّكُمُ اللَّهُ رَبُّ الْعَالَمِينَ

STEM High School for Boys - 6th of October

